Lesson 2

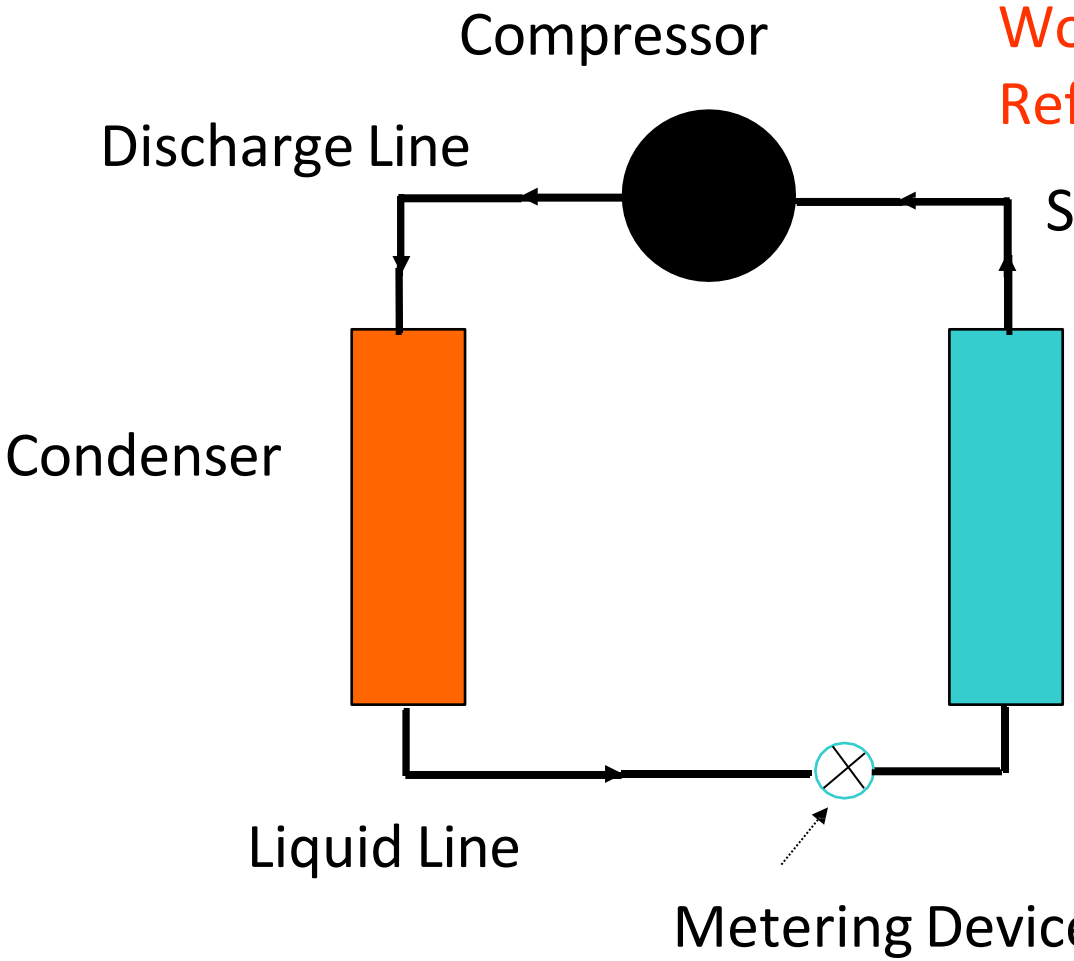
AIR-CONDITIONING (REFRIGERATION) EQUIPMENT

Objectives

- Describe the construction of the different types of compressors, metering devices, evaporators, condensers and refrigerant circuit accessories.

- Describe the construction and maintenance of room air conditioners, split units, air-cooled packaged units, water cooled packaged units and water-cooled chillers.

Vapour Compression Refrigeration System



Working Fluid - Refrigerant

- Performance of each component influenced by other components
- Fitted with many accessories

Air-conditioning (Refrigeration) Equipment

- Window (Room) Air Conditioners
- Air-cooled Split Units
- Air-cooled Packaged Units
- Water-cooled Packaged Units
- VRF Units

- Air-cooled Chillers
- Water-cooled Chillers

Compressor - Classification

- Method of compression:

Type	Capacity Range
Reciprocating	5 to 500 kW
Centrifugal	450 kW to 20 MW
Rotary - Vane	5 – 35kW
- Screw	450 kW to 1500 kW
- Scroll	5 – 50 kW

Compressor - Classification

Construction

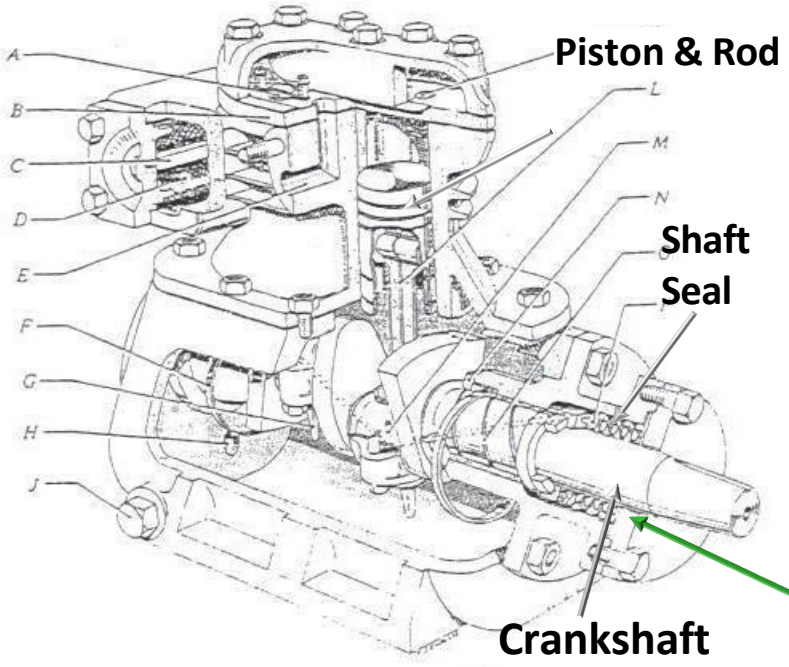
- Open type
- Hermetic
- Serviceable Hermetic (Semi-hermetic)





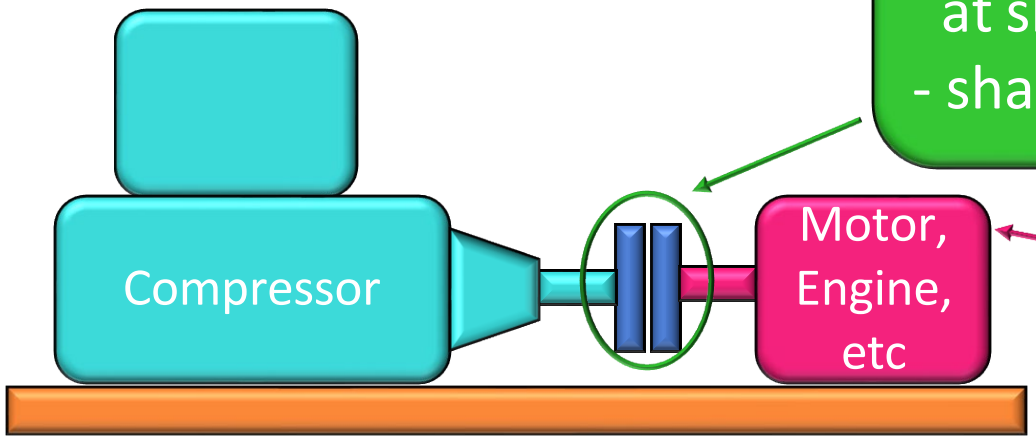


Compressors – Open Type





Main disadvantages
- refrigerant leakage at shaft seals
- shaft alignment,

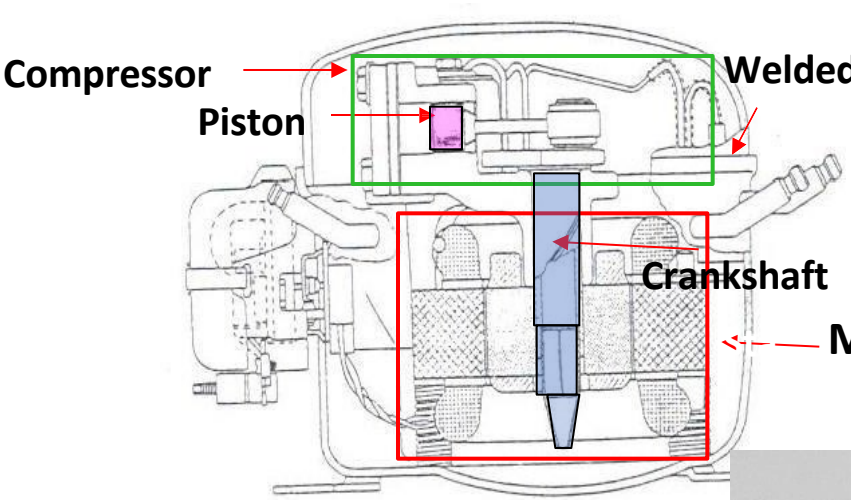


Separate external drive

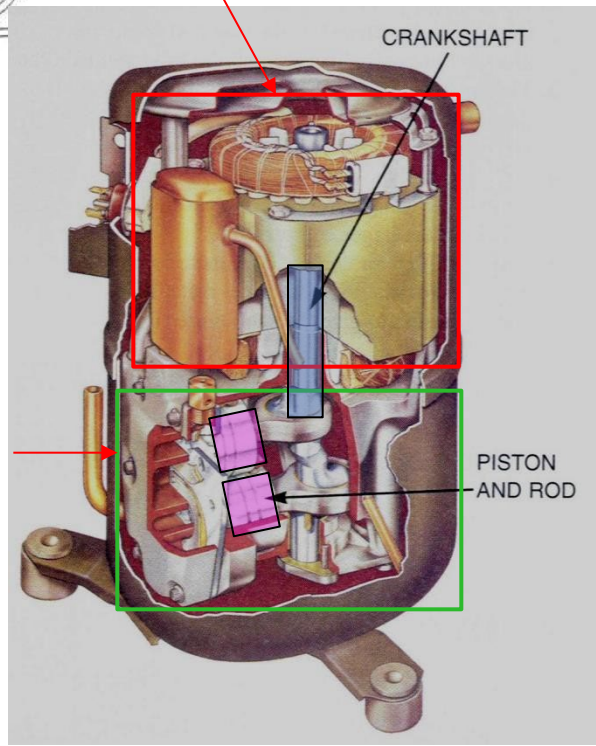
Compressors

Open	Hermetic	Serviceable Hermetic
Separate drive	Motor is an integral part of compressor.	
Bolted construction	Completely welded construction	Bolted construction - easier dismantling for repairs
	Small capacities	Medium to large capacities

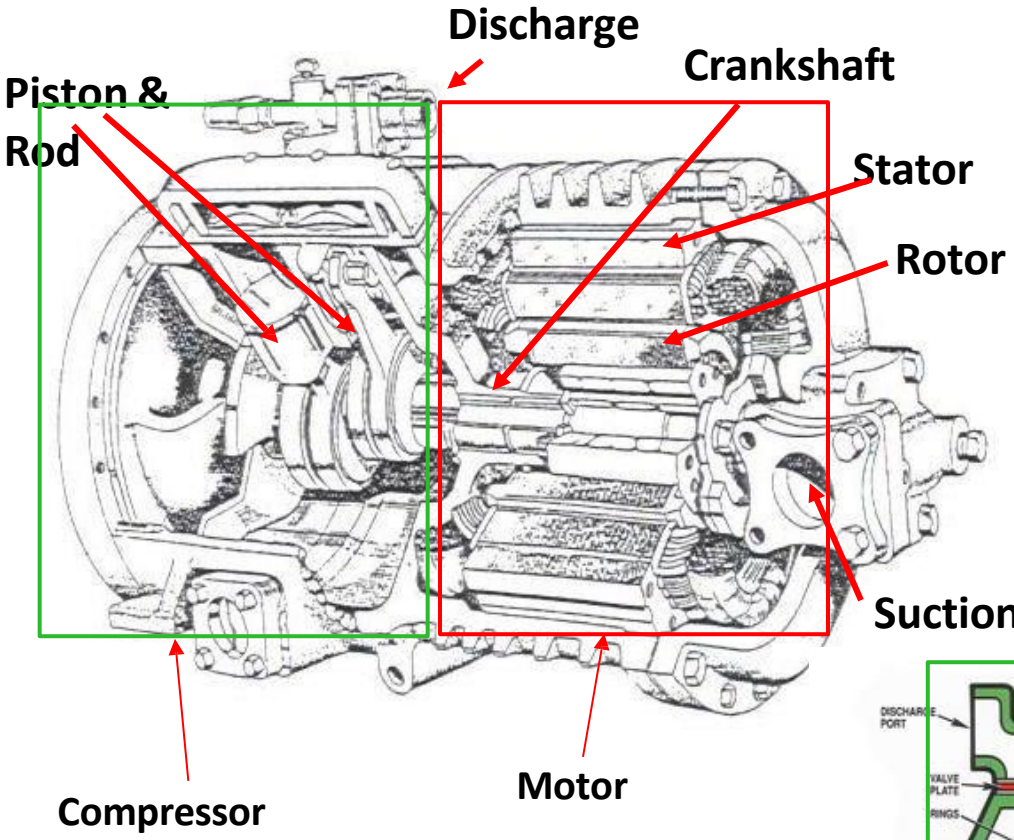
Hermetic Compressors



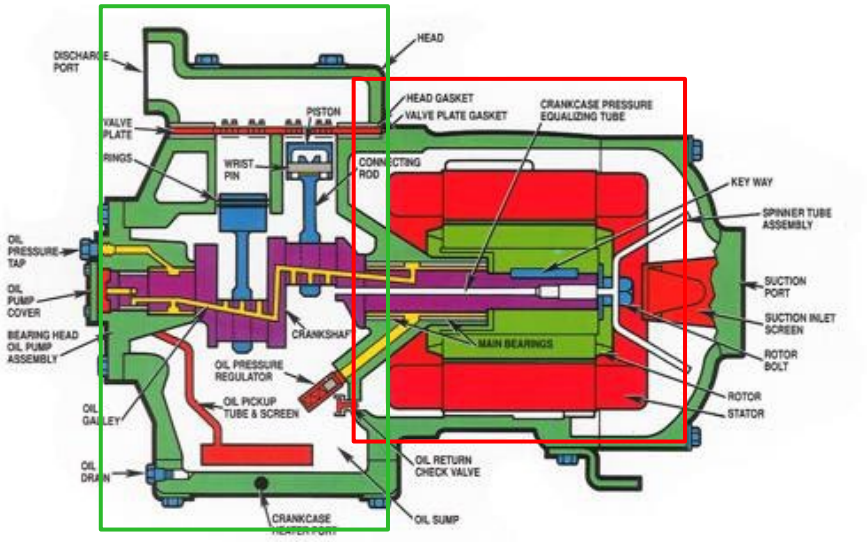




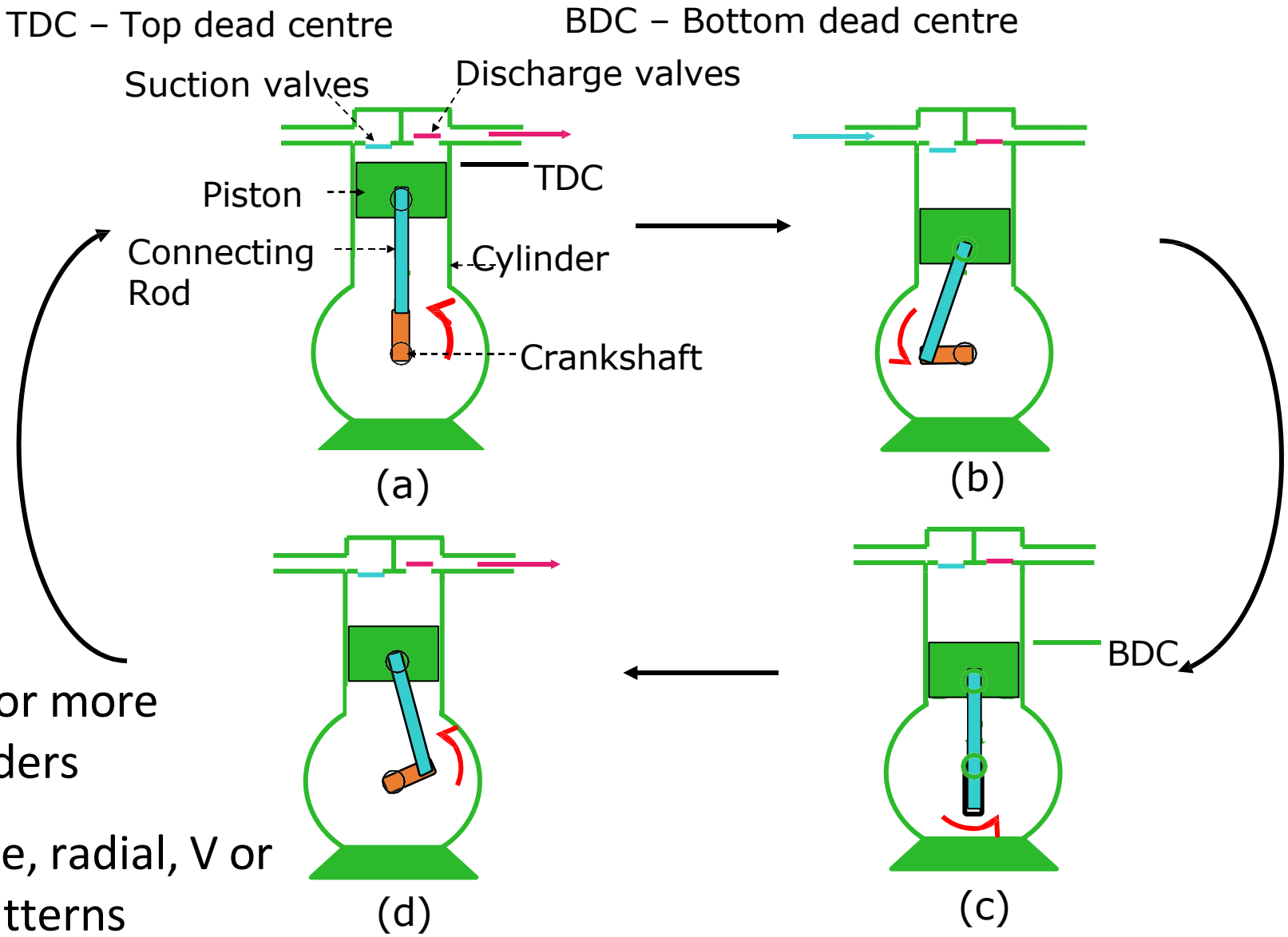
Semi-Hermetic Compressors







Reciprocating Compressors



- One or more cylinders
- In-line, radial, V or W patterns

Reciprocating Compressors



Reciprocating Compressors

Capacity Control

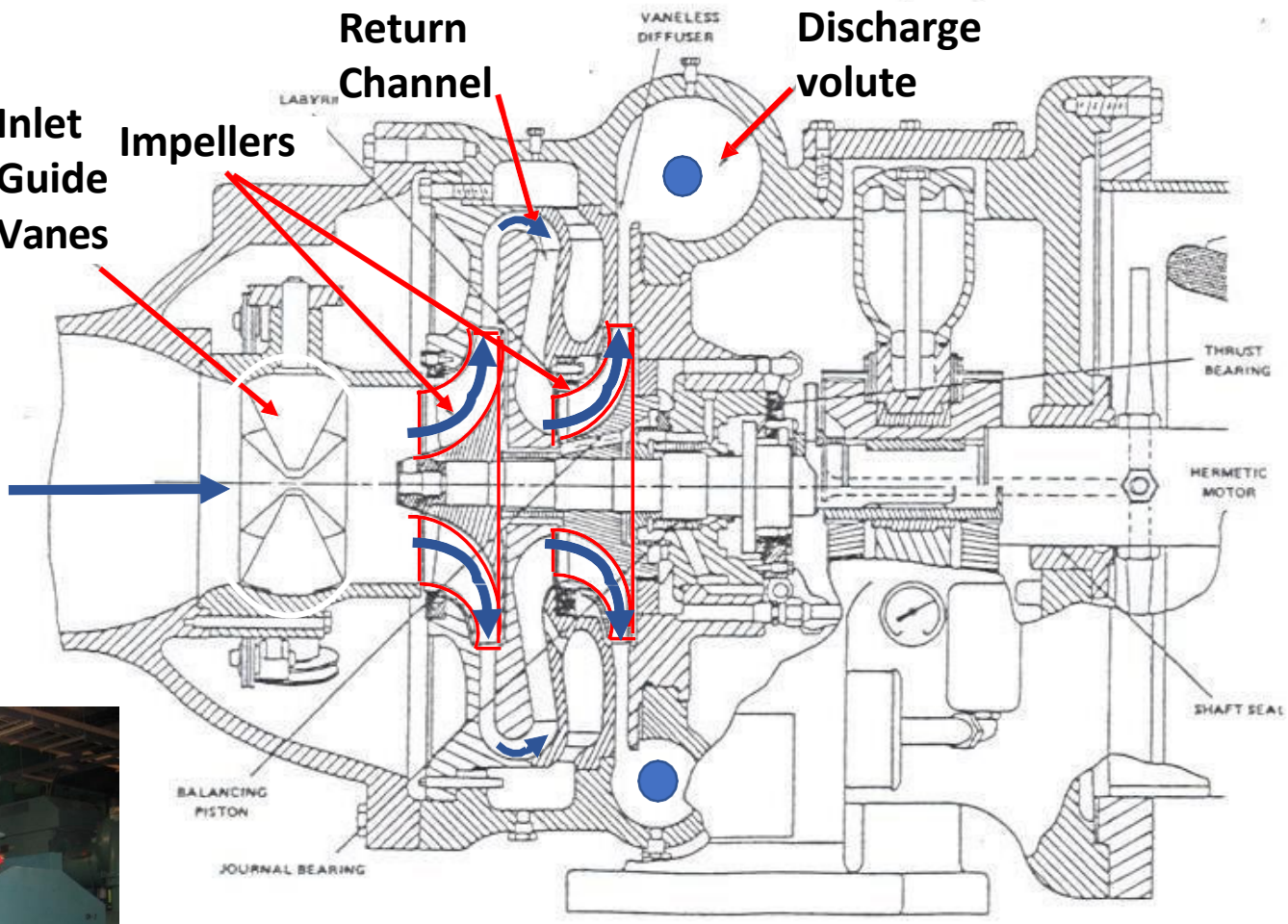
- On-OFF Control
- Unloading of one or more cylinders of a multi-cylinder compressor
- Speed Control
- Multiple compressors

Centrifugal Chillers

- Low pressure, low temperature drawn into eye of impeller
- Forced radially outwards by centrifugal force at high pressure and velocity
- Collects in casing (or volute)
- Casing converts kinetic energy to pressure energy
- Magnetic bearings – no lubrication system

Centrifugal Compressors

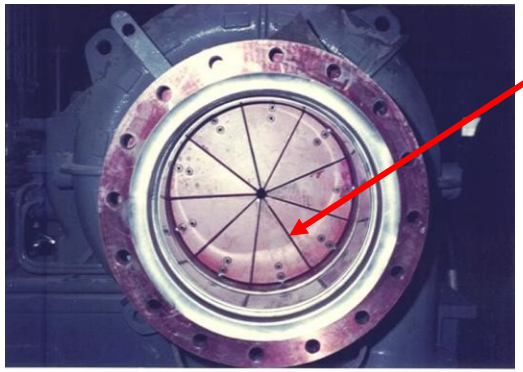




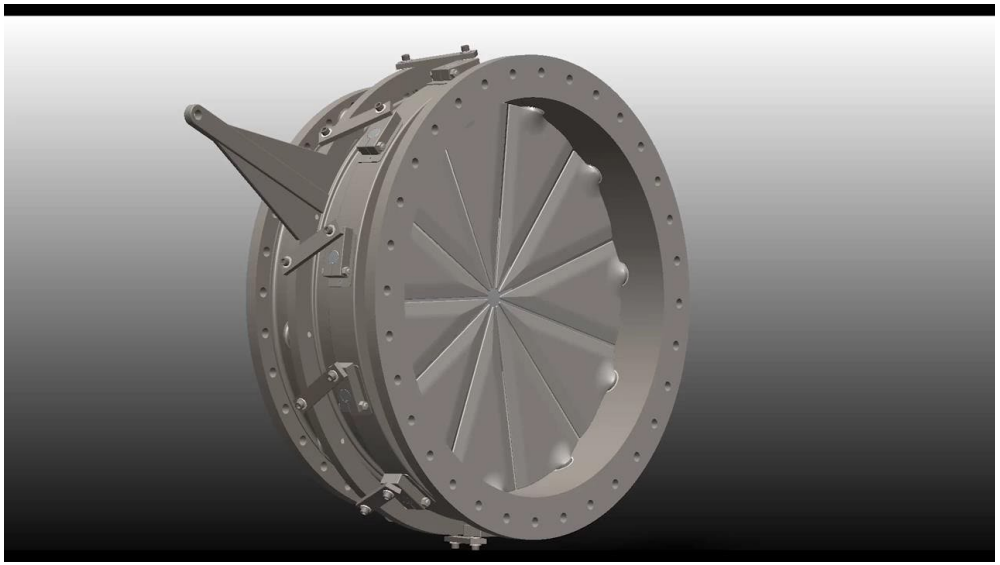
Centrifugal Compressor

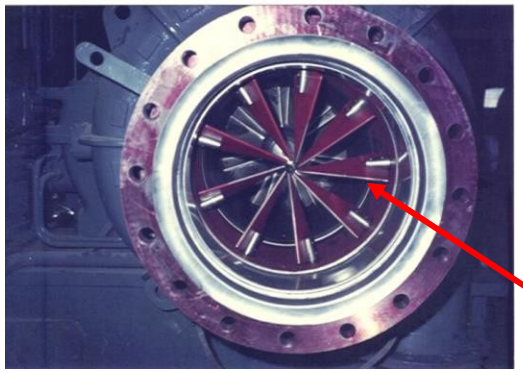


Impeller & Variable Inlet Guide Vanes



Vanes fully closed





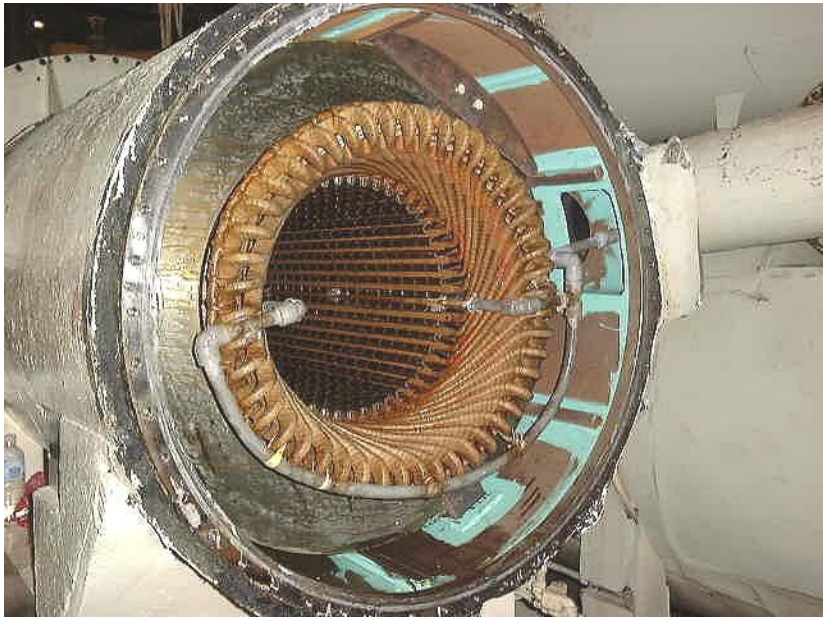
Vanes almost fully open





Rotor & Stator





Centrifugal Compressors



Centrifugal Compressors – Magnetic Bearings

McQuay® International

a member of DAIKIN group

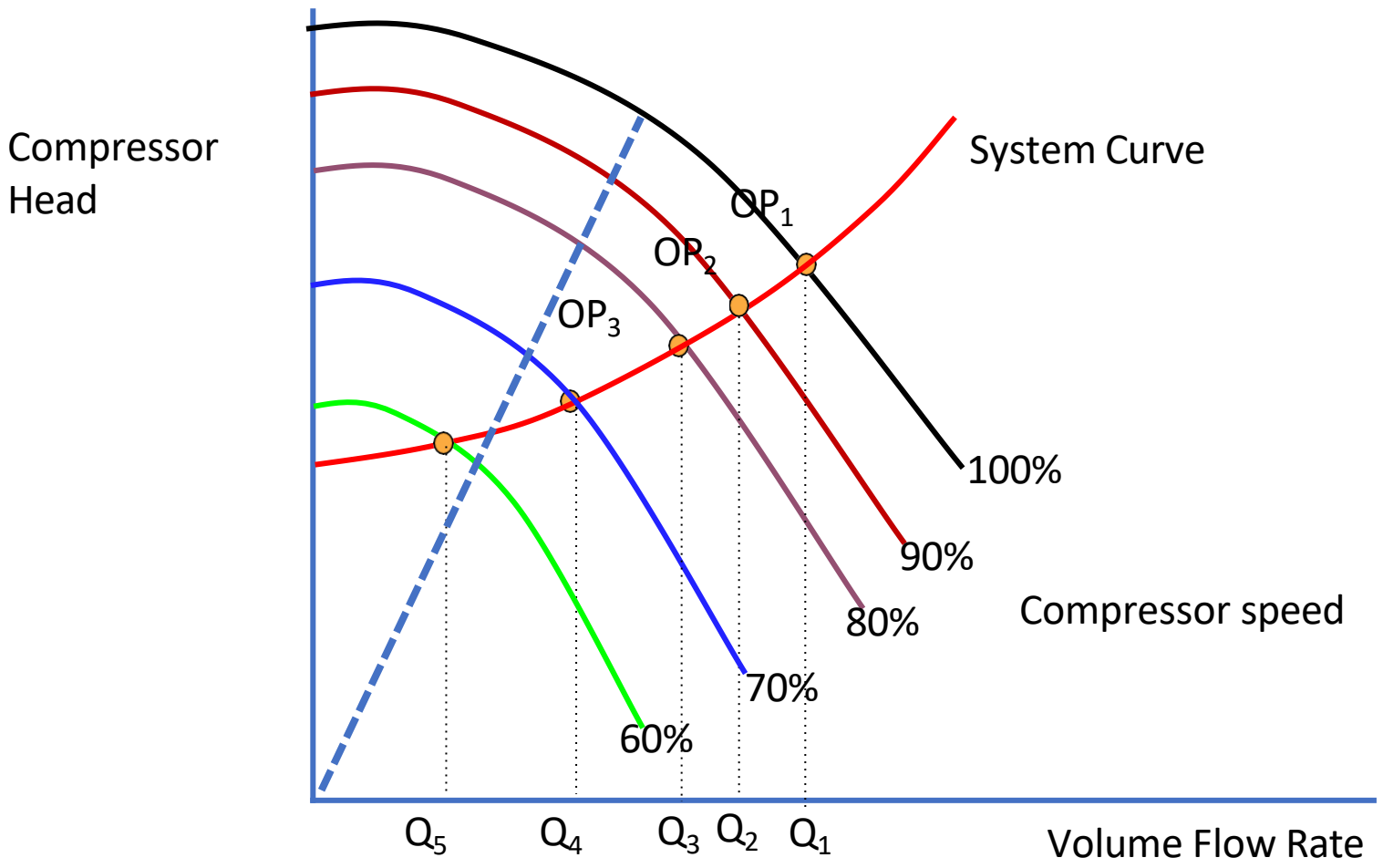
Centrifugal Compressors

Capacity Control

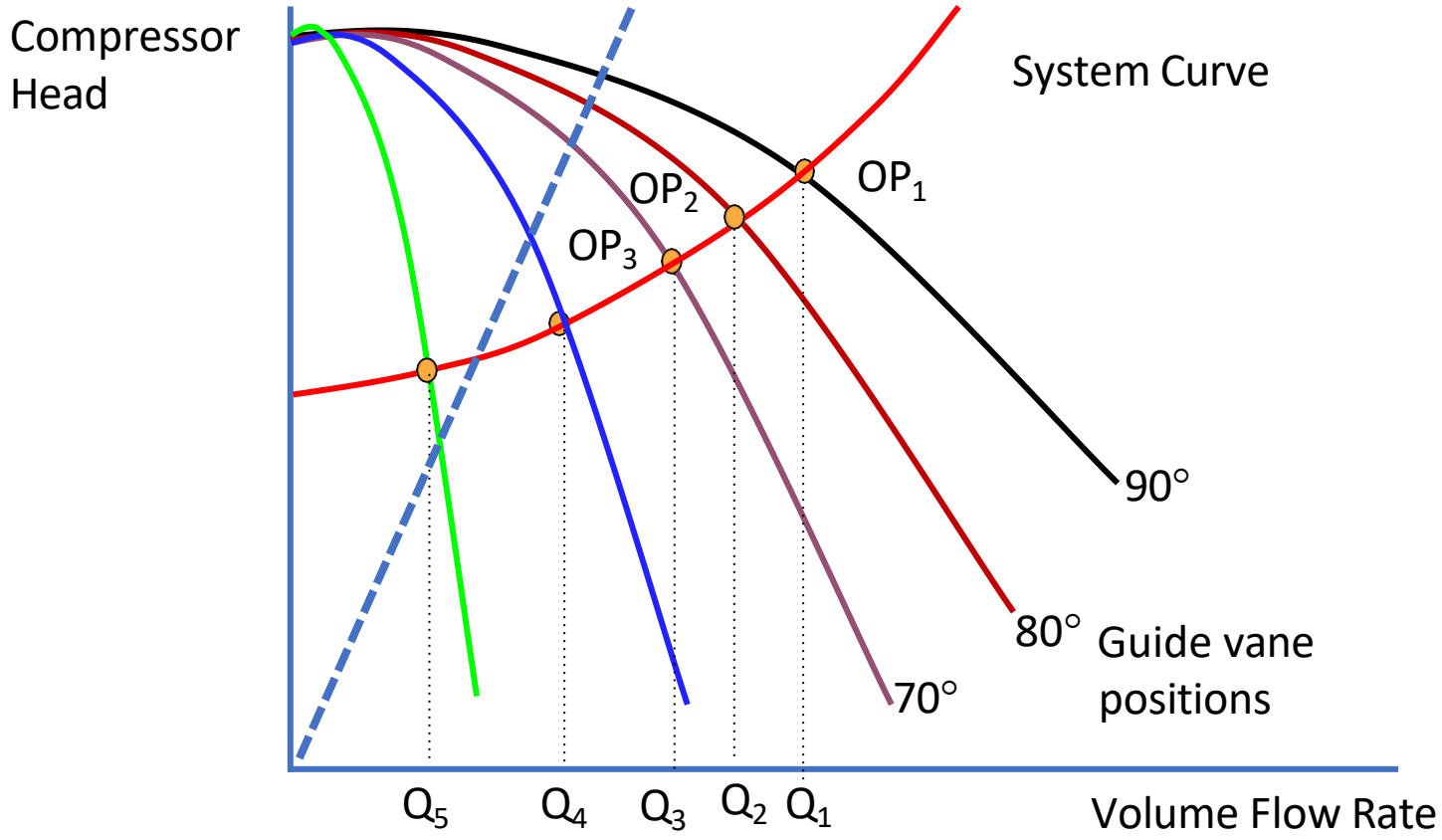
Speed Control

Variable Inlet Guide Vanes

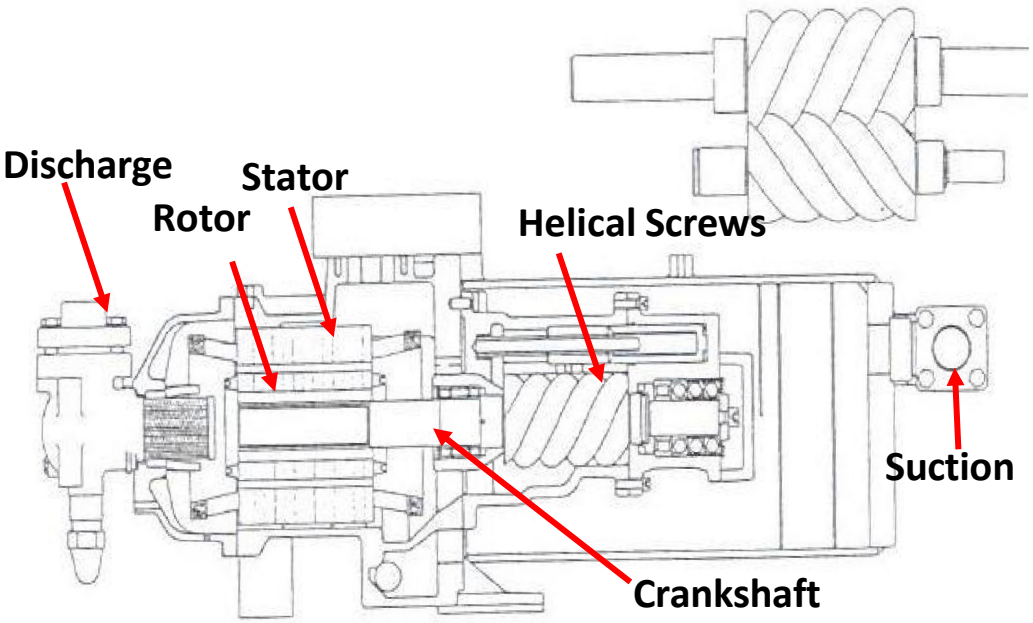
Speed Control

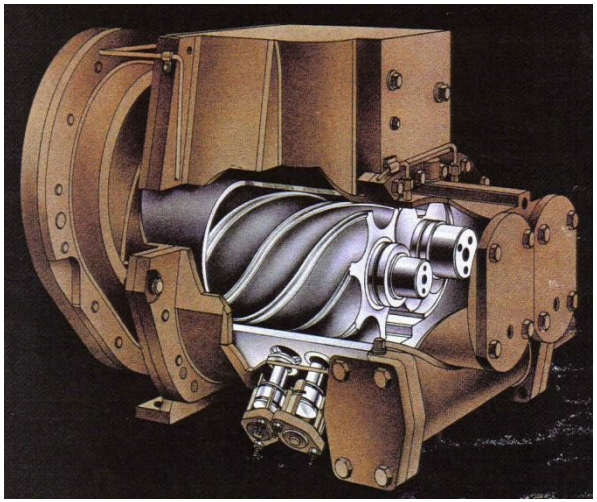


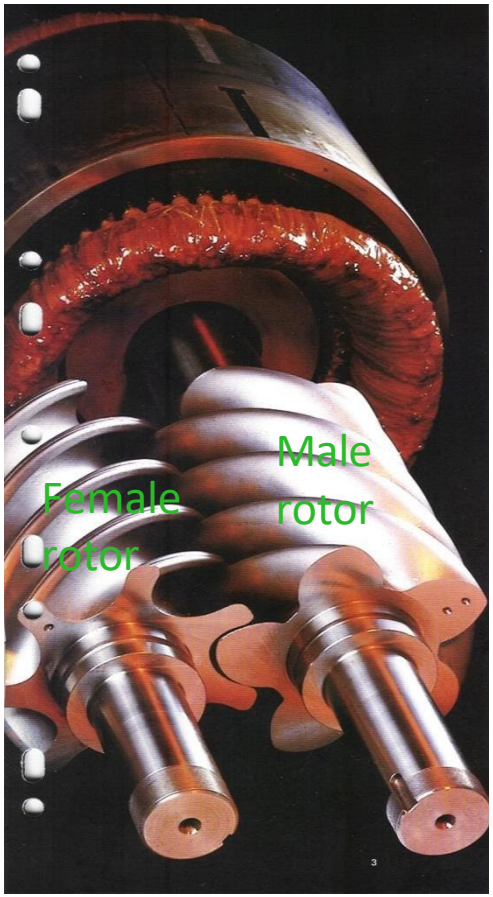
Variable Inlet Guide Vanes



Screw Compressors







Screw Compressors

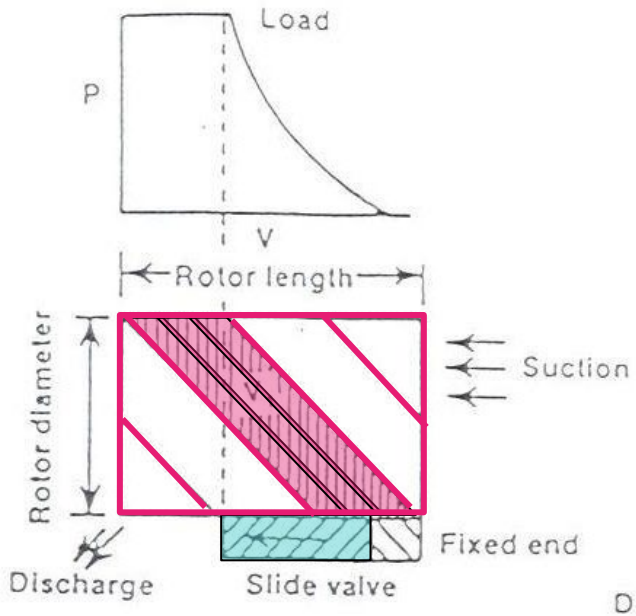
Screw Compressors



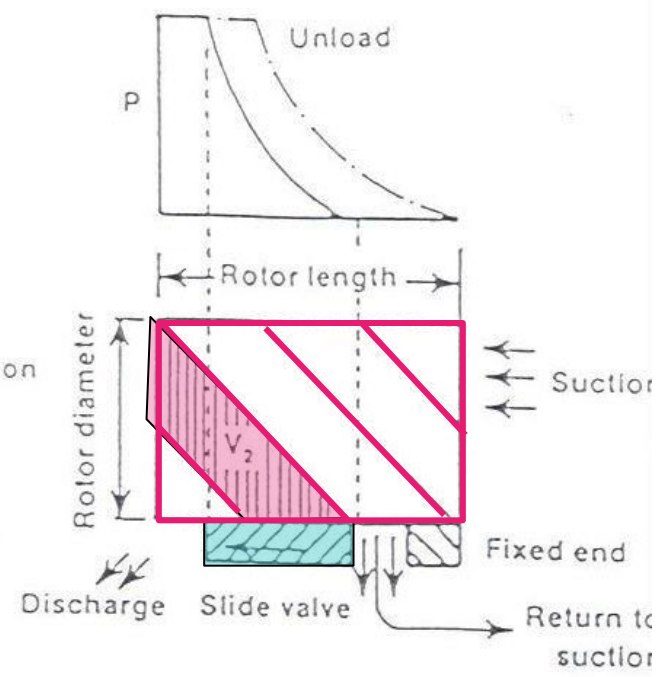
Screw Compressors

Capacity Control

Slide Valve

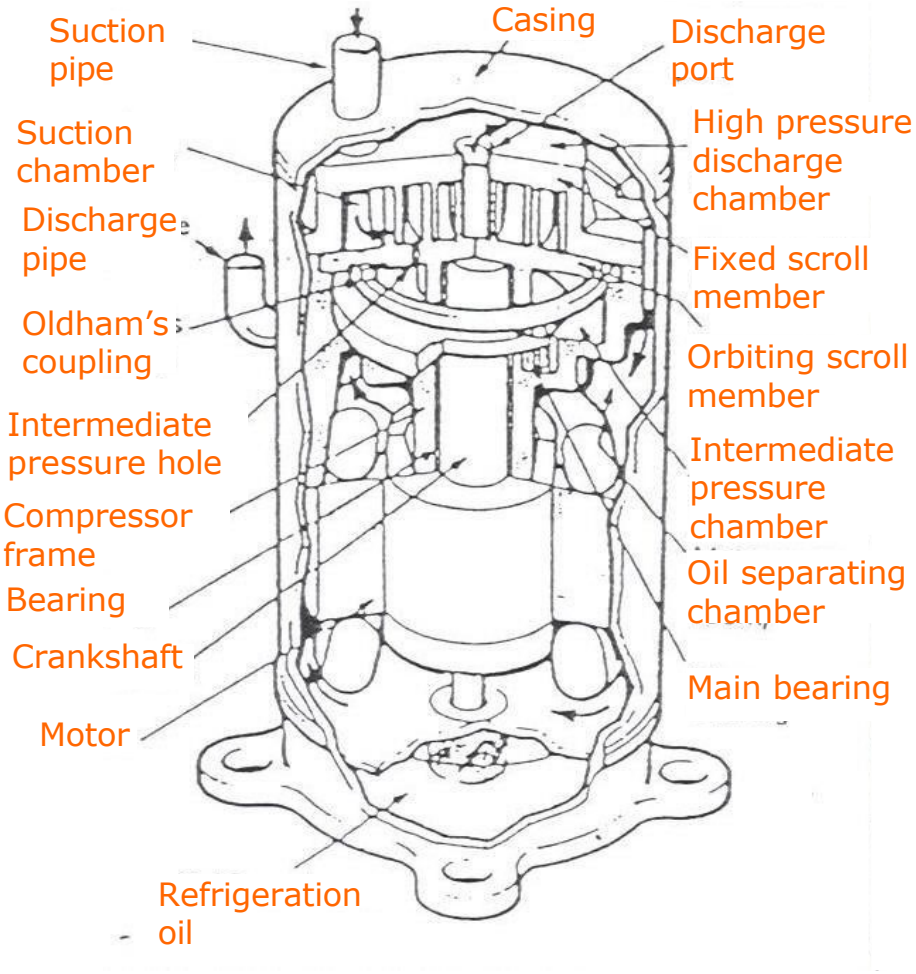


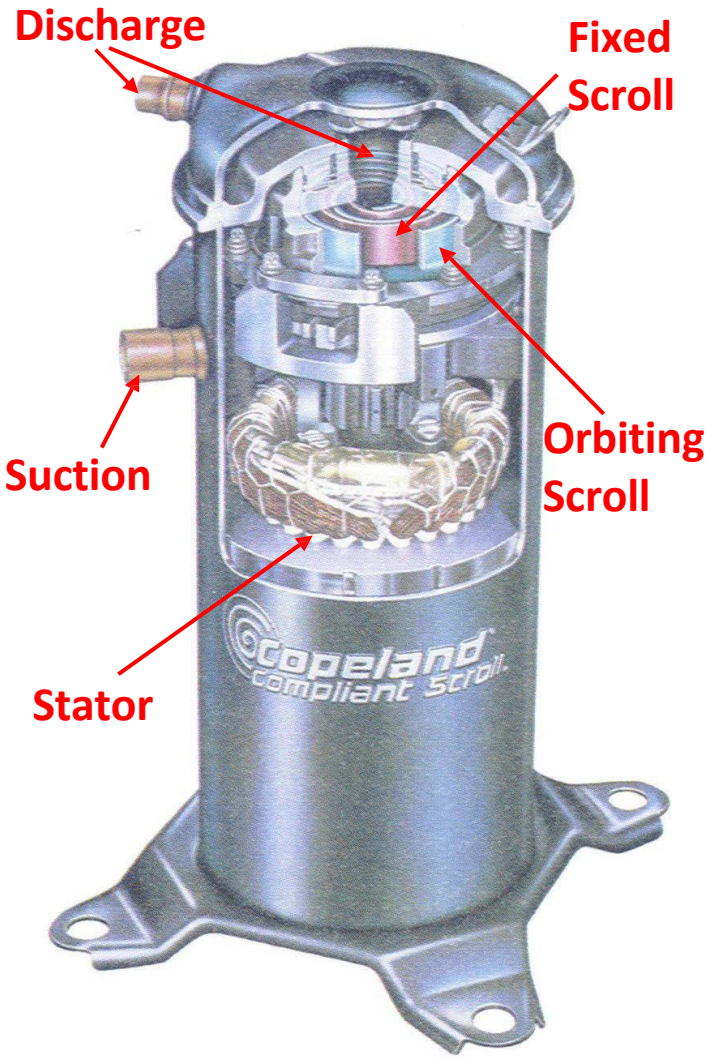
Speed Control



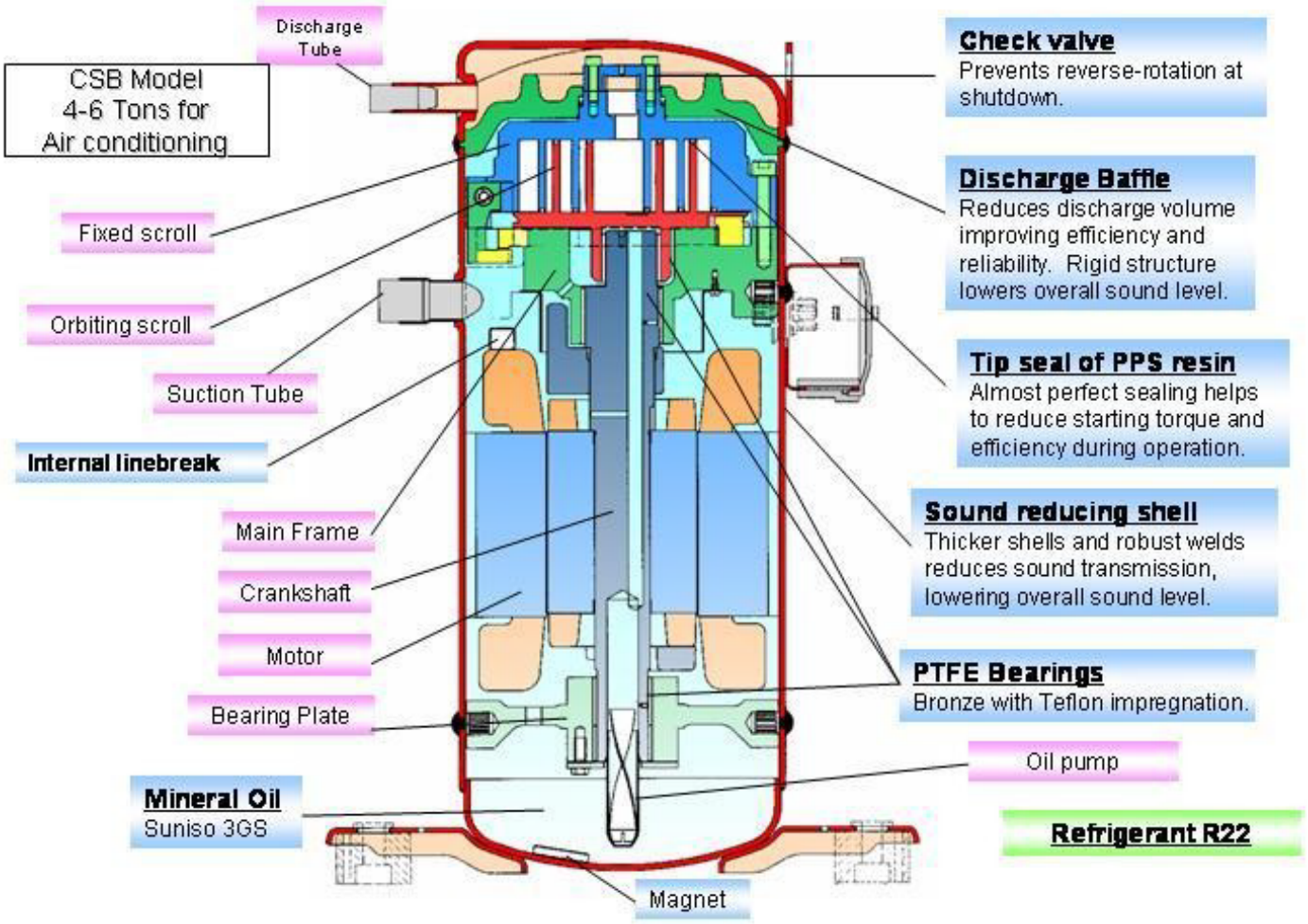
Scroll Compressors







Scroll Compressors

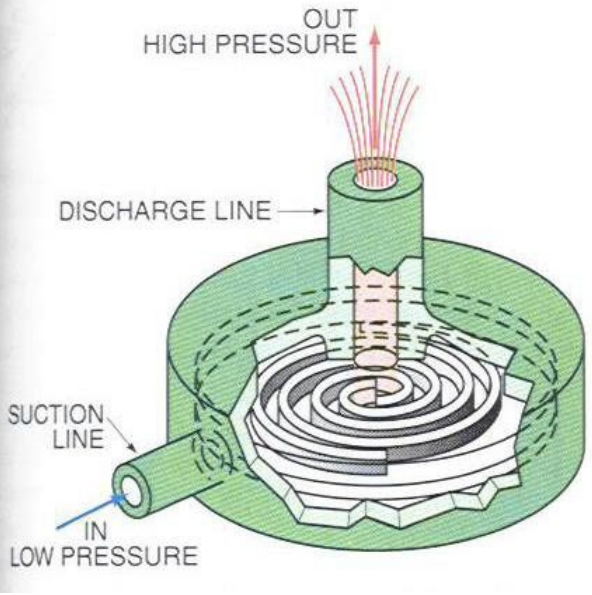


Scroll Compressors

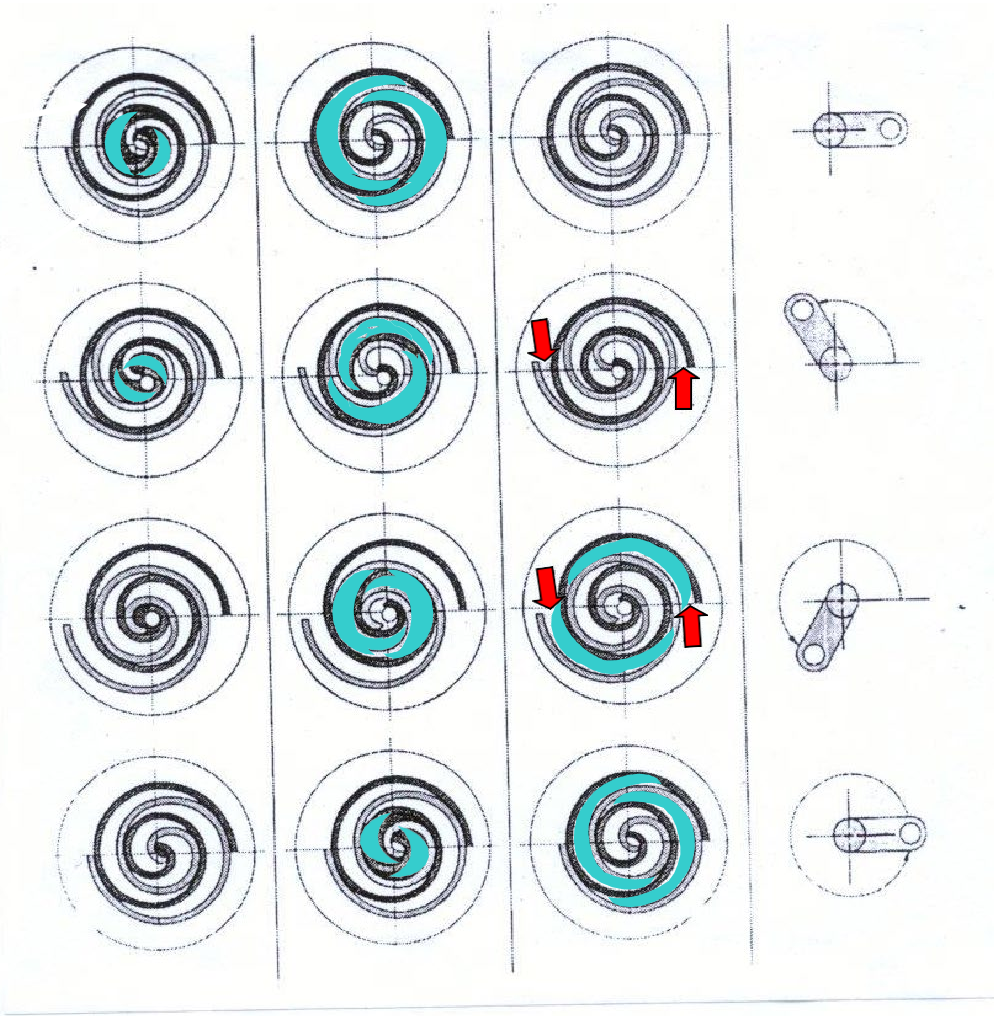


Fixed Scroll

Orbiting Scroll

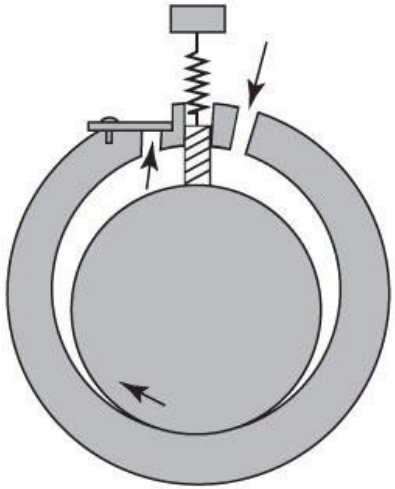


3rd Orbit 2nd Orbit 1st Orbit

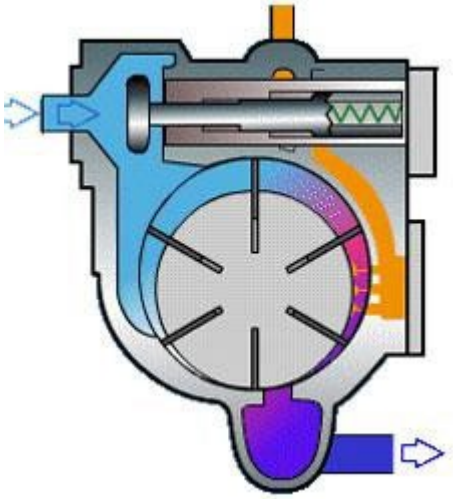


Vane Compressors

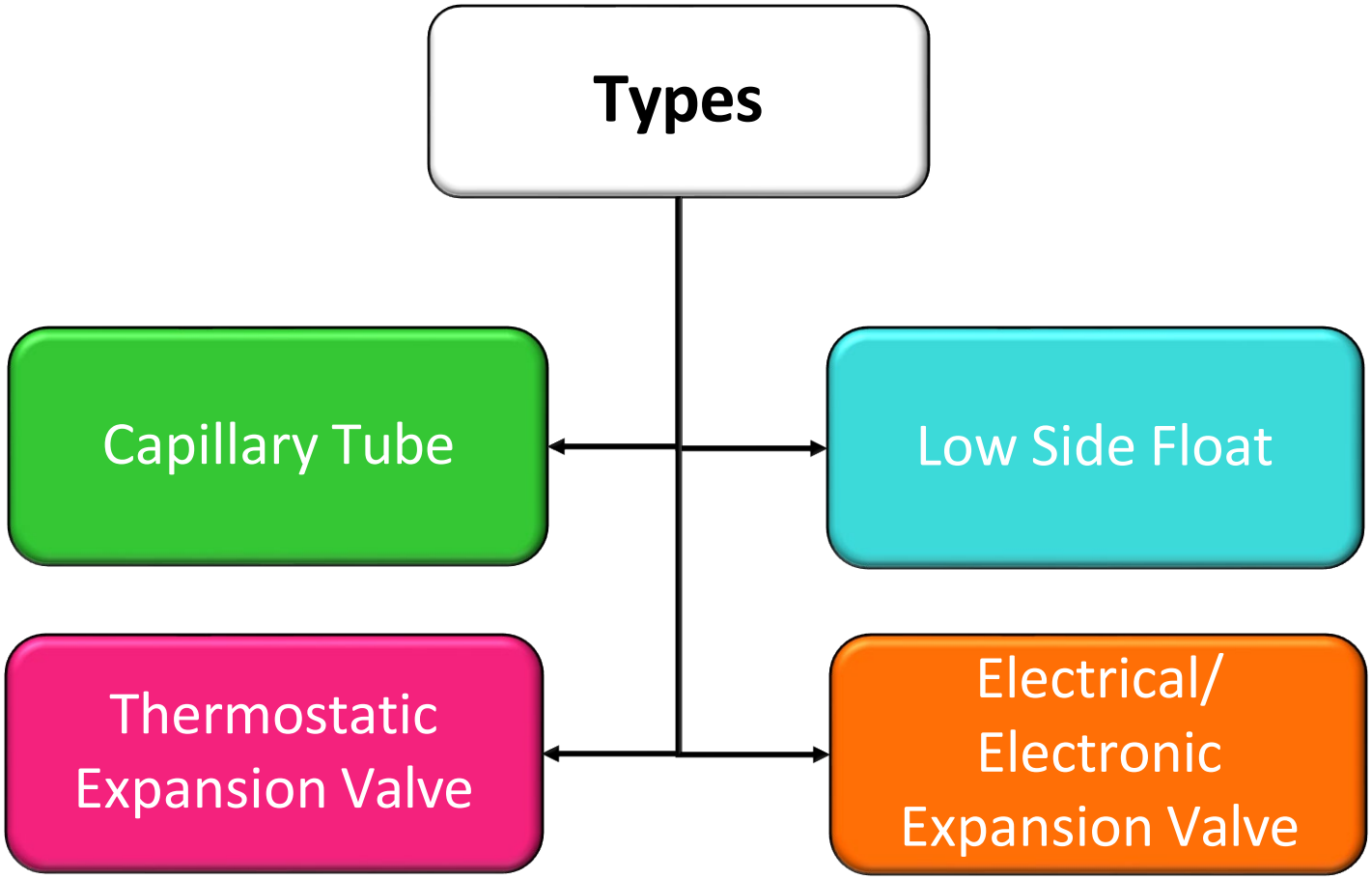
Roller



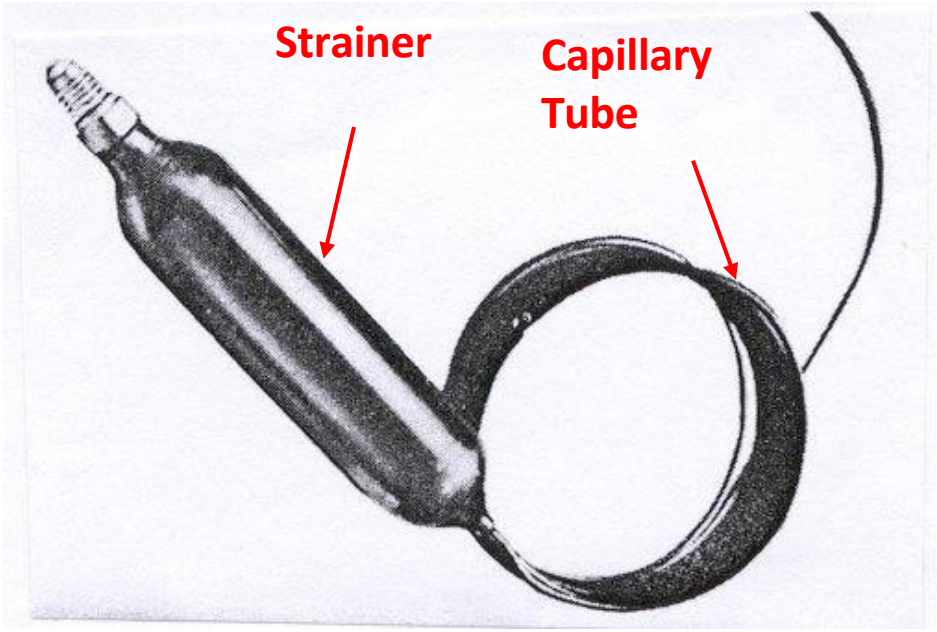
Multi-Vane



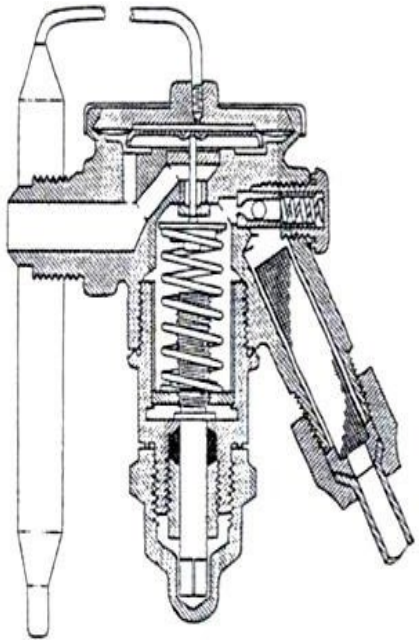
Metering Devices

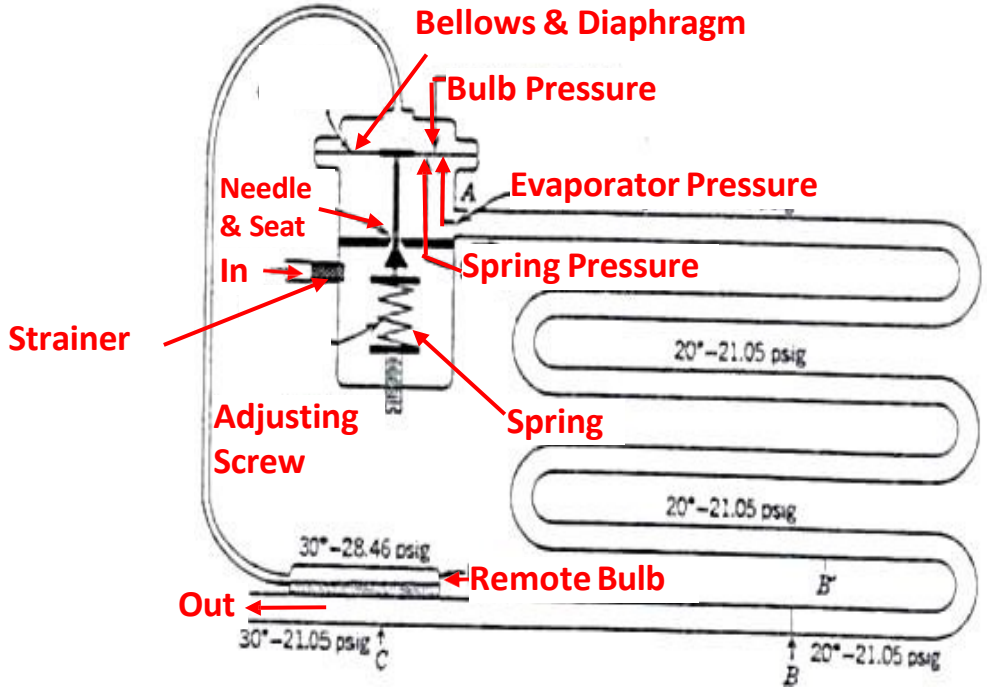


Capillary Tube

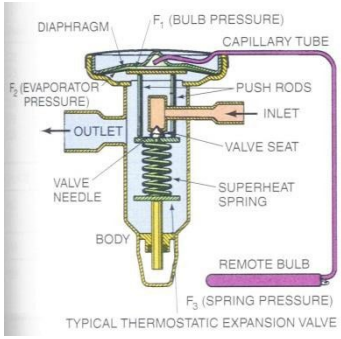


Thermostatic Expansion Valve

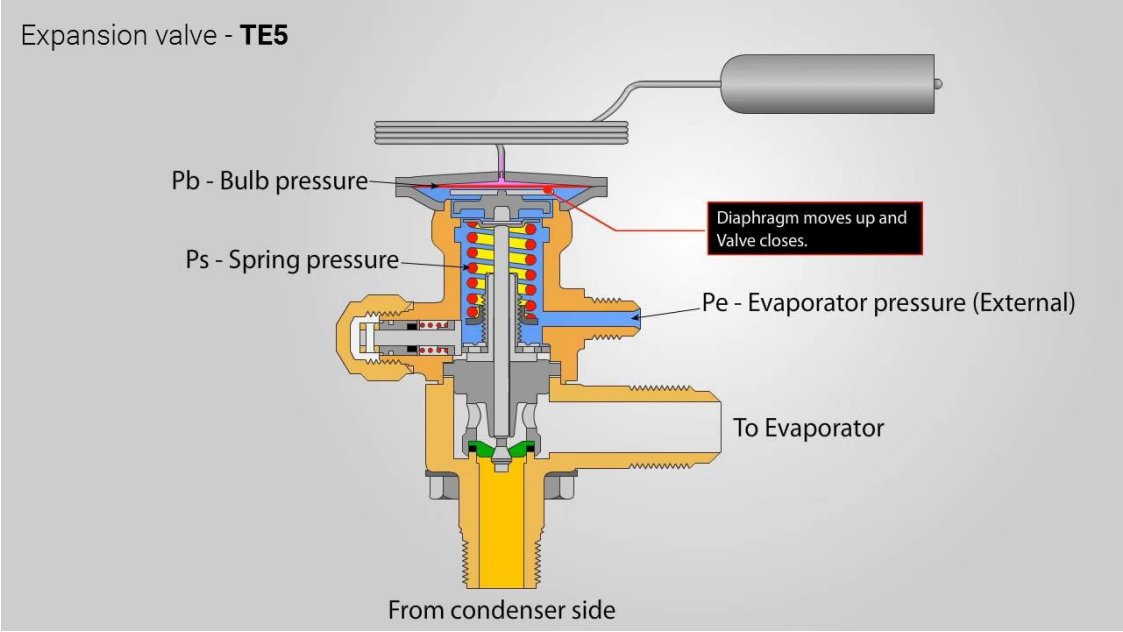


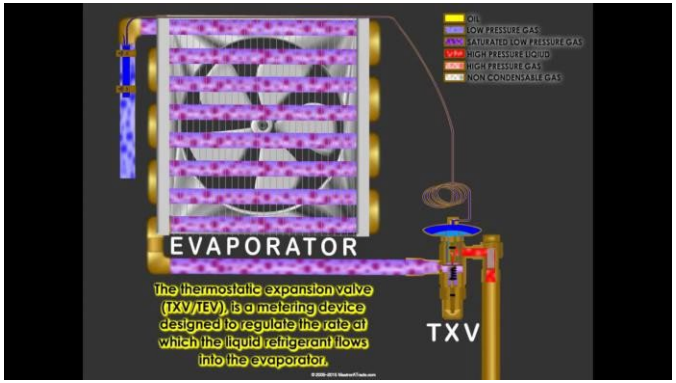




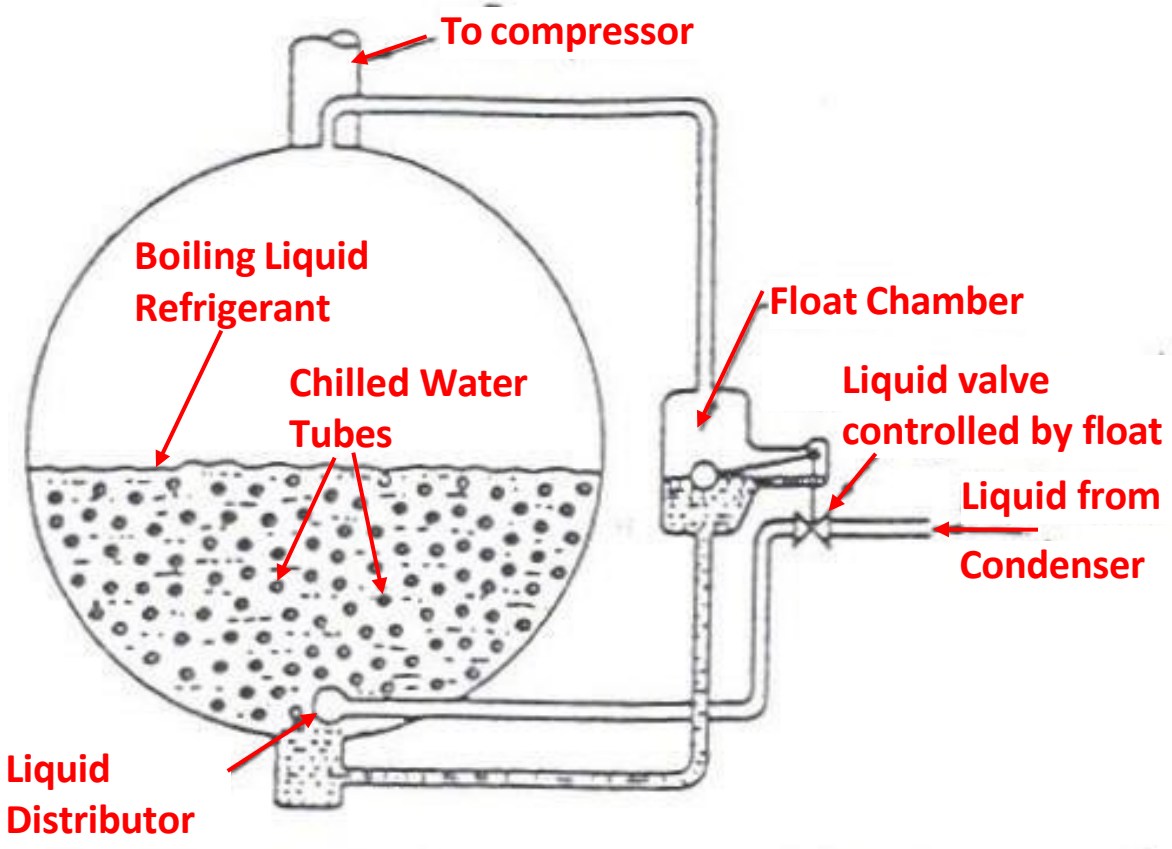


Thermostatic Expansion Valve

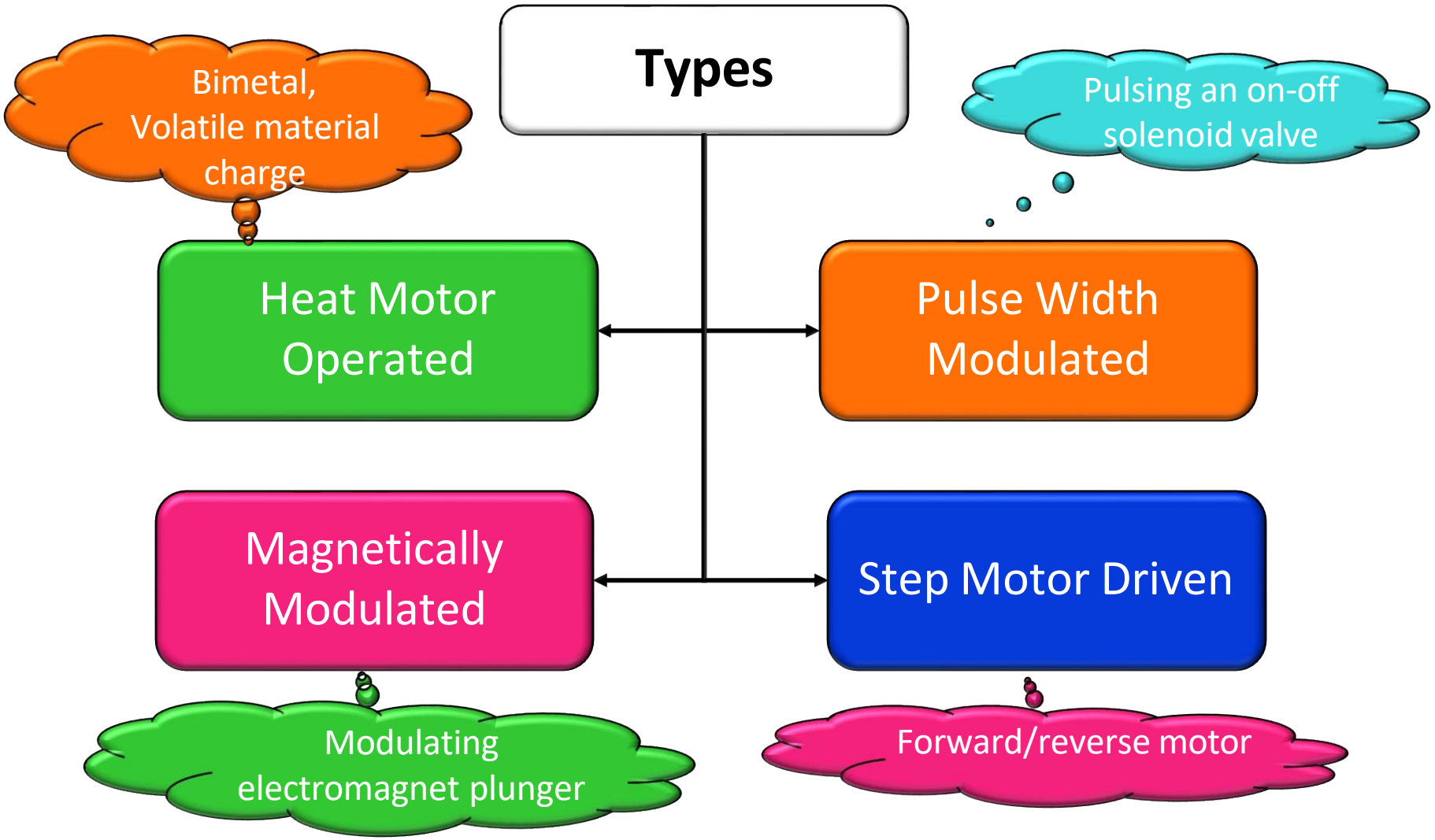




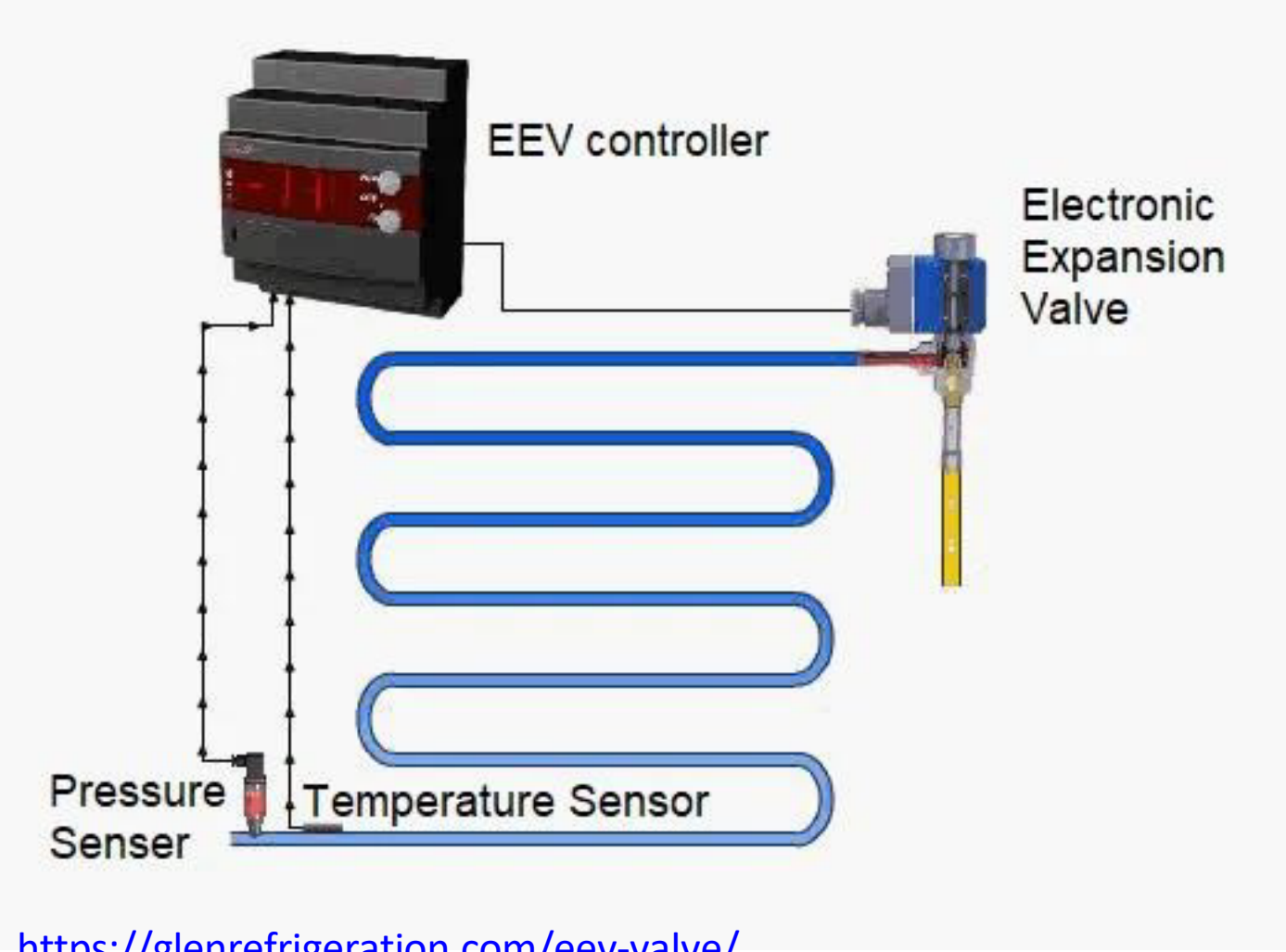
Low Side Float



Electrical/Electronic Expansion Valve



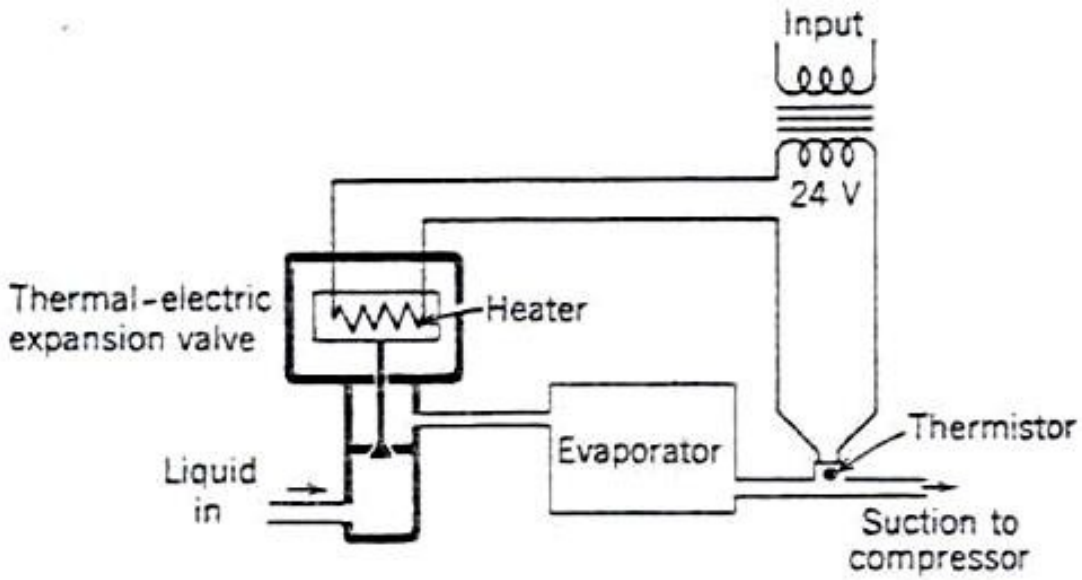
Electrical/Electronic Expansion Valve

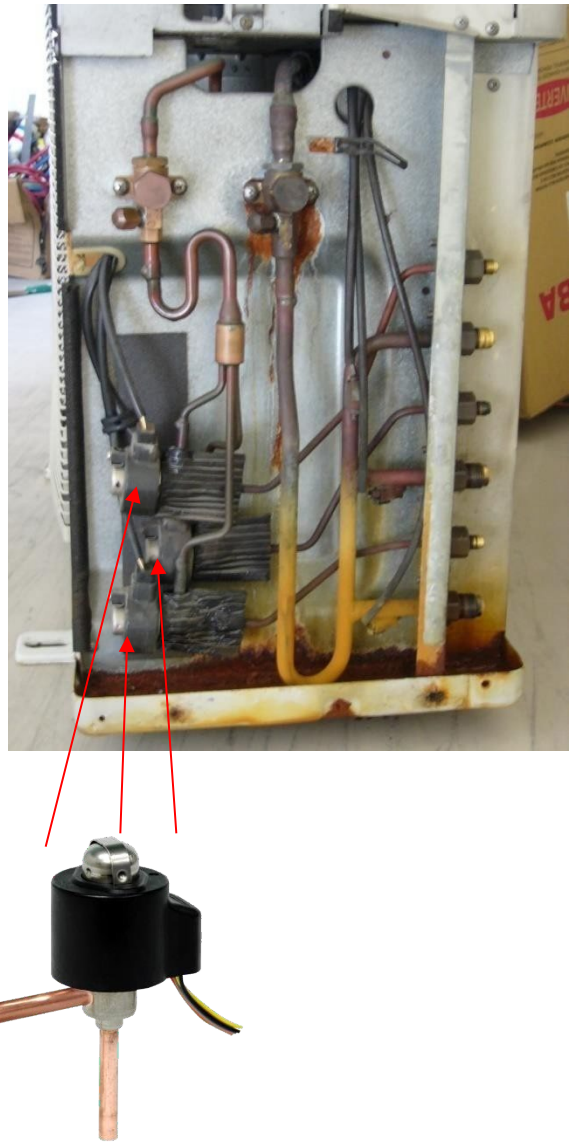


https://glenrefrigeration.com/eev-valve/

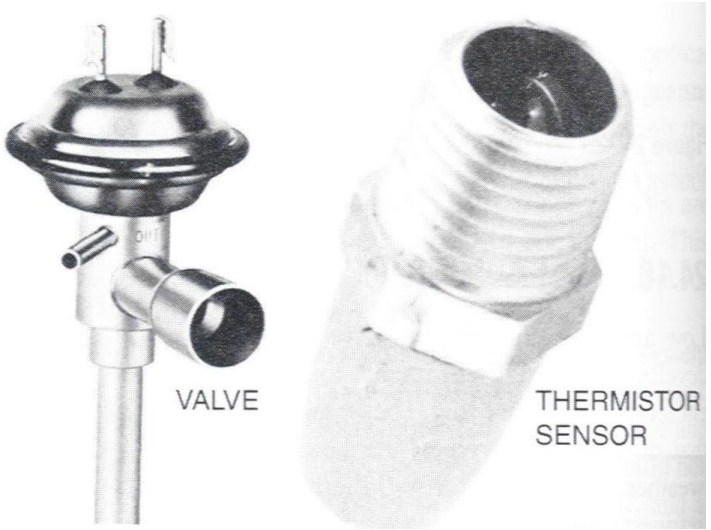
Electronic Expansion Valve

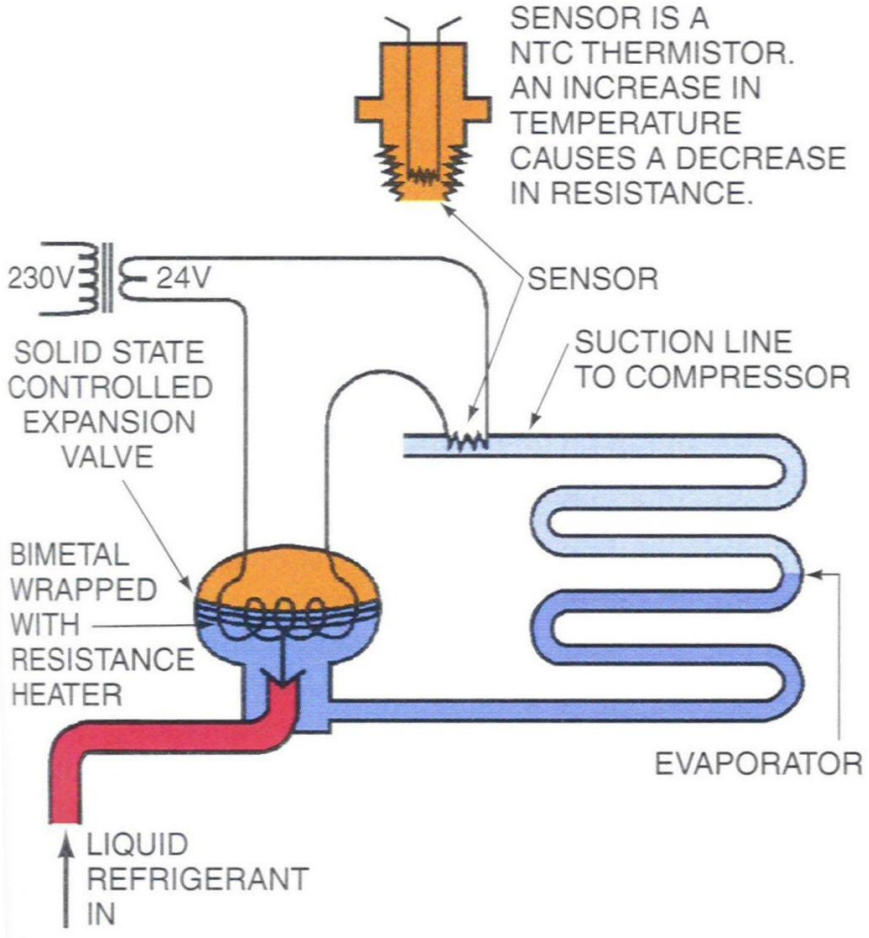
- Maintain **degree of superheat** of refrigerant vapour at outlet of evaporator.



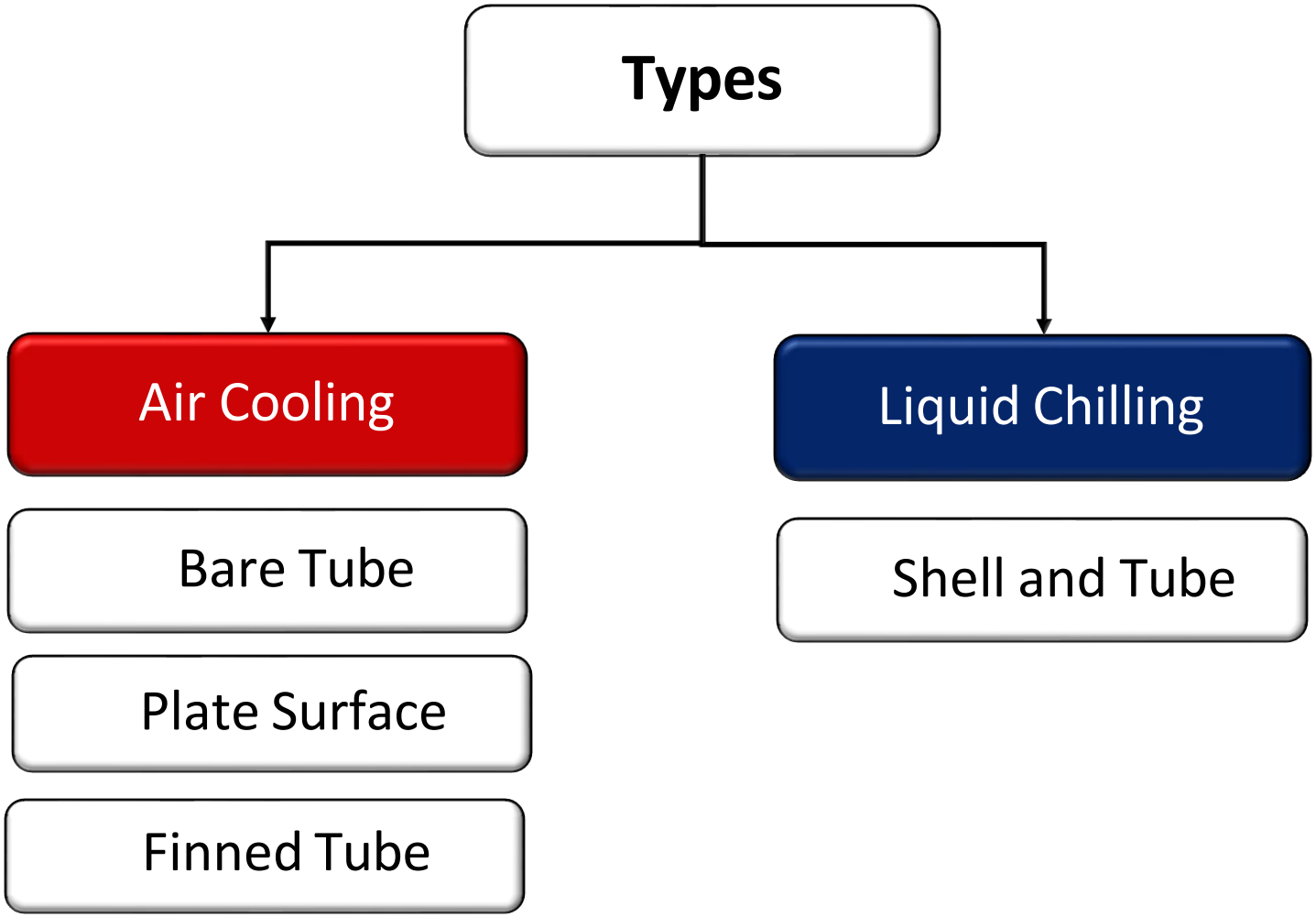


Electronic Expansion Valve

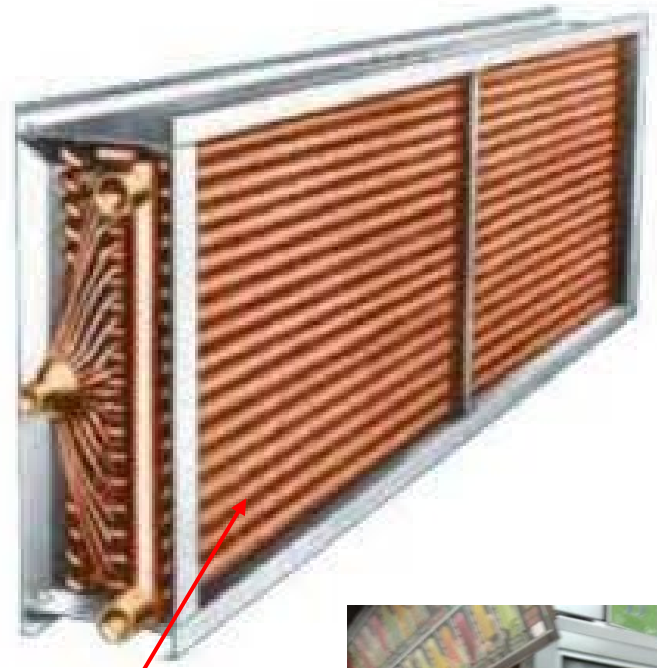


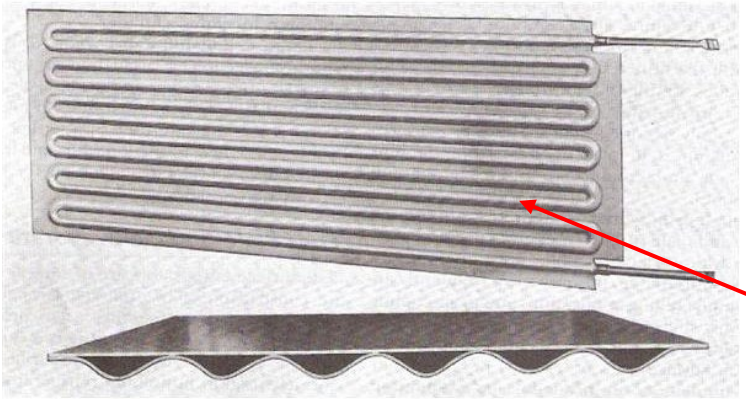


Evaporators



Bare Tube and Plate Evaporators





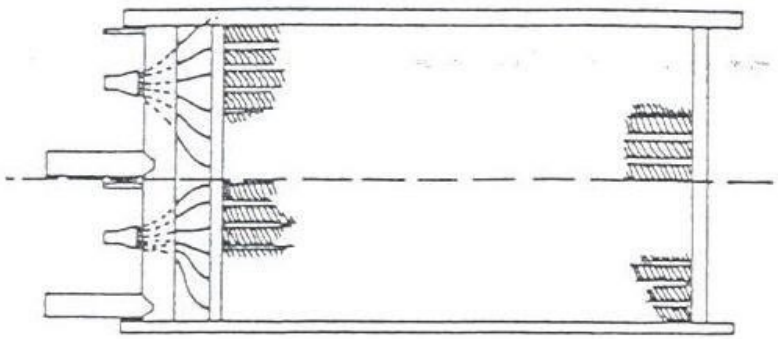
Plate

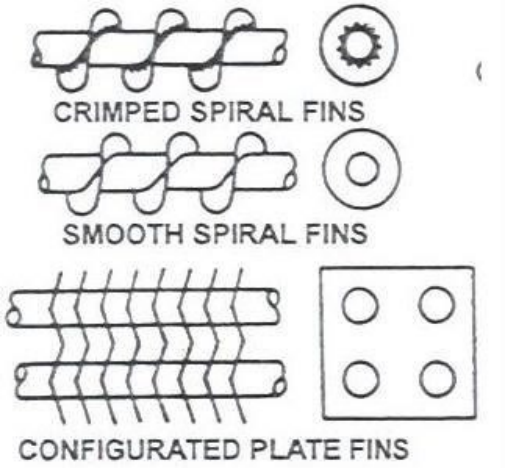
Bare tube





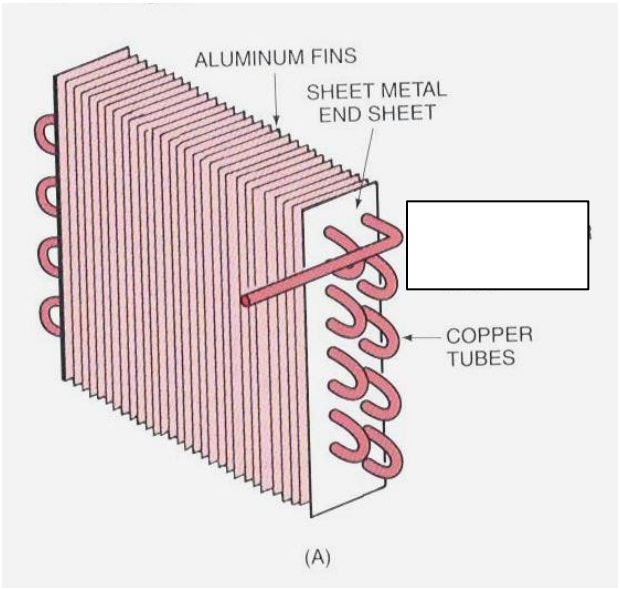
Finned Tube Evaporators



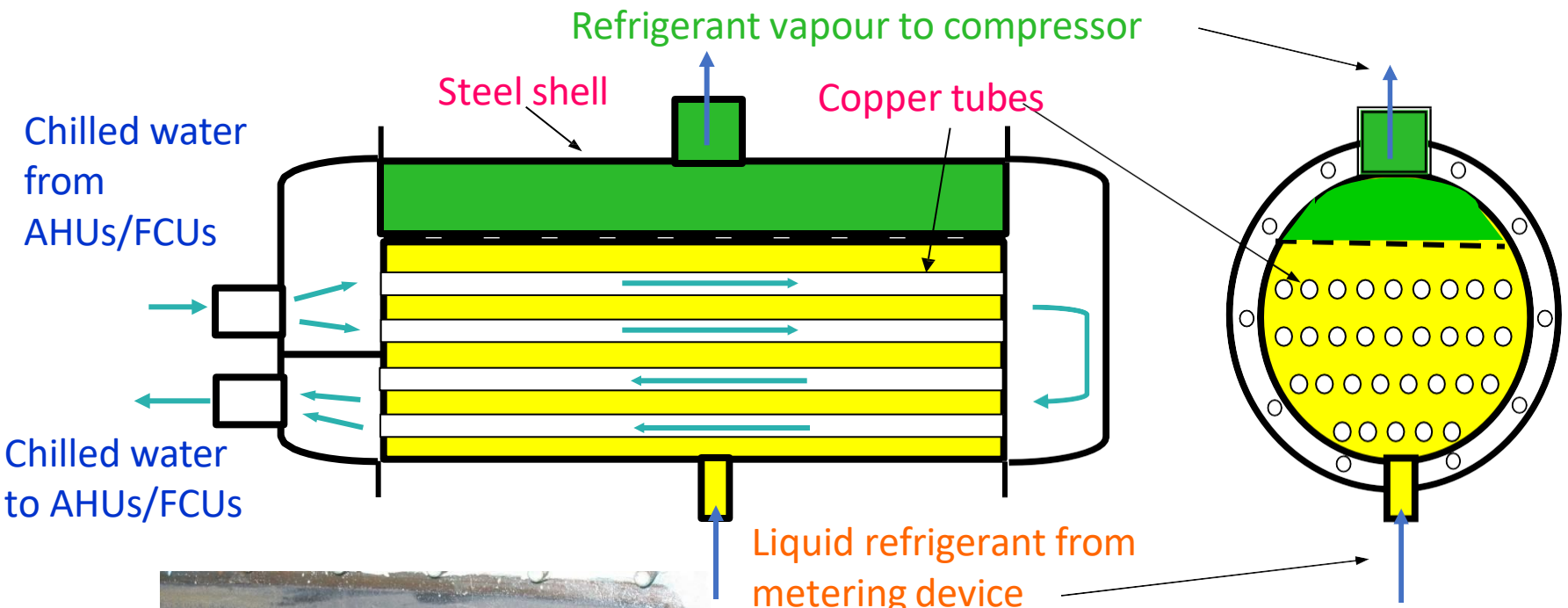


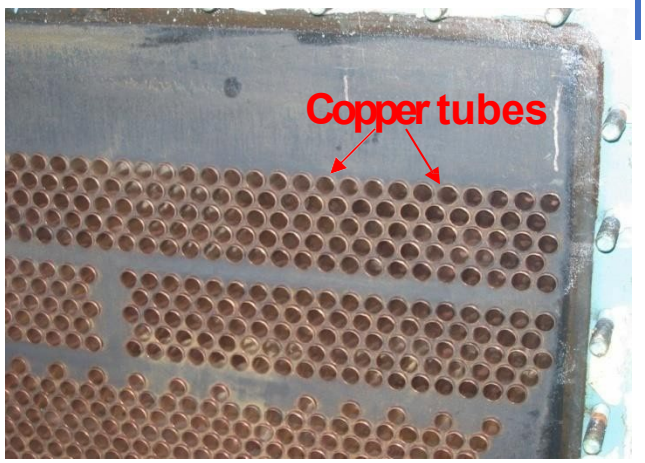


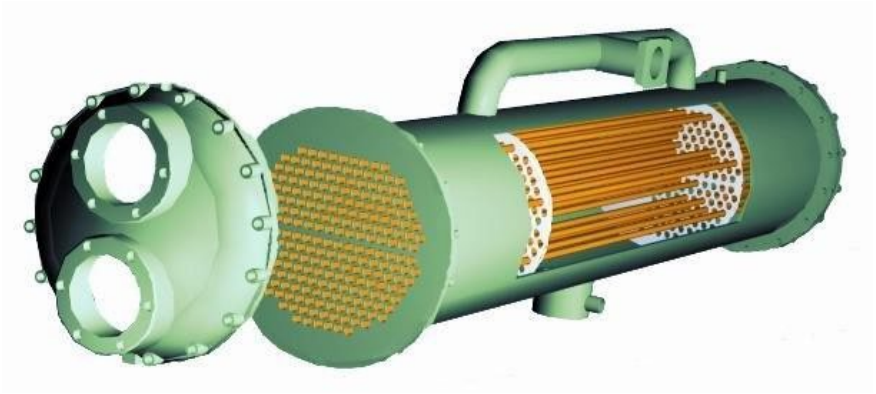




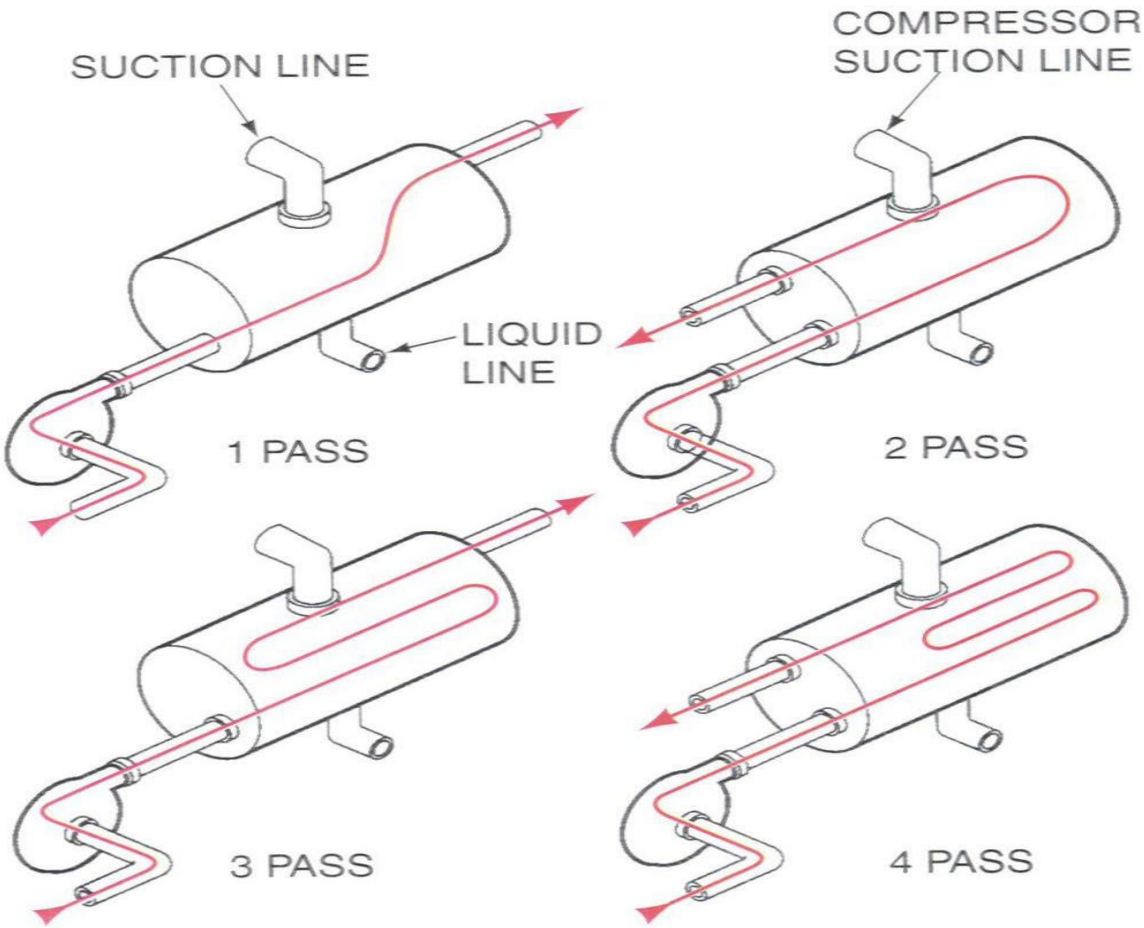
Liquid Chilling Evaporator - Shell and Tube







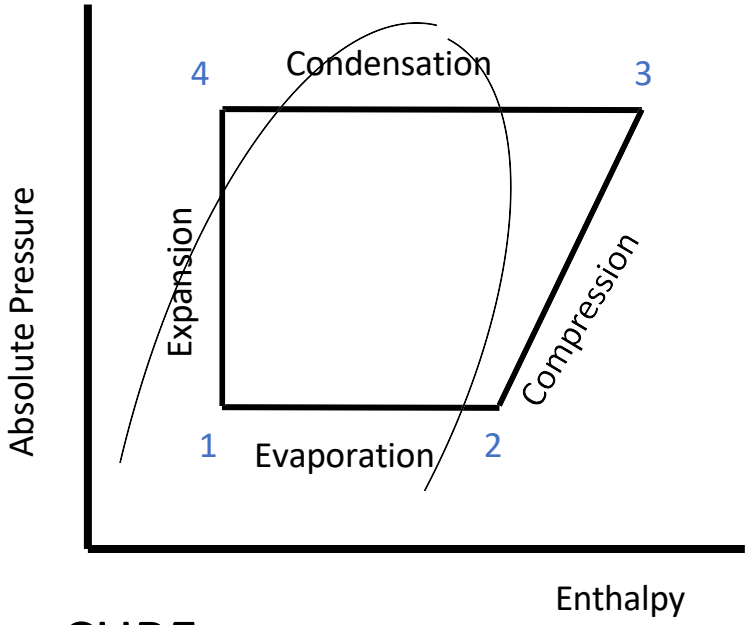
Number of Passes



Condensers

- Condenser Heat Rejection Rate
  = Refrigerating Capacity +
  Compressor Power

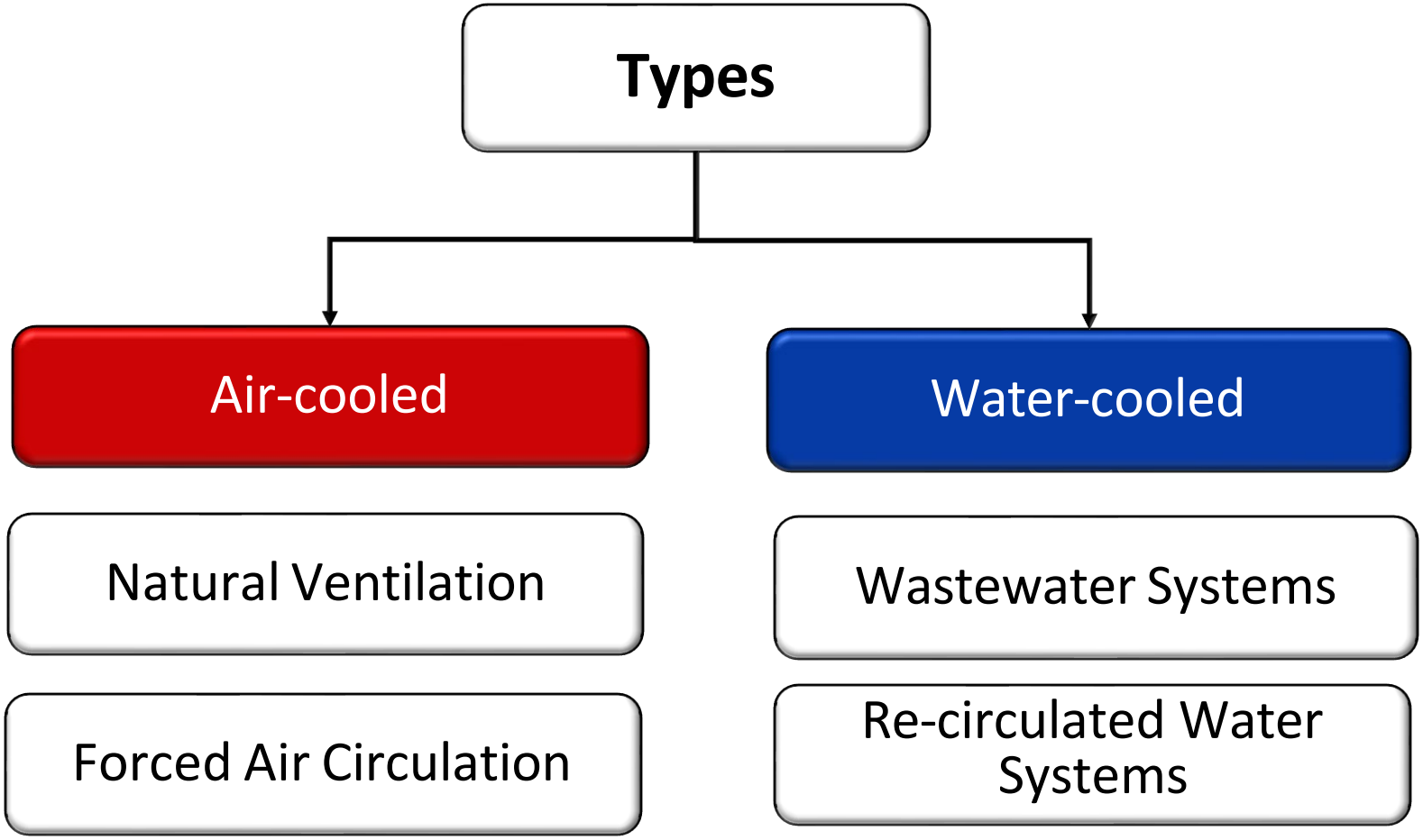
  $Q_r = Q_e + P$



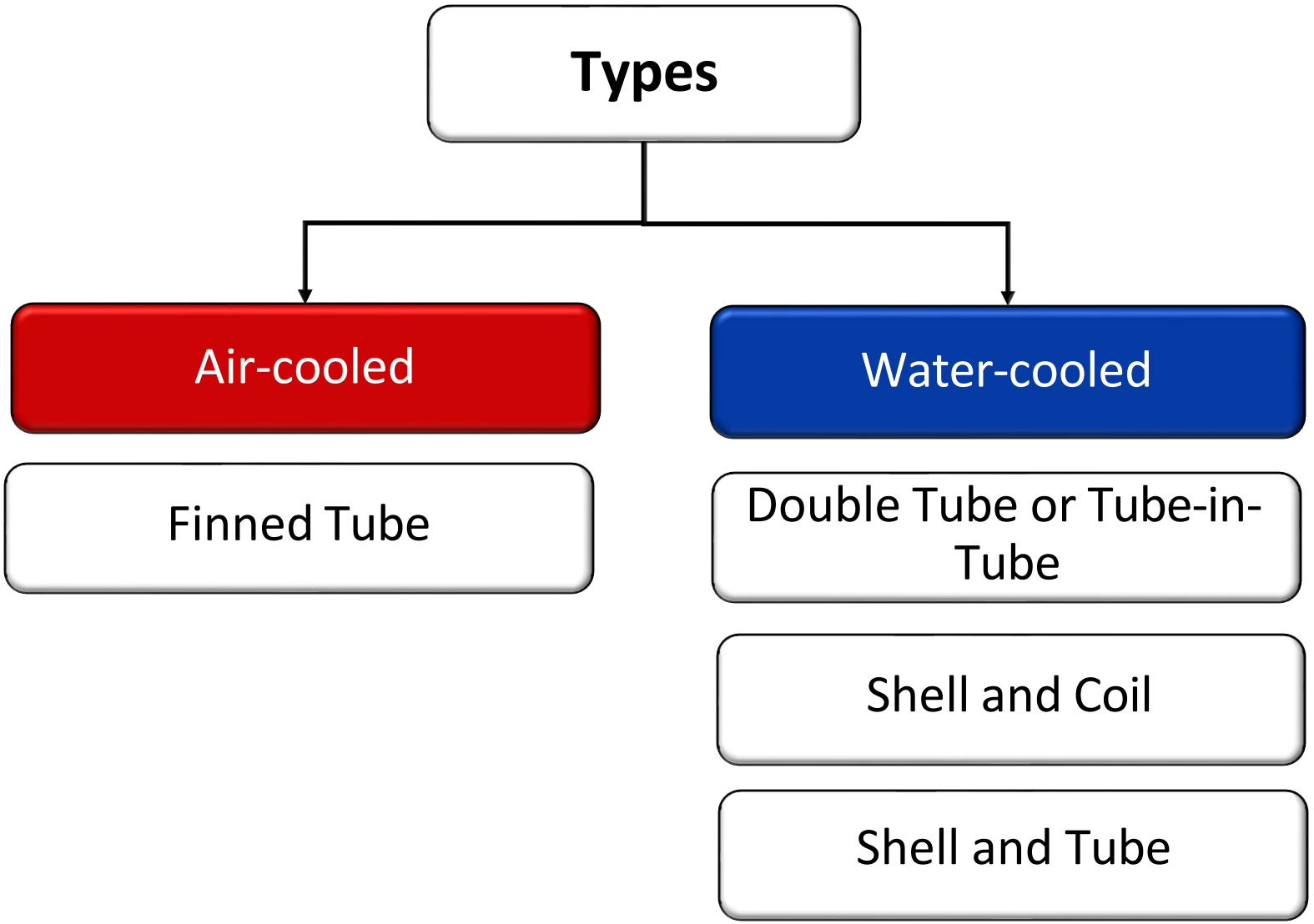
- Condenser Heat Rejection Factor, CHRF

  = $\frac{\text{Condenser Heat Rejection}}{\text{Refrigerating Effect}}$

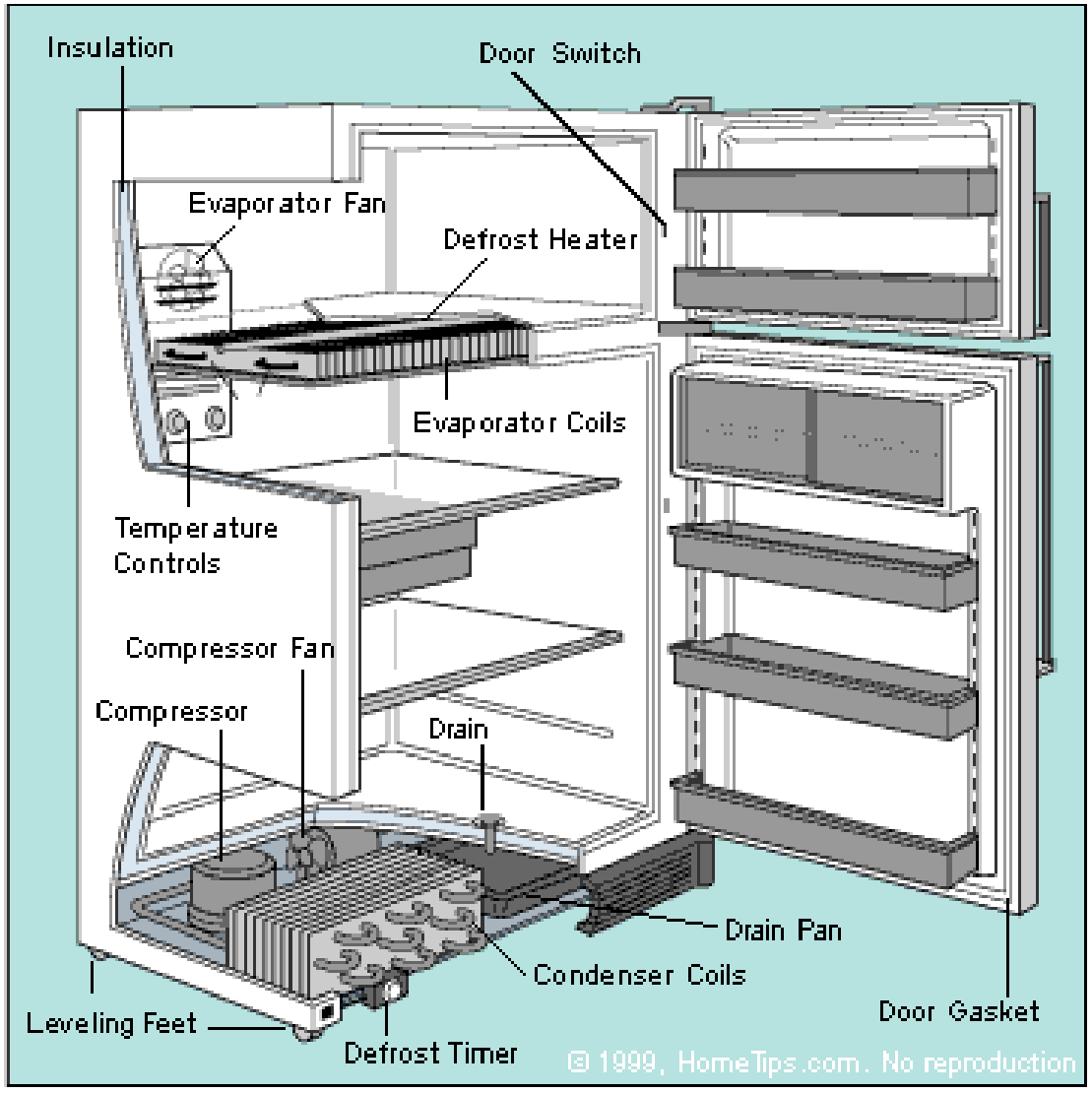
Condensers


Types
Air-cooled
Water-cooled
Natural Ventilation
Forced Air Circulation
Wastewater Systems
Re-circulated Water Systems

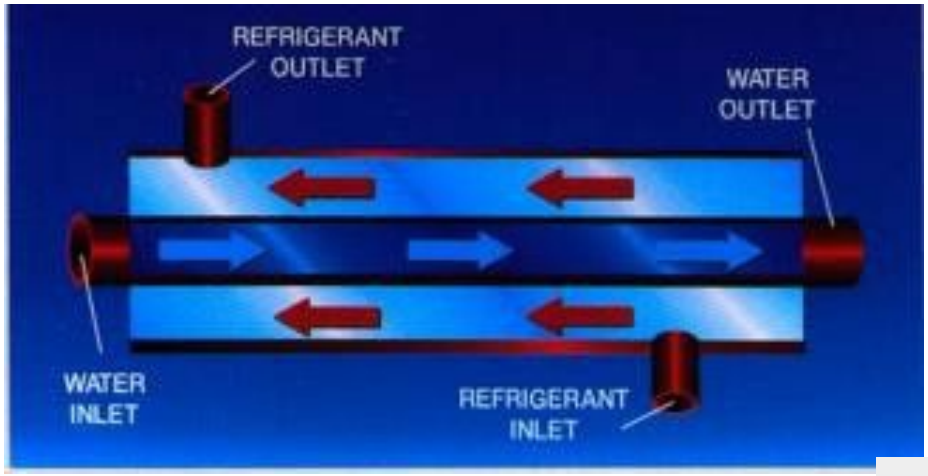
Condensers


Types
Air-cooled
Water-cooled
Finned Tube
Double Tube or Tube-in-Tube
Shell and Coil
Shell and Tube

Refrigerator

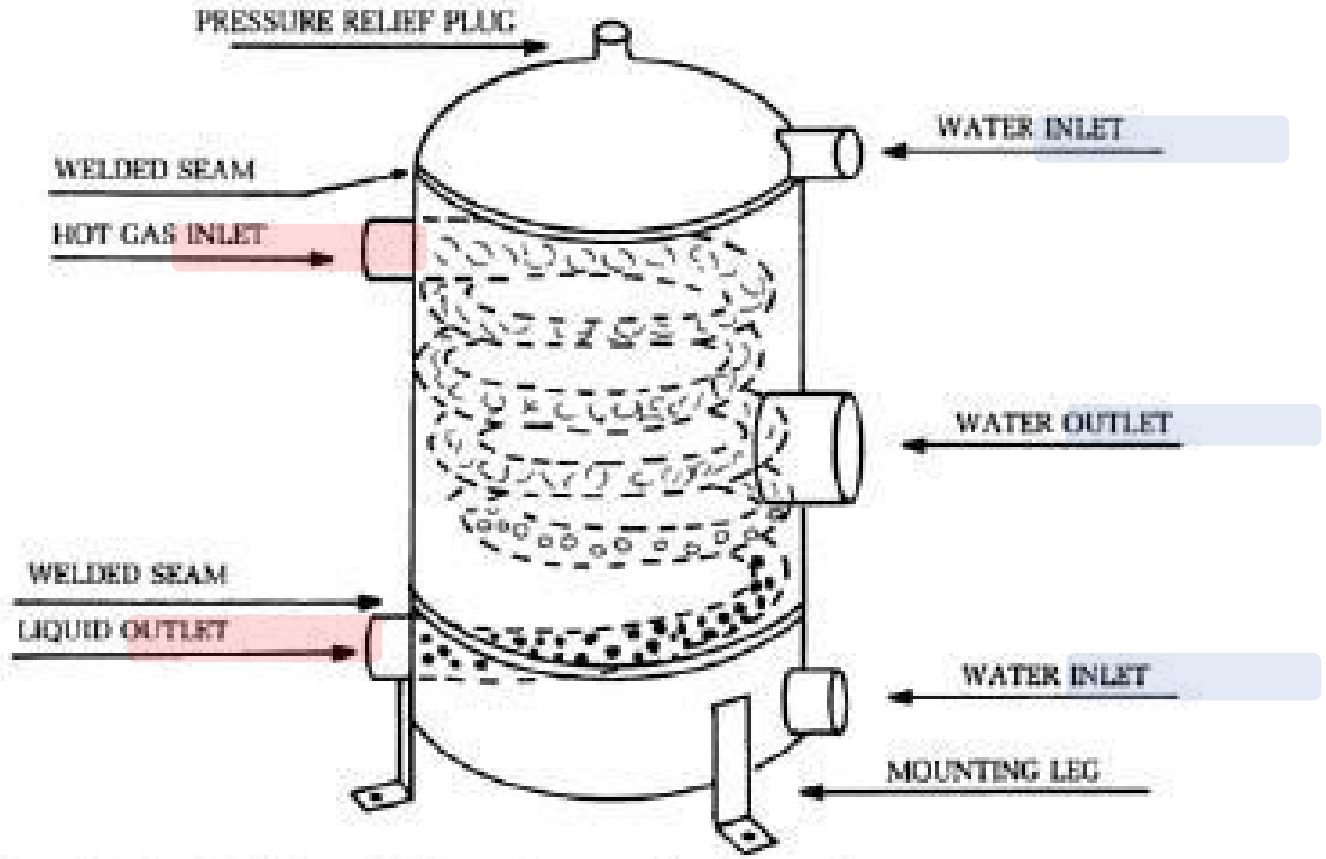


Tube-in-Tube Condensers

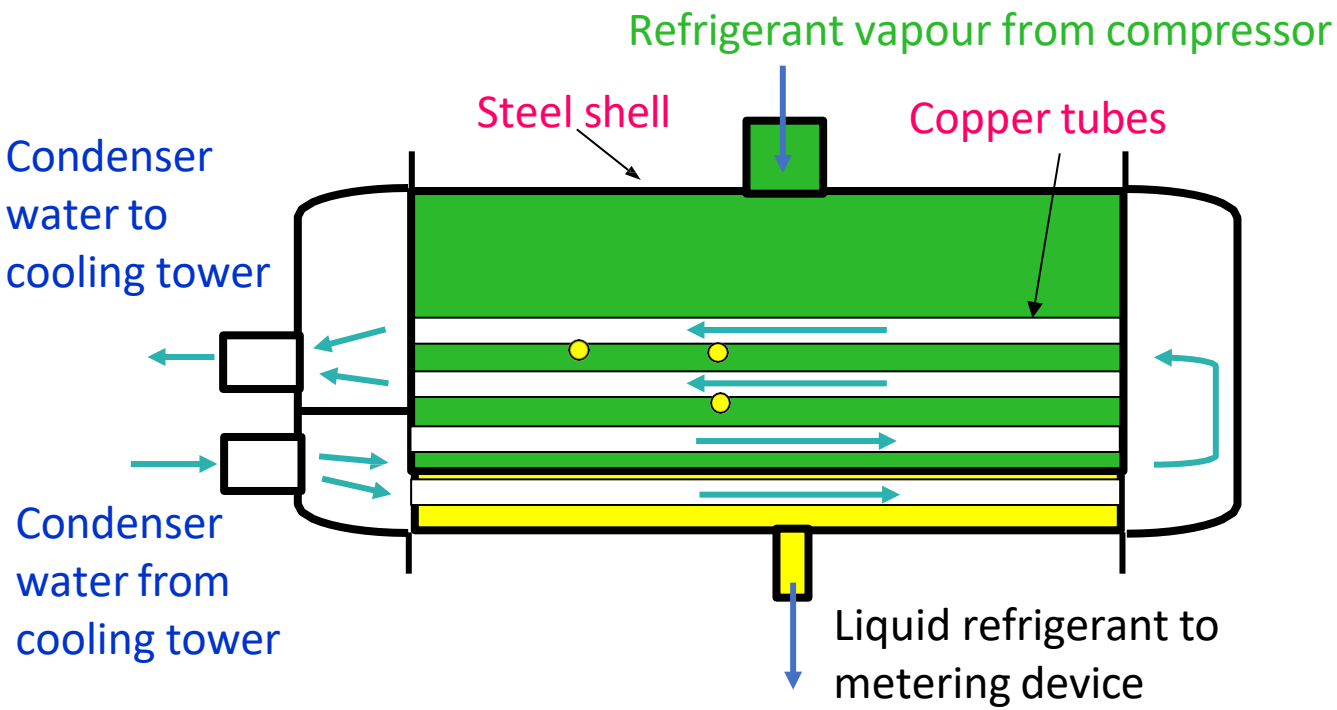




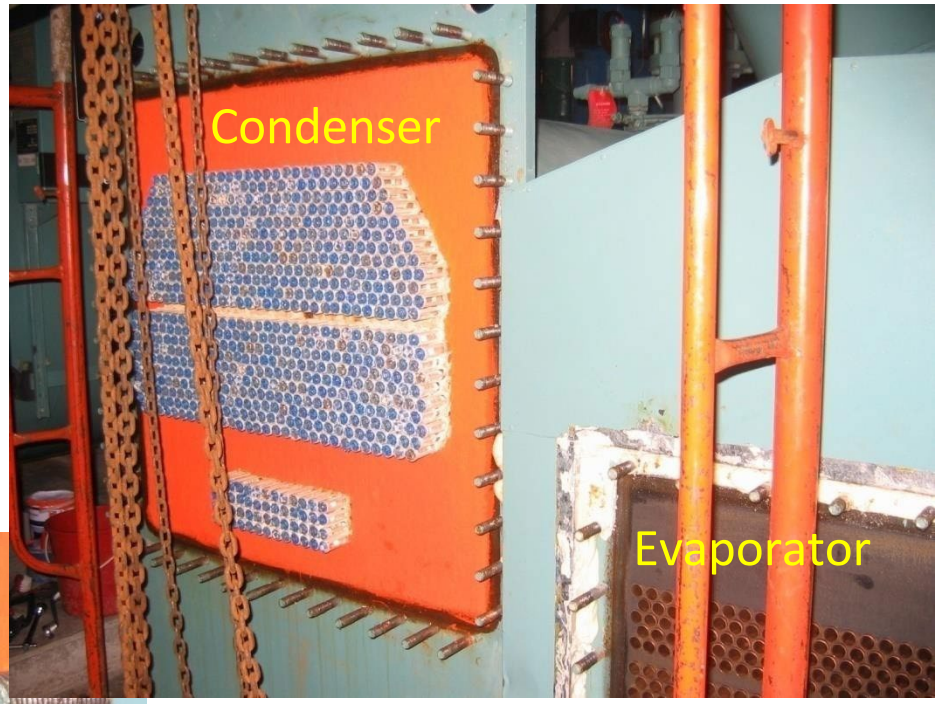
Shell and Coil Condensers

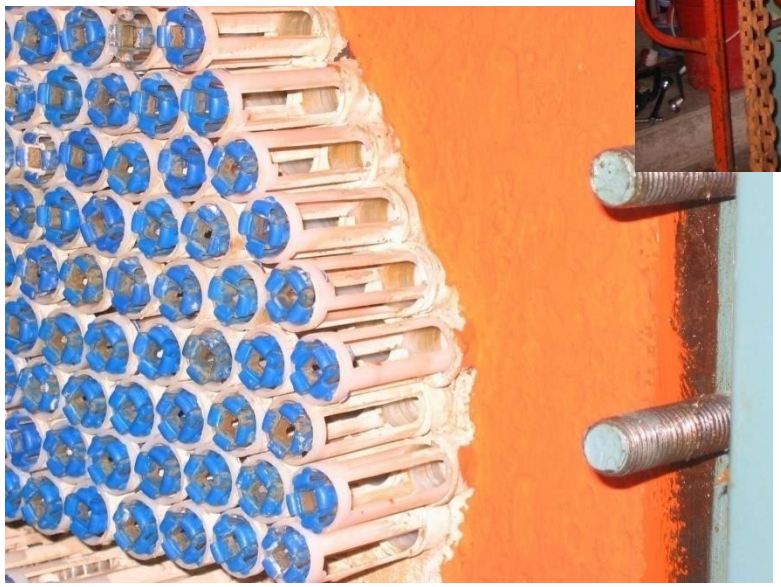


Shell and Tube Condenser



Condenser

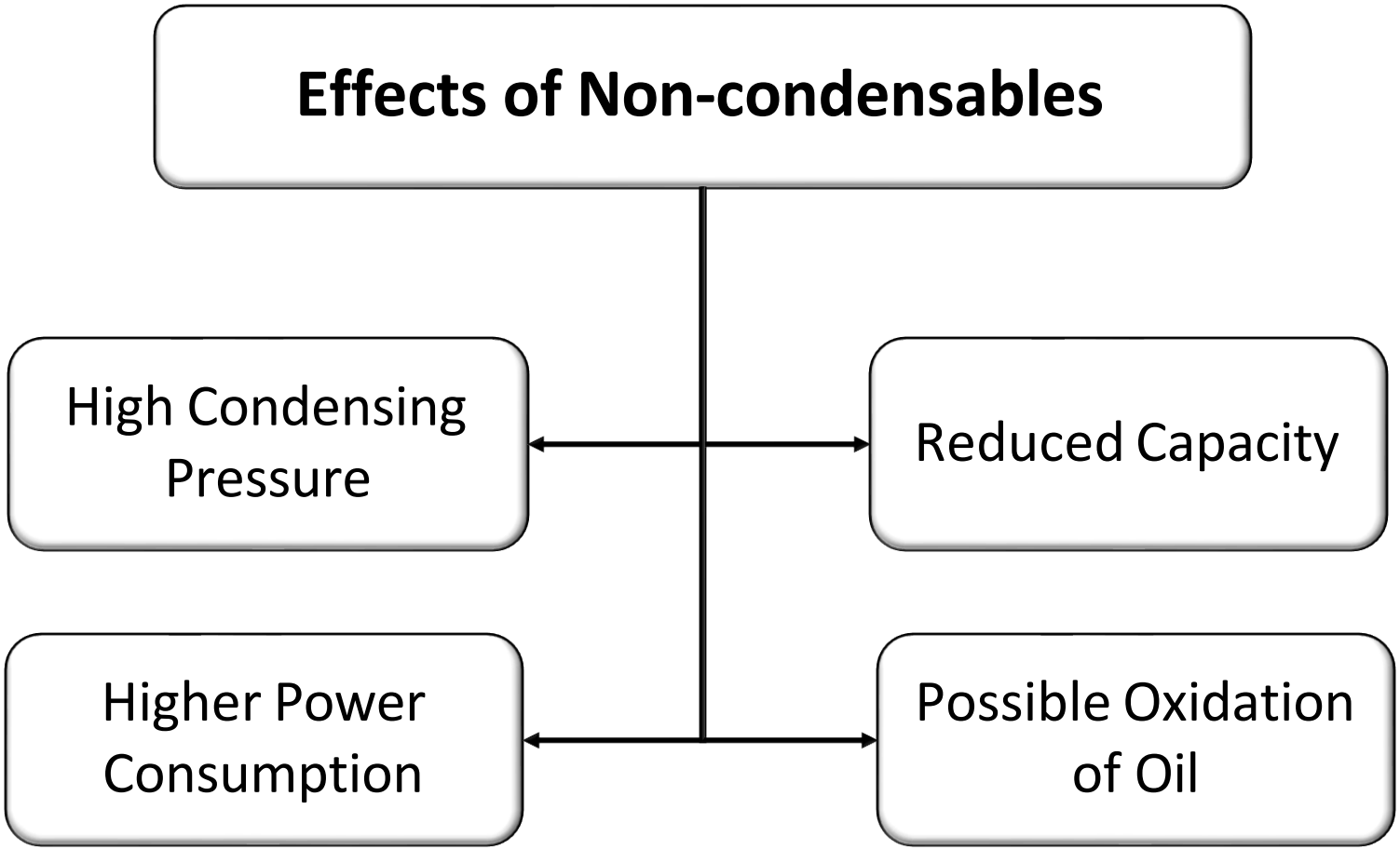




WSA

Innova-ball

Condensers

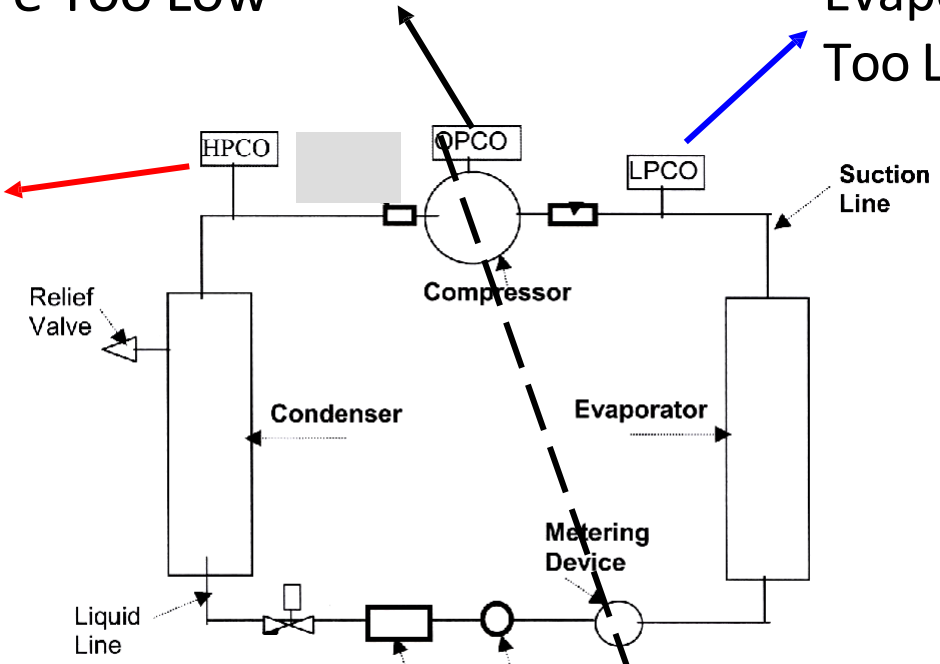

Effects of Non-condensables
High Condensing Pressure
Reduced Capacity
Higher Power Consumption
Possible Oxidation of Oil

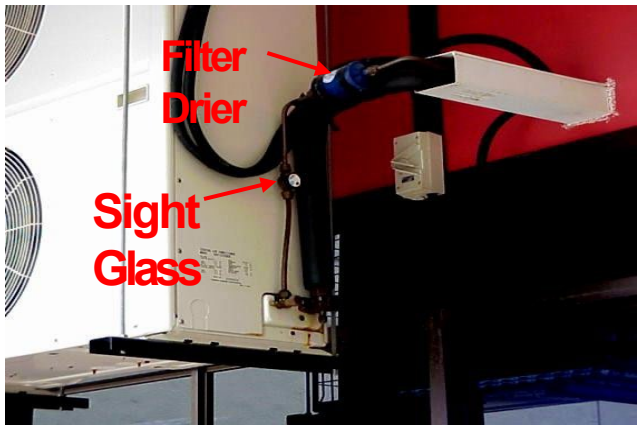
Accessories

Shuts Down Compressor if Oil Pressure Too Low

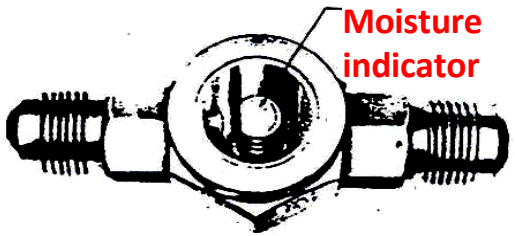
Shuts Down Compressor if Evaporating Press/ Temp Too Low

Shuts Down Compressor if Condensing Press/Temp To High





Filters & Absorbs Water



Check State of Refrigerant & Presence of water

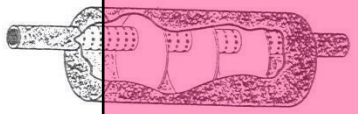
Accessories

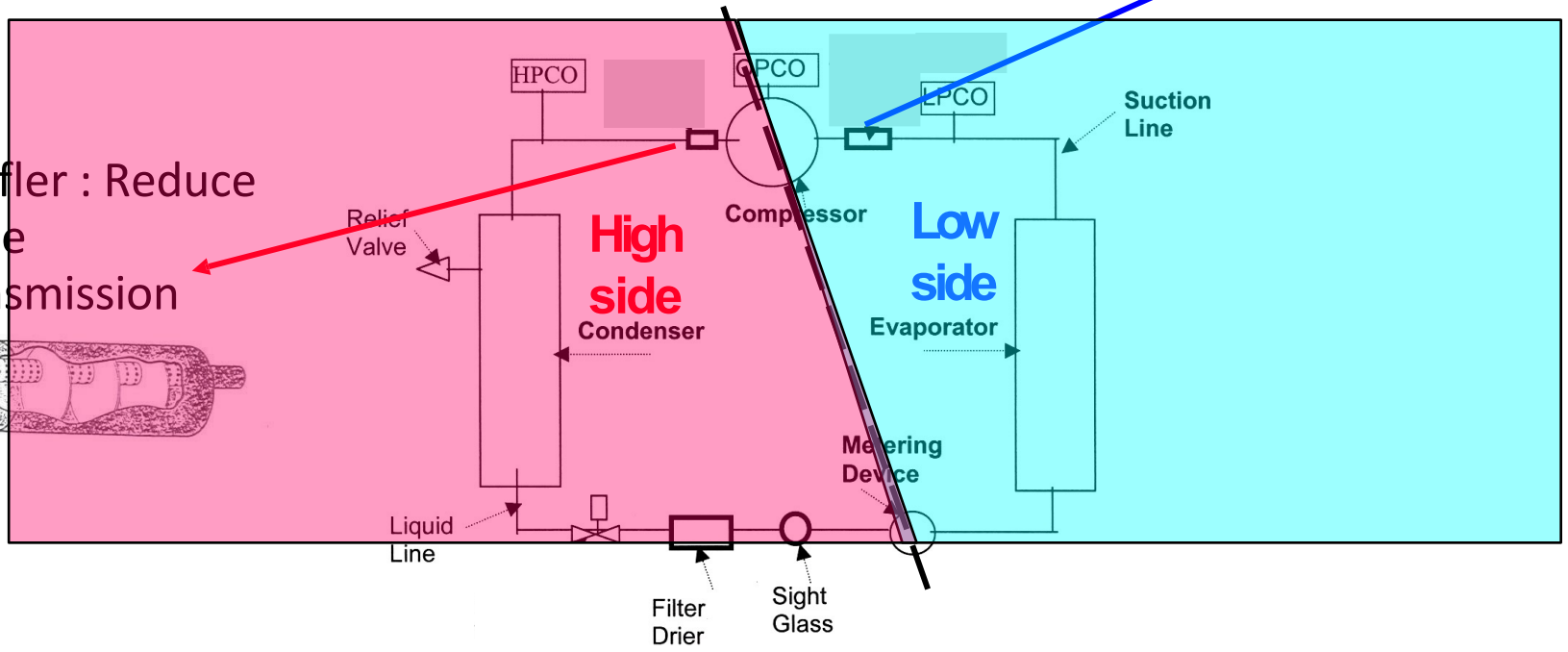
Keep oil warm during OFF cycle to reduce amount of refrigerant absorb by oil - avoid excessive loss of oil during compressor startup - compressor seizure.



Accumulator : Prevent Entry of Liquid Refrigerant into Compressor

Muffler : Reduce Noise Transmission





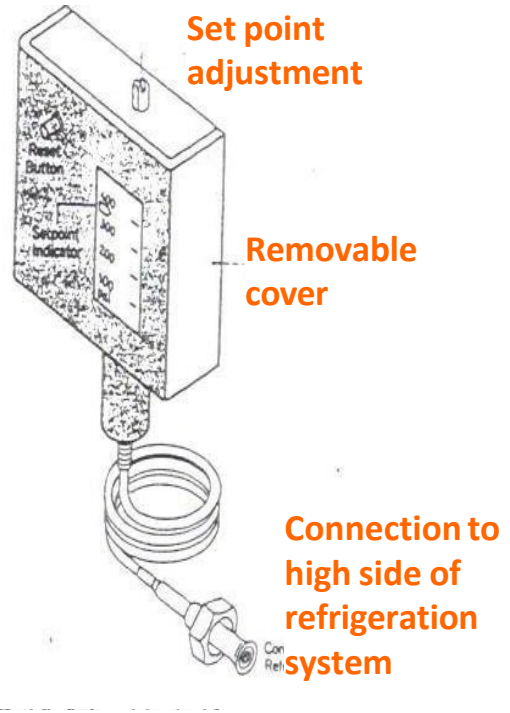
Accessories

- **High Pressure Cut-out**

  Location - High side

  Function - Shuts Down Compressor if Condensing Pressure/Temperature Too High



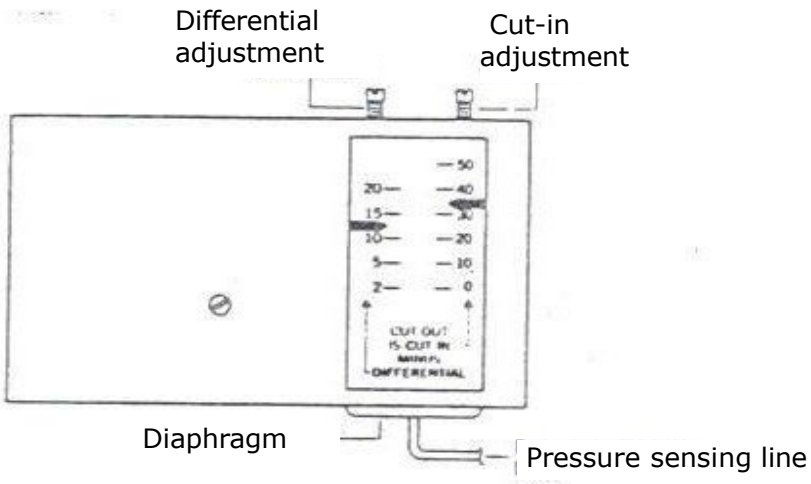


Accessories

- **Low Pressure Cut-out**

  Location - Low side

  Function - Shuts Down Compressor if Evaporating Pressure/Temperature Too Low





Accessories

- **Oil Pressure Cut-out**

  Location - Lubricating Oil Circuit

  Function - Shuts Down Compressor if Oil Pressure Too Low

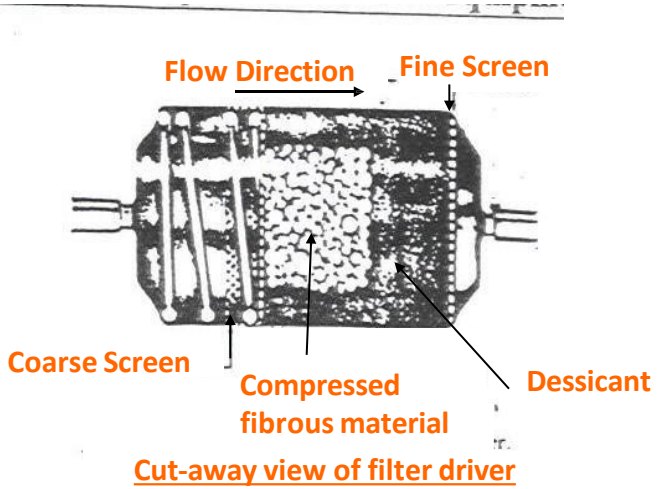
Accessories

- **Filter Driers**

  Location - Liquid Line

  Function - Filters and Absorbs & Water





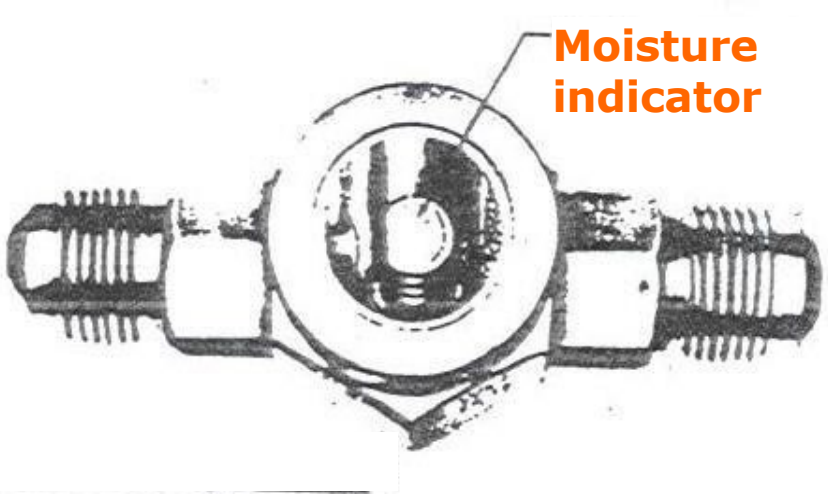
Cut-away view of filter driver

Accessories

- Sight Glass

  Location - Liquid Line

  Function - Check State of Refrigerant and Presence of water



Accessories

- **Suction Line Accumulator**

  Location - Suction Line or Compressor Suction

  Function - Prevent Entry of Liquid Refrigerant into Compressor

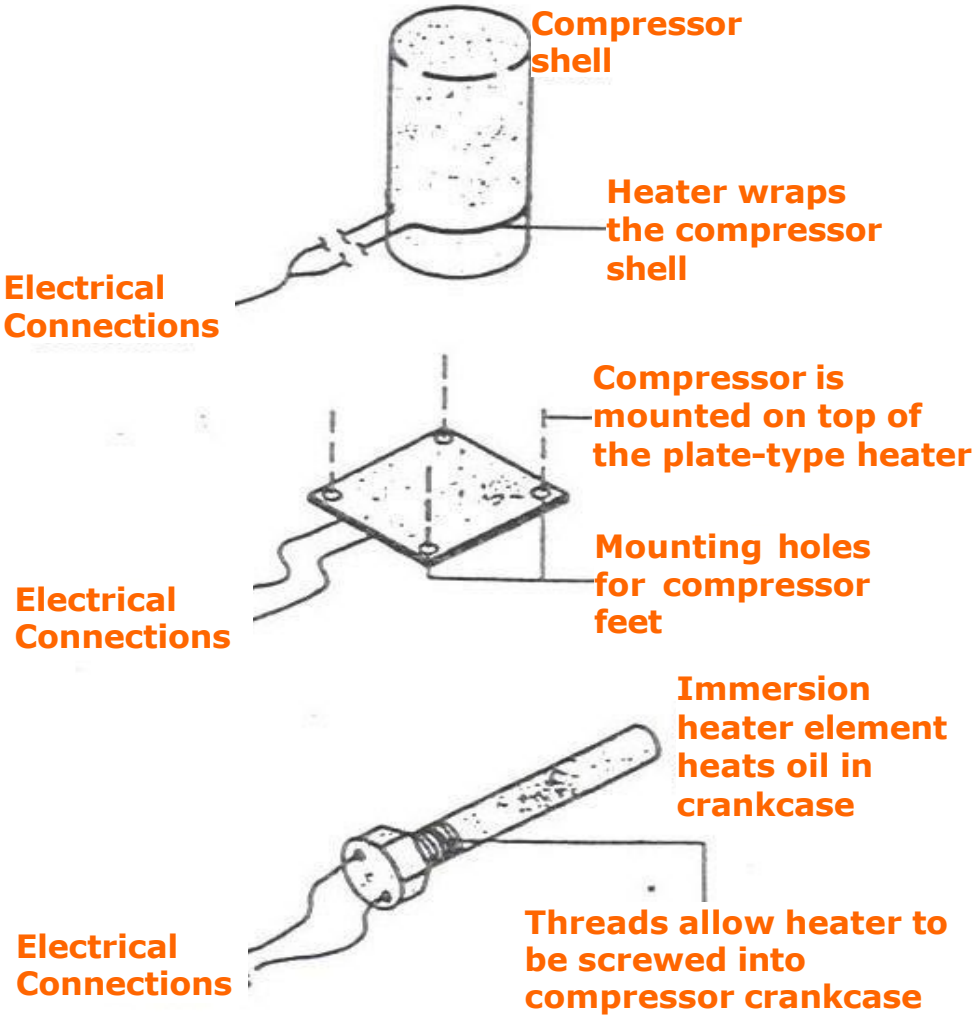
Accessories

- **Crankcase Heater/Oil Heater**

  Location - Compressor or Separate Lubricating Oil Circuit

  Function - Keep oil sufficiently warm during the OFF cycle to reduce amount of refrigerant absorb by oil

Accessories

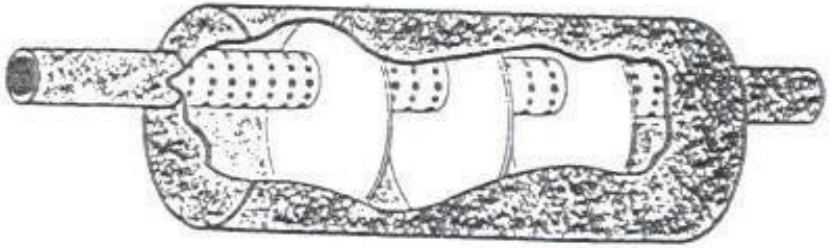


Accessories

- **Muffler**

  Location - Discharge Line

  Function - Reduce Noise Transmission



Summary

- Types of compressors by method of compression and construction, operation and capacity control
- Types of metering devices
- Types of evaporators
- Types of condensers
- Refrigerant circuit accessories

Room Air-conditioner

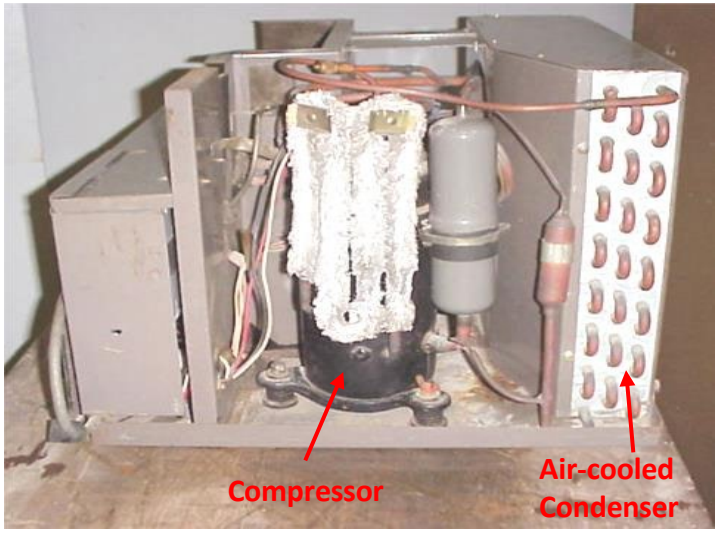
- Components
  - Rotary or scroll hermetic compressor
  - Finned tube air-cooling evaporator coil
  - Finned tube air-cooled condenser coil
  - Capillary tube metering device
  - Double shafted motor to drive centrifugal evaporator blower and propeller condenser fan

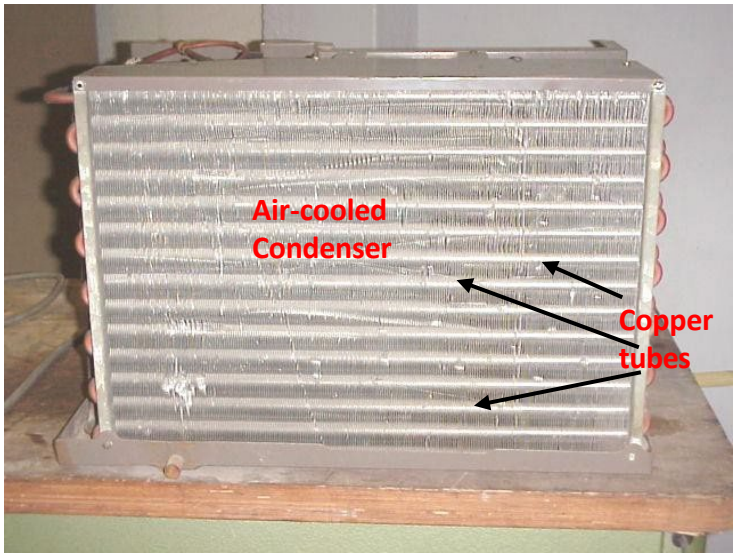
Room Air-conditioner

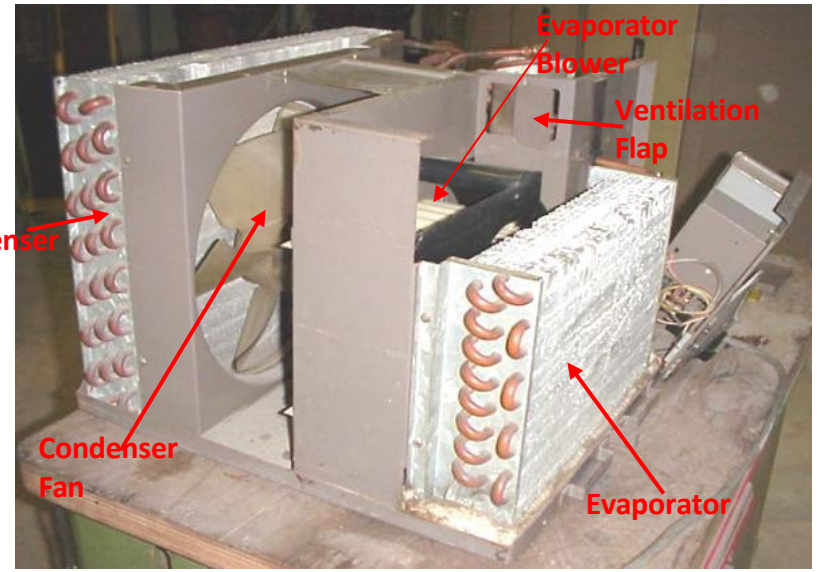
- Components
  - Thermostatic control for compressor
  - Other controls and electrical components
  - Air filters
  - Condensate drain pan
  - Casing
  - Remote control

Room Air-conditioner









Room Air-conditioner

- **P**reventive Maintenance
  - Clean filter and front panel
  - Clean condensate drain pan and drainpipe
  - Check electrical control function
  - Check running current and on and off-coil air temperatures
  - Check evaporator and condenser coils
  - Clean fan blades
  - Straighten/comb fins where necessary



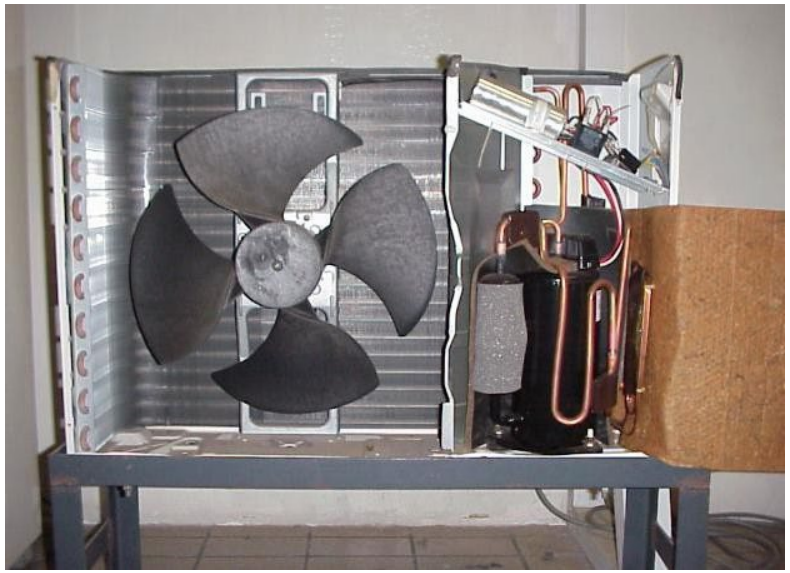


Air-cooled Split Unit

- Components

  Condensing Unit

  - Rotary or scroll hermetic compressor
  - Finned tube air-cooled condenser coil
  - Propeller type condenser fan with motor
  - Electronic expansion valve



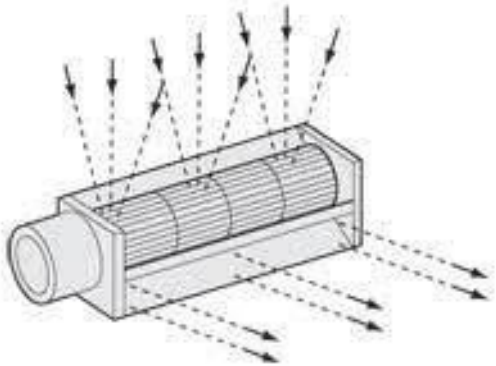
Air-cooled Split Unit

- Components

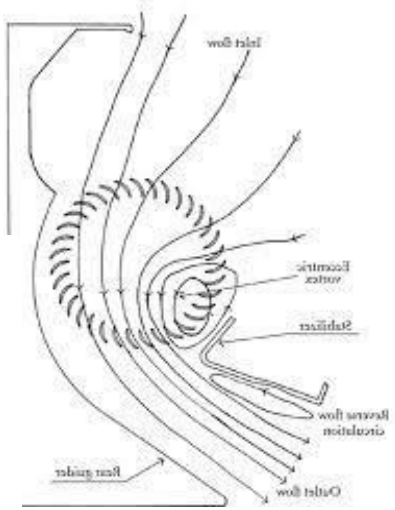
  Fan Coil Unit

  - Finned tube air-cooling evaporator coil
  - Cross flow type evaporator fan with motor
  - Air filters
  - Condensate drain pan
  - Remote control





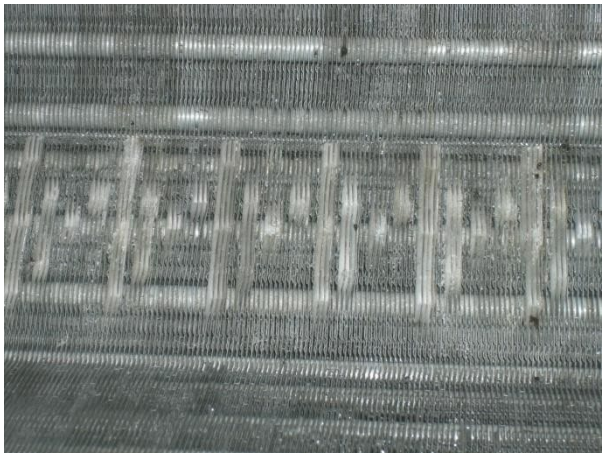
Cross Flow Fans



Air-cooled Split Unit









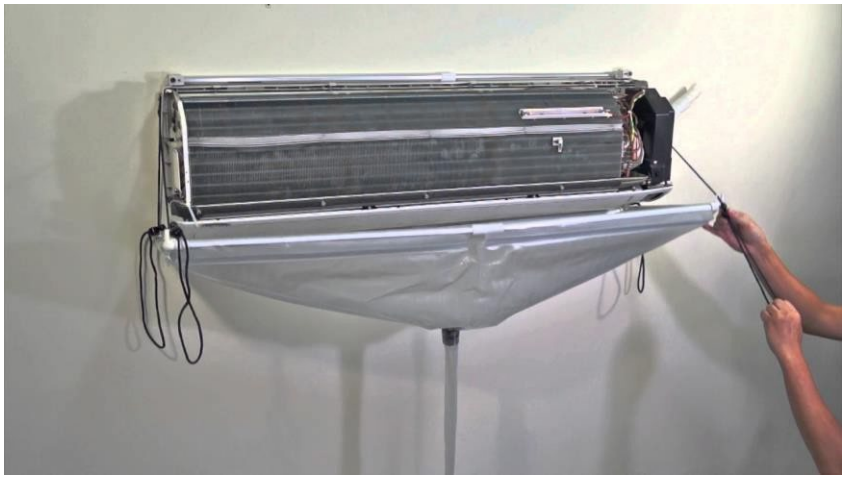




Air-cooled Split Unit









Fan Coil Unit

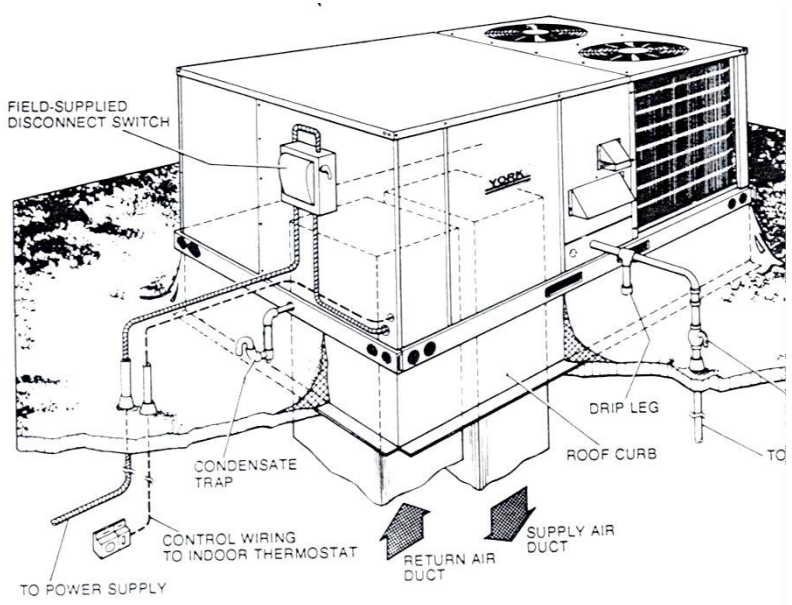


Air-cooled Package Unit

- Components
  - Scroll type hermetic or semi-hermetic compressor
  - Finned tube air cooling evaporator
  - Finned tube air-cooled condenser
  - Capillary or thermostatic expansion valve or electronic expansion valve
  - Propeller type condenser fans
  - Centrifugal type evaporator blowers with belt drive

Air-cooled Package Unit

- Components
  - Thermostatic control based on compressor
  - Air filters
  - Condensate drain pan
  - Safety controls
  - Electrical power and control components



Air-cooled Package Unit

- Maintenance
  - Check for unusual fan, motor and compressor noise.
  - Check for obstructions to air flow at condenser, evaporator, outdoor air intake.
  - Check for signs of refrigerant leak.
  - Clean and replace filters, clean outdoor air intake.
  - Inspect and clean condensate pan and drainpipe.
  - Check belts and pulleys for wear, tension and alignment.

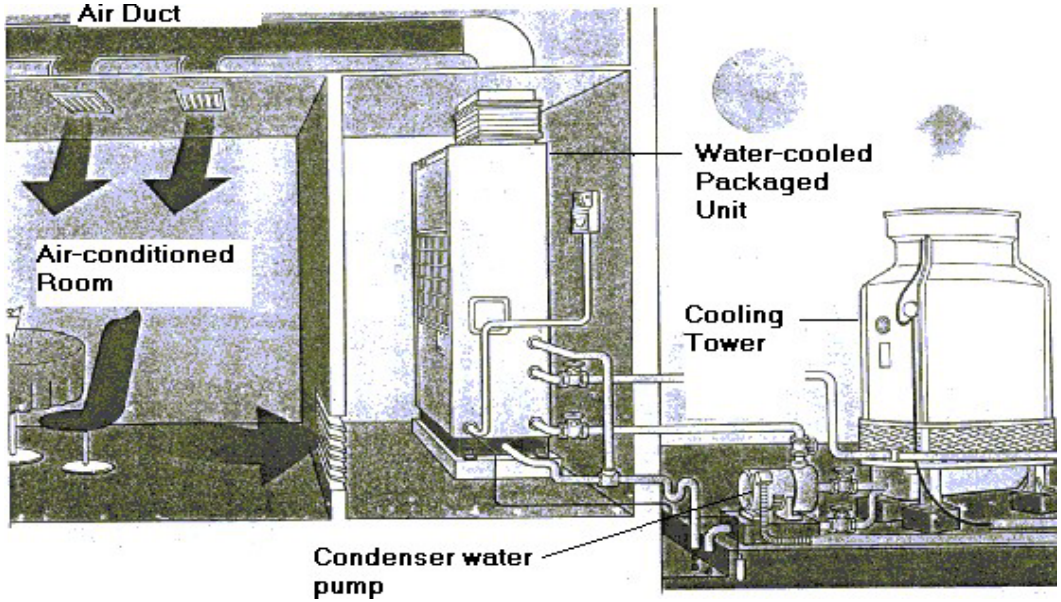
Air-cooled Package Unit

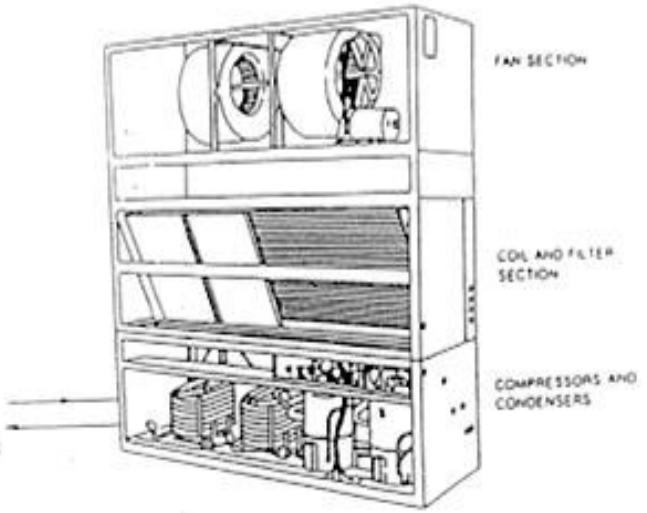
- Maintenance
  - Clean fan motors and blades.
  - Lubricate fan and motor bearings, if necessary.
  - Clean condenser and evaporator coils.
  - Straighten and comb bent fins
  - Check controls and reset, if necessary.
  - Clean contacts of contactors.
  - Log current, suction pressure, condensing pressure, oil pressure when possible.

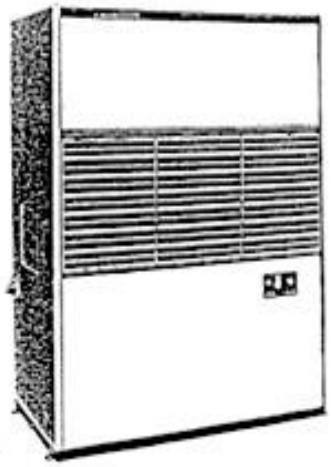
Water-cooled Package Unit

- Components
  - Same as ACPU except air-cooled condenser is replaced with shell and tube water-cooled condenser.
  - Interlock normally provided with cooling tower & condenser water pump.

Water-cooled Package Unit







Water-cooled Package Unit

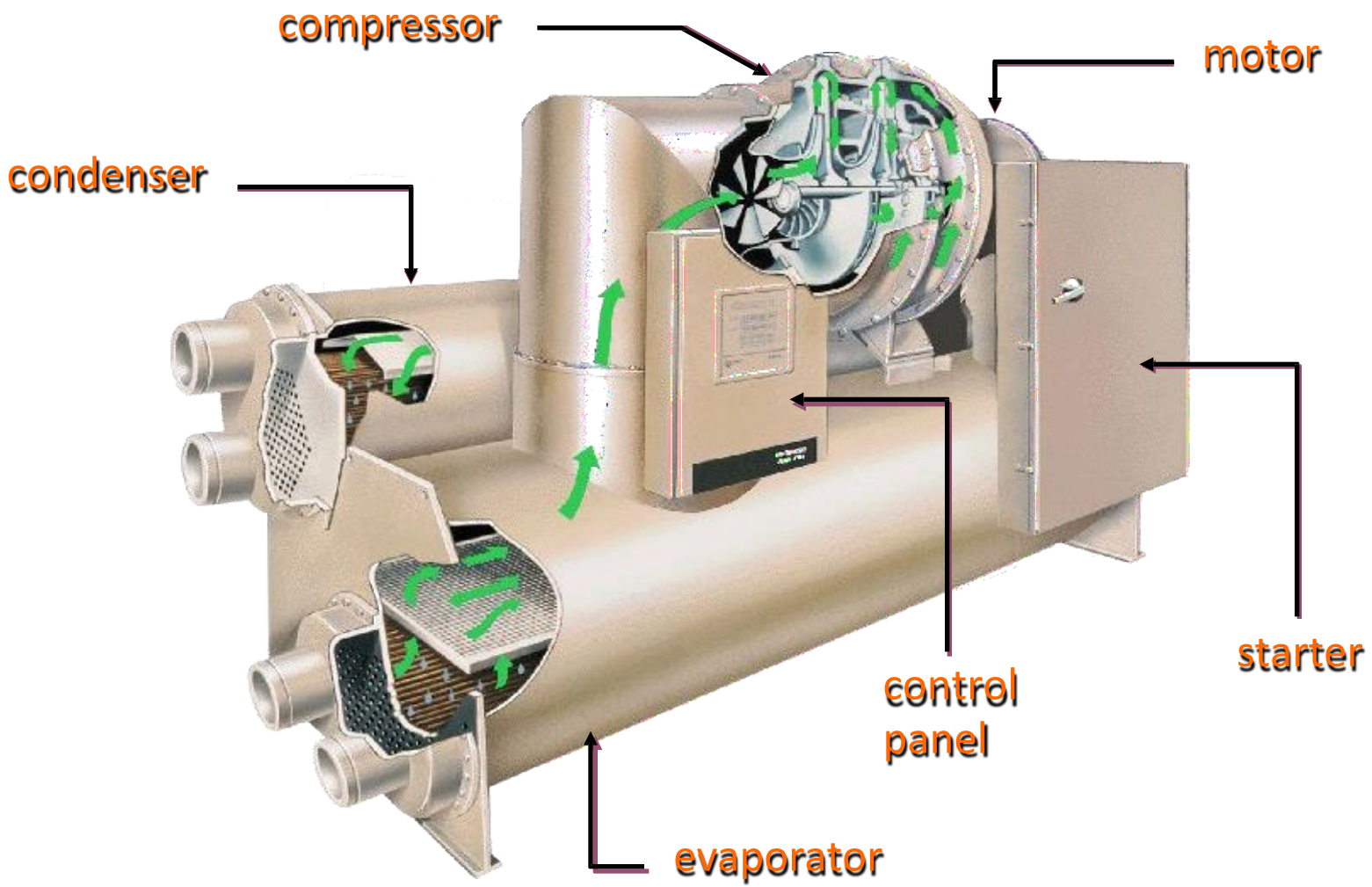
- Maintenance
  - All maintenance functions as for air-cooled packaged unit except the ones related to air-cooled condenser plus
  - Cleaning of condenser tubes - water side.
  - Maintenance of condenser water pump.
  - Maintenance of the cooling tower.

Water-cooled Chiller

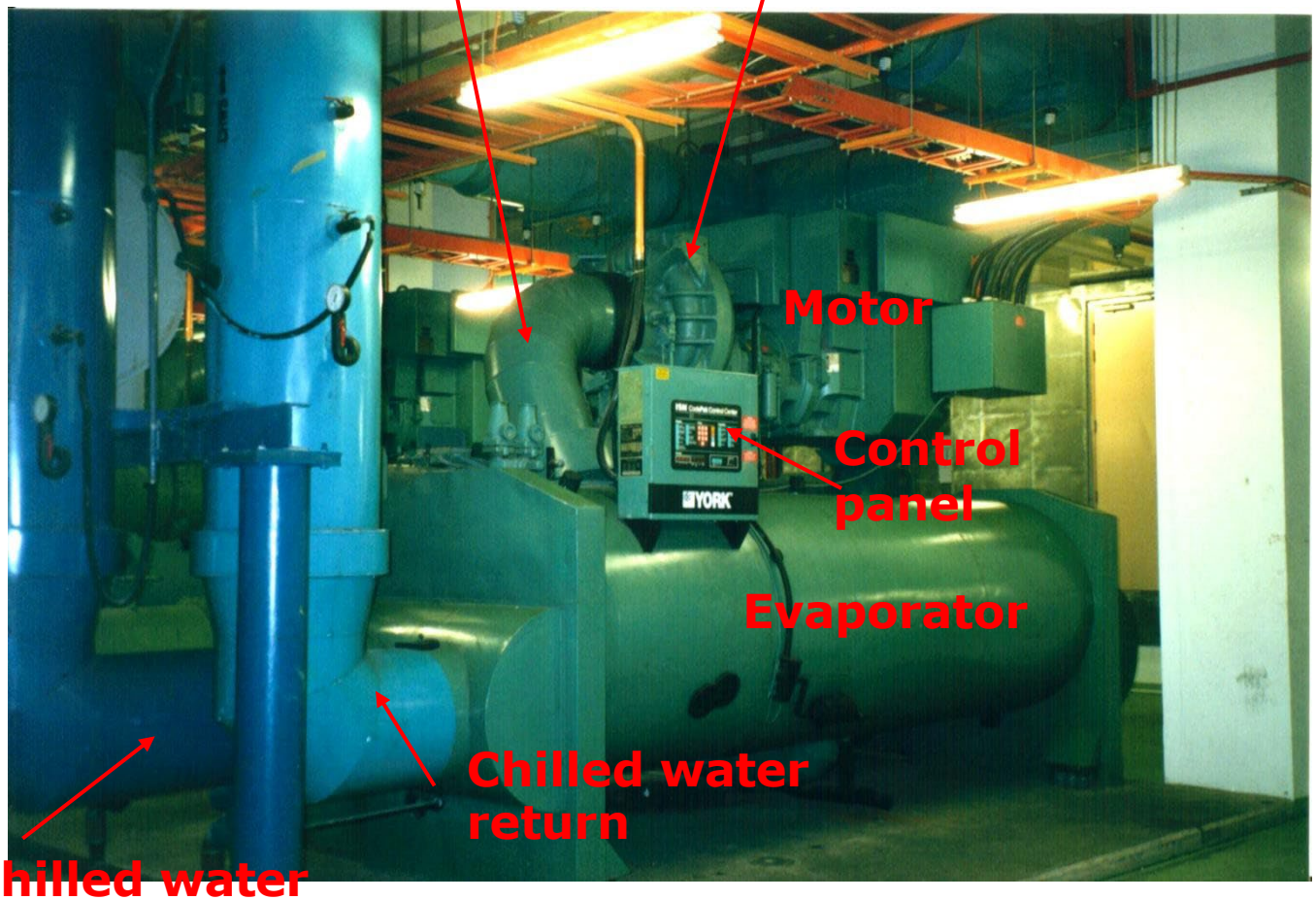
Water-cooled Chiller

- Major Components
  - An open or semi-hermetic centrifugal compressor (or screw compressor) with capacity control
  - A shell-and-tube water-cooled condenser
  - Flooded type shell-and-tube evaporator
  - Oil tank, oil pump, oil filter, oil cooler & oil heater.
  - Electrical Control Panel
  - Safety controls
  - Inter cooler or flash chamber for two-stage compressor

Water-cooled Chiller



Water-cooled Chiller



Manual Condenser Cleaning

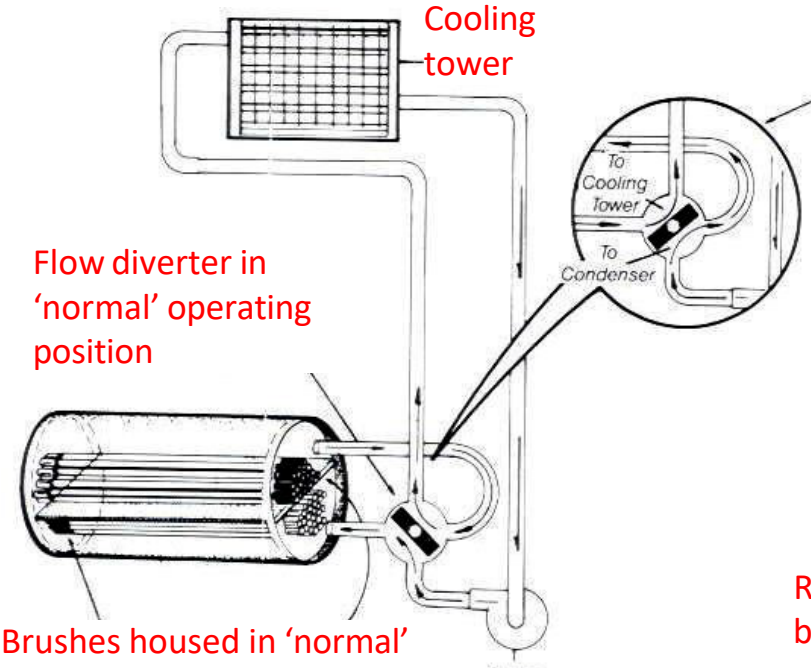


GOODWAY

Condenser Cleaning - Chemical



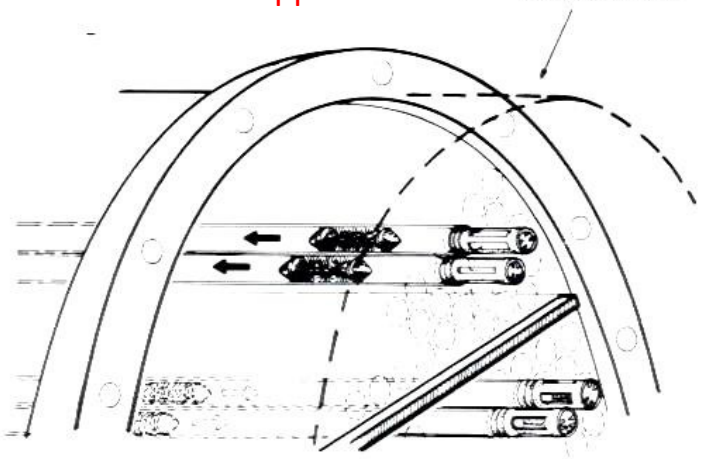
Automatic Brush Cleaning



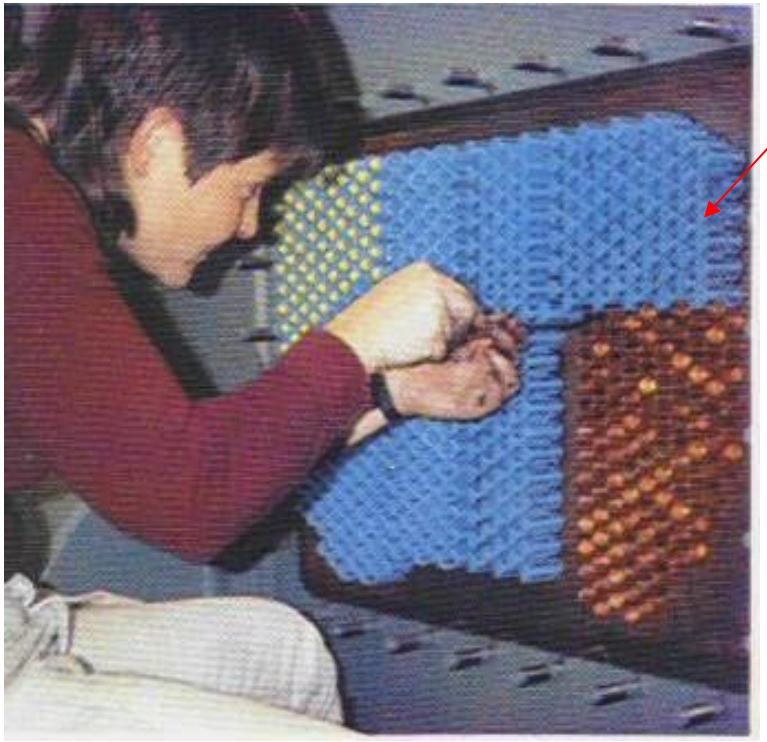
Flow diverter in opposite position reverses water flow through condenser



Reversed water flow expels brushes from baskets advancing them to their housings at the opposite end of condenser

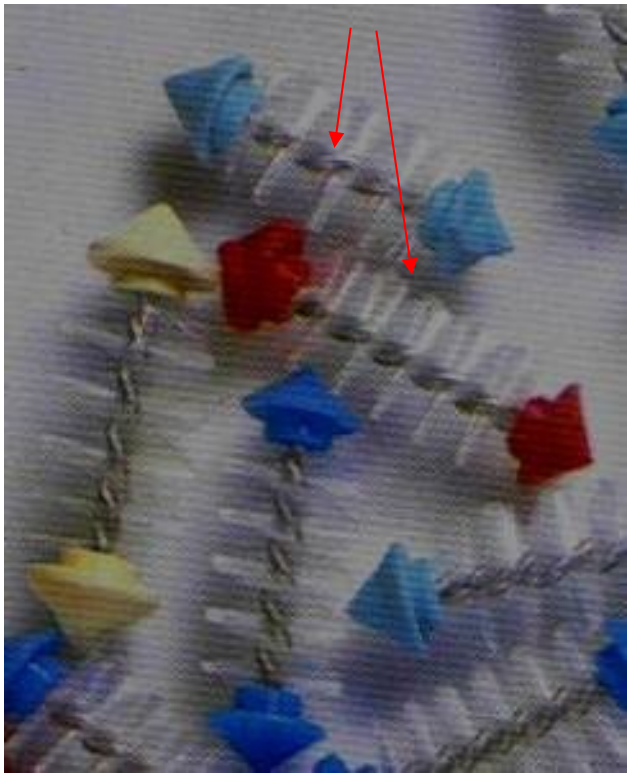


Automatic Brush Cleaning



Baskets

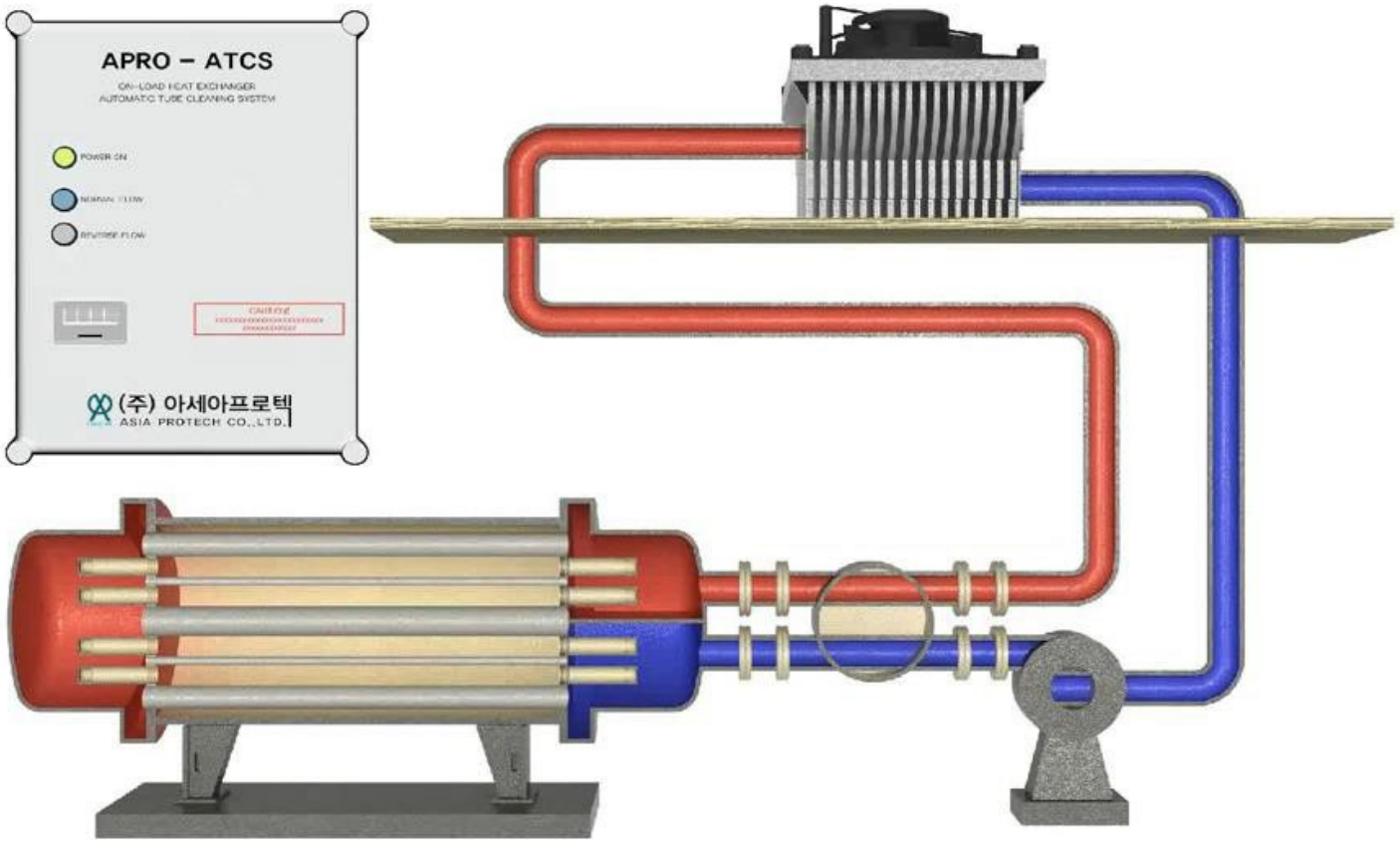
Brushes



WSA

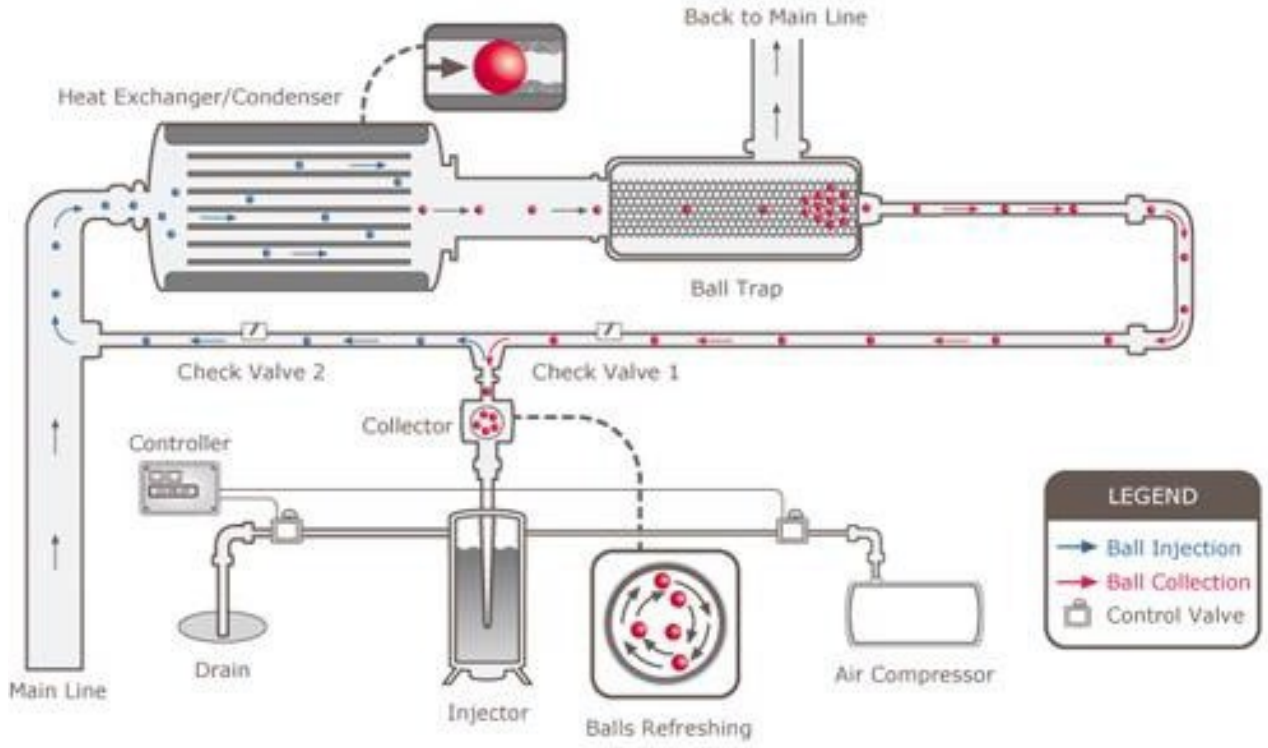
CQM-ball

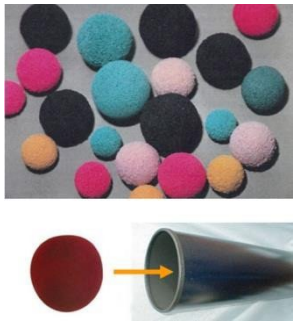
Automatic Brush Cleaning Systems



Automatic Condenser Cleaning Systems

- Important to operate chillers with tubes free of or with minimal scale deposits.





Automatic Ball Cleaning System

Water-cooled Chiller





Factory Installed Safety Control

- High Pressure Cut-out
- Low Pressure Cut-out
- Low Oil Pressure Cut-out
- Low chilled water temp cut-out (Anti-freeze Cut-out)
- High Motor Temperature Cut-out
- Low Oil Sump Temperature Cut-out
- High Oil Temperature Cut-out
- Anti-cycling timer
- Current Demand Limiter
- Pressure Relief Valves

Field Installed Safety Controls

- Chilled Water Flow Switch
- Condenser Water Flow Switch
- Motor Over-current
- Control Interlocks

Logging of Chiller Operating Parameters

- Chilled water inlet & outlet temps
- Chilled water inlet & outlet pressures
- Condenser water inlet & outlet temps
- Condenser water inlet & outlet pressures
- Lubrication oil level, temp & pressure
- Refrigerant level, discharge & suction pressures
- Motor voltages and running currents
- Purge unit refrigerant level, discharge and suction pressures (for chillers fitted with purge unit)
- Ambient dry-bulb and wet-bulb temperatures

Starting and Shutting-Down Sequence

- Starting sequence:
  1. Start cooling tower(s)
  2. Start condenser water pump(s)
  3. Start chilled water pump(s)
  4. Start water chiller(s)
- The shutting-down sequence is the reverse of the above.

Maintenance

- Daily Inspection
  - Log operating parameters.

    Chilled water inlet & outlet temps/press

    Condenser water inlet and outlet temps and press.

    Lubrication oil level, temp, pressure.

    Refrigerant level, evaporating pressure and condensing pressure.

    Motor running current.

Maintenance

- Daily Inspection

  - Chemical treatment as necessary
  - Check for any unusual noise and vibration.
  - Check for water, oil and refrigerant leaks

Maintenance

- Every 6 months

  - Check and lubricate Inlet Guide Vane linkages.

  - Clean strainers in water systems.

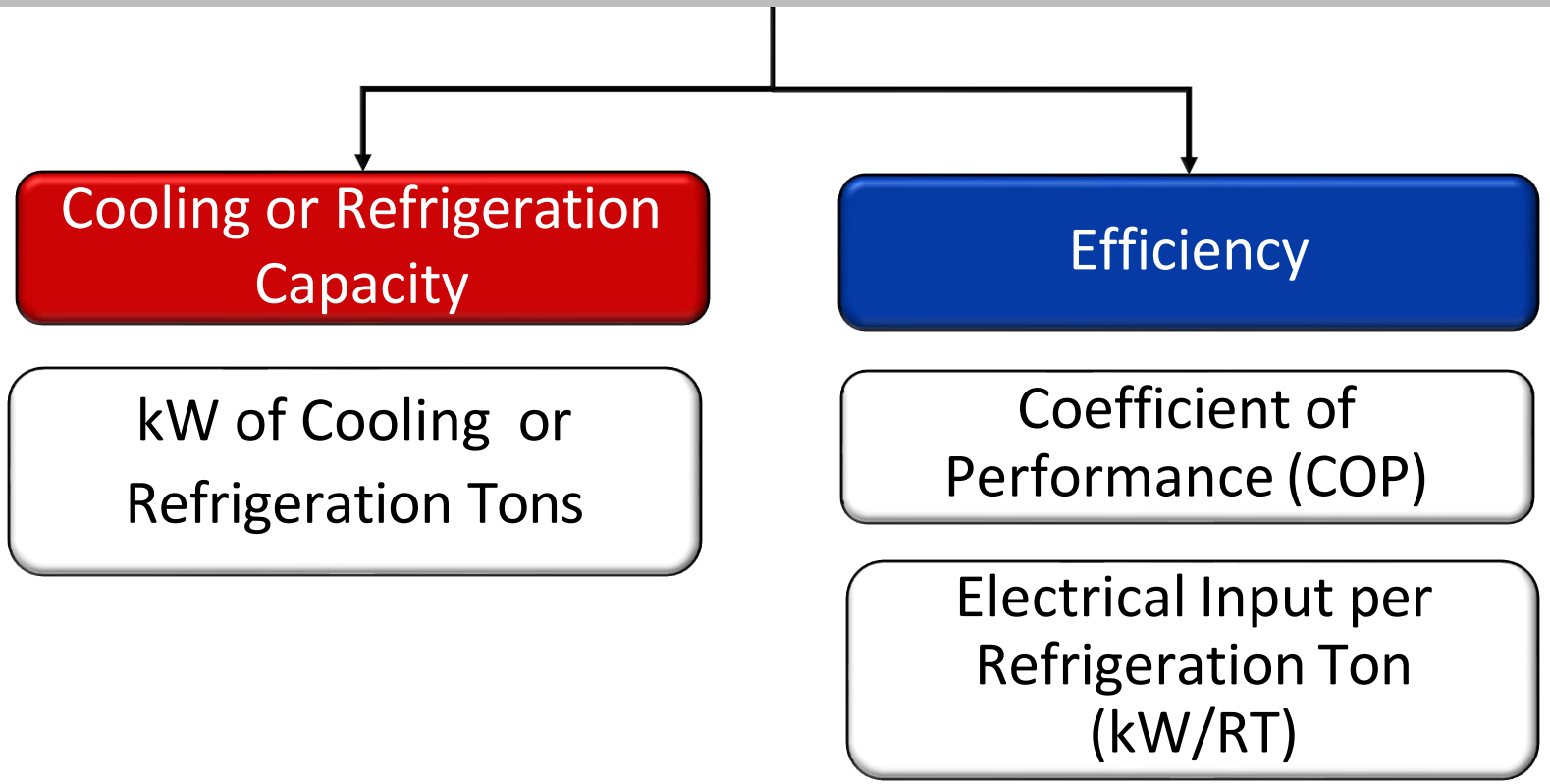
Maintenance

- Yearly
  - Check and adjust control and safety devices and set-points.
  - Check inlet guide vane movement and lubricate linkage.
  - Check oil condition using an oil acid test kit and replace oil if necessary.
  - Inspect and clean cooler tubes.
  - Inspect and clean condenser tubes.

Maintenance

- Yearly
  - Overhaul purge unit.
  - Replace lubrication oil and oil filter.
  - Replace refrigerant filter.
  - Check connectivity of electric circuit
  - Check motor insulation.

Chiller Performance



- A **refrigeration ton** represents the rate of heat energy absorbed when a ton (1000 kg) of ice melts at 0°C over a 24-hour period.
- 1 RT = 3.517 kW of Cooling

Evaporator/Condenser Approach

- Cleanliness of evaporator and condenser affect performance of chillers.

- Monitored by evaporator approach and condenser approach.

Evaporator Approach

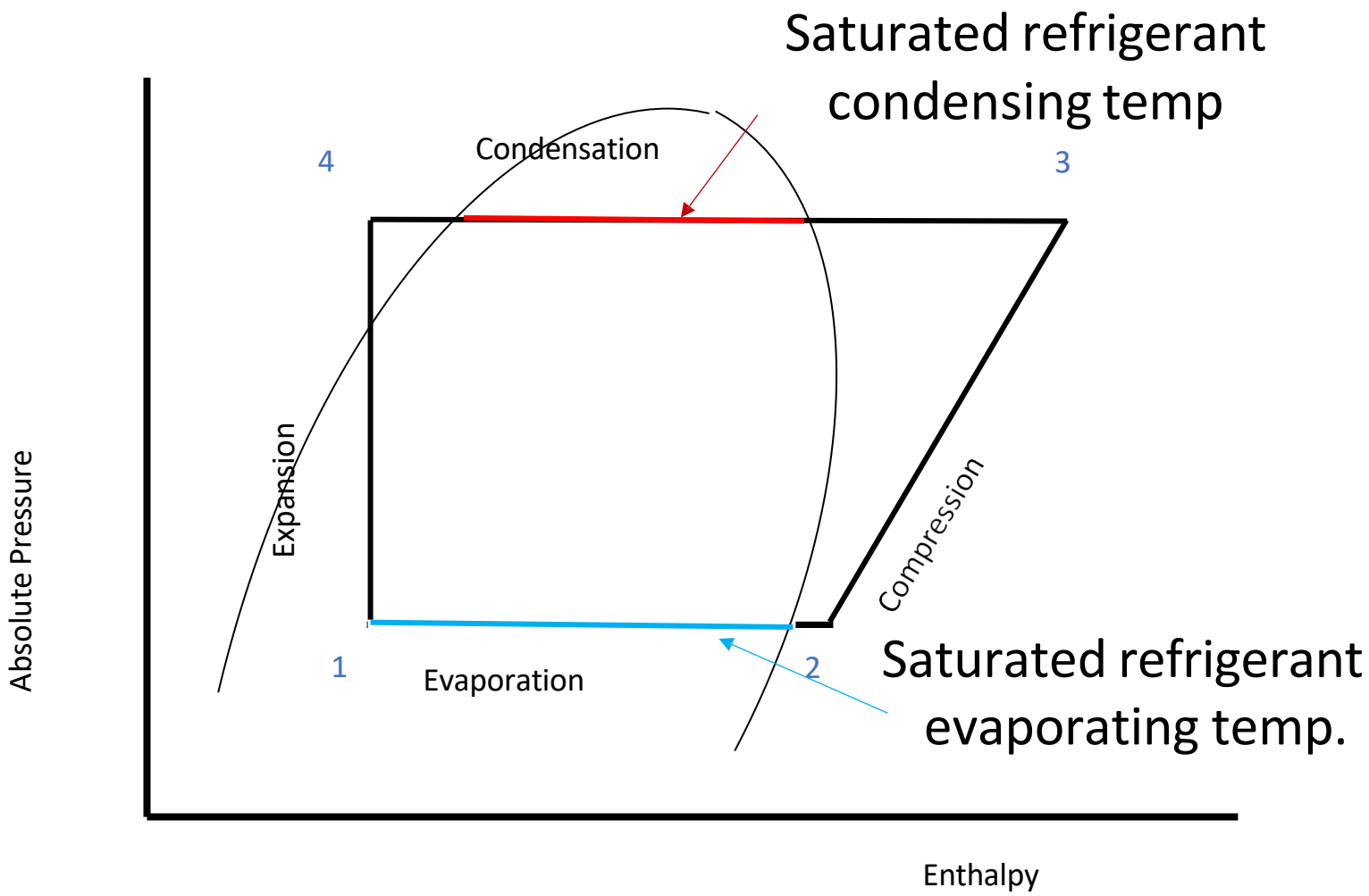
= Leaving (Supply) chilled water temp. – Saturated refrigerant evaporating temp.

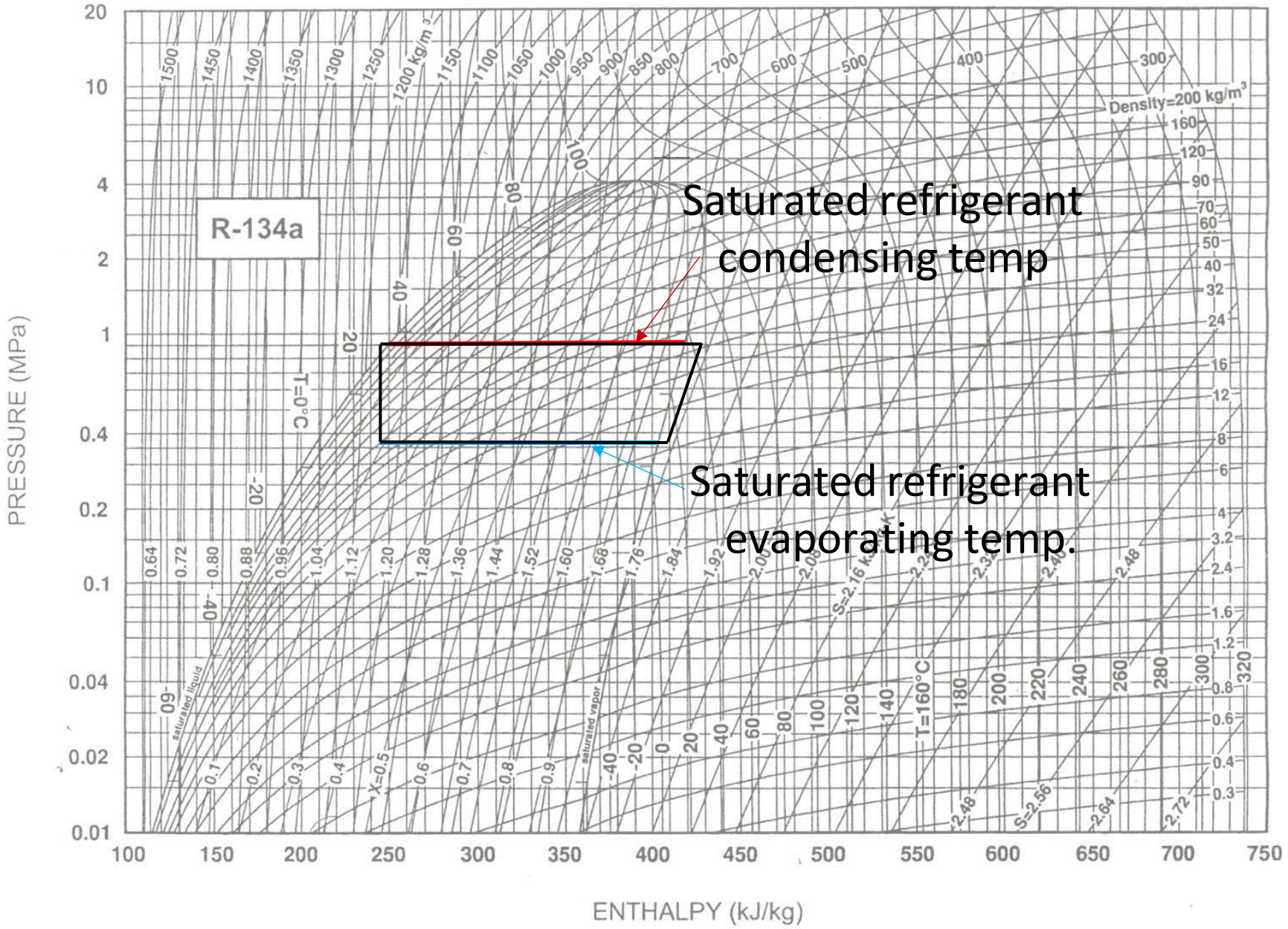
Condenser Approach

= Saturated refrigerant condensing temp. – Leaving (Return) condenser water temp.

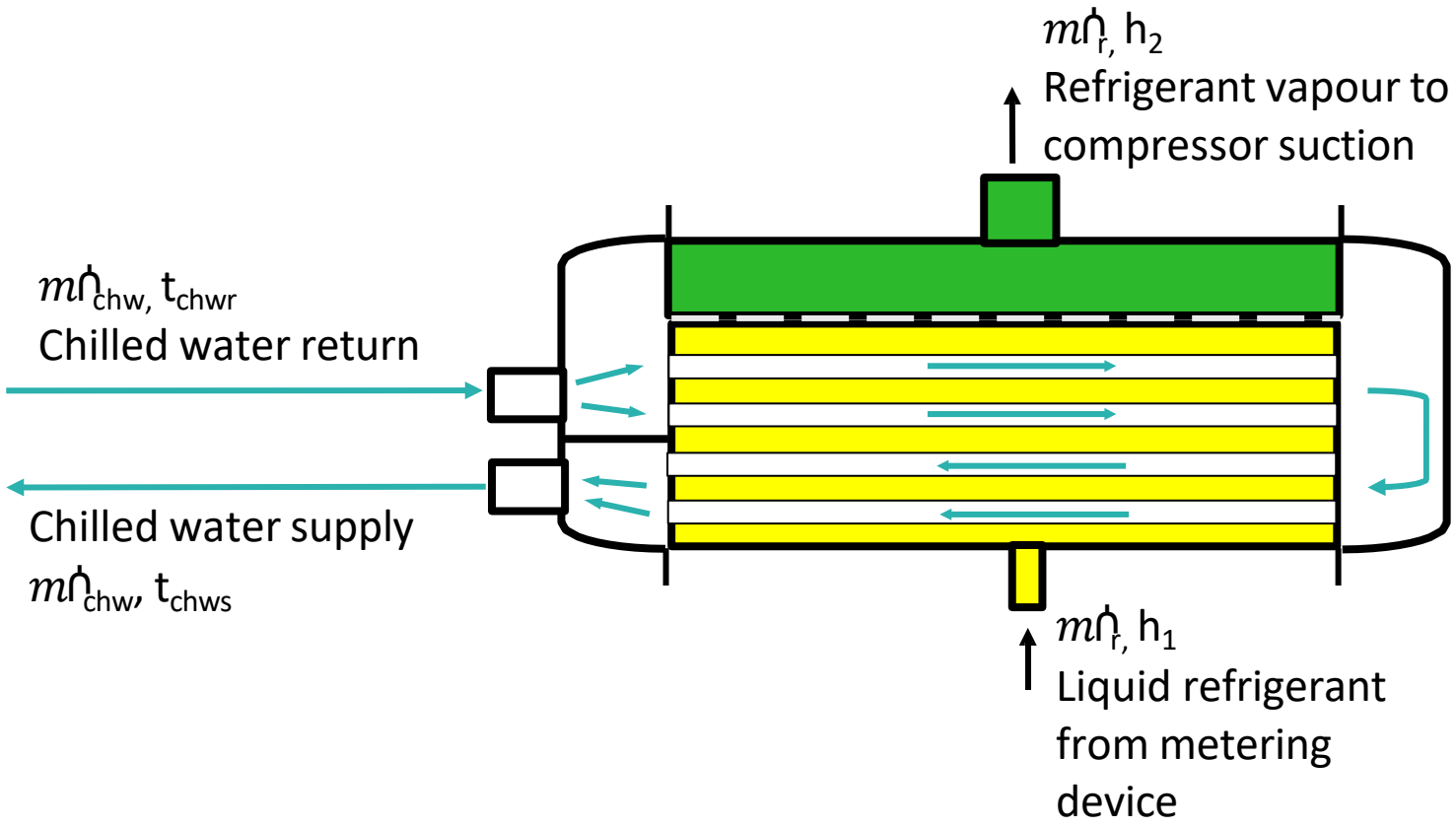
- Approach should be between 1°C and 2°C.
- Smaller the approach, more efficient the heat transfer.

Evaporator/Condenser Approach




R-134a
Saturated refrigerant condensing temp
Saturated refrigerant evaporating temp.
PRESSURE (MPa)
ENTHALPY (kJ/kg)
Density=200 kg/m³
T=160°C
saturated liquid
saturated vapor

Chiller Cooling Capacity



Chiller Cooling Capacity

Chiller cooling capacity in kW

$= \dot{m}_{chw}\, c_w\, (t_{chwr} - t_{chws})$ or $\rho_w \dot{V}_{chw}\, c_w\, (t_{chwr} - t_{chws})$ kW

Chiller cooling capacity in RT = $\dfrac{\text{Cooling Capacity in kW}}{3.517}$ RT

Where : $\dot{m}_{chw}$ = mass flow rate of chilled water, kg/s

c_w = specific heat capacity of water = 4.2 kJ/kgK

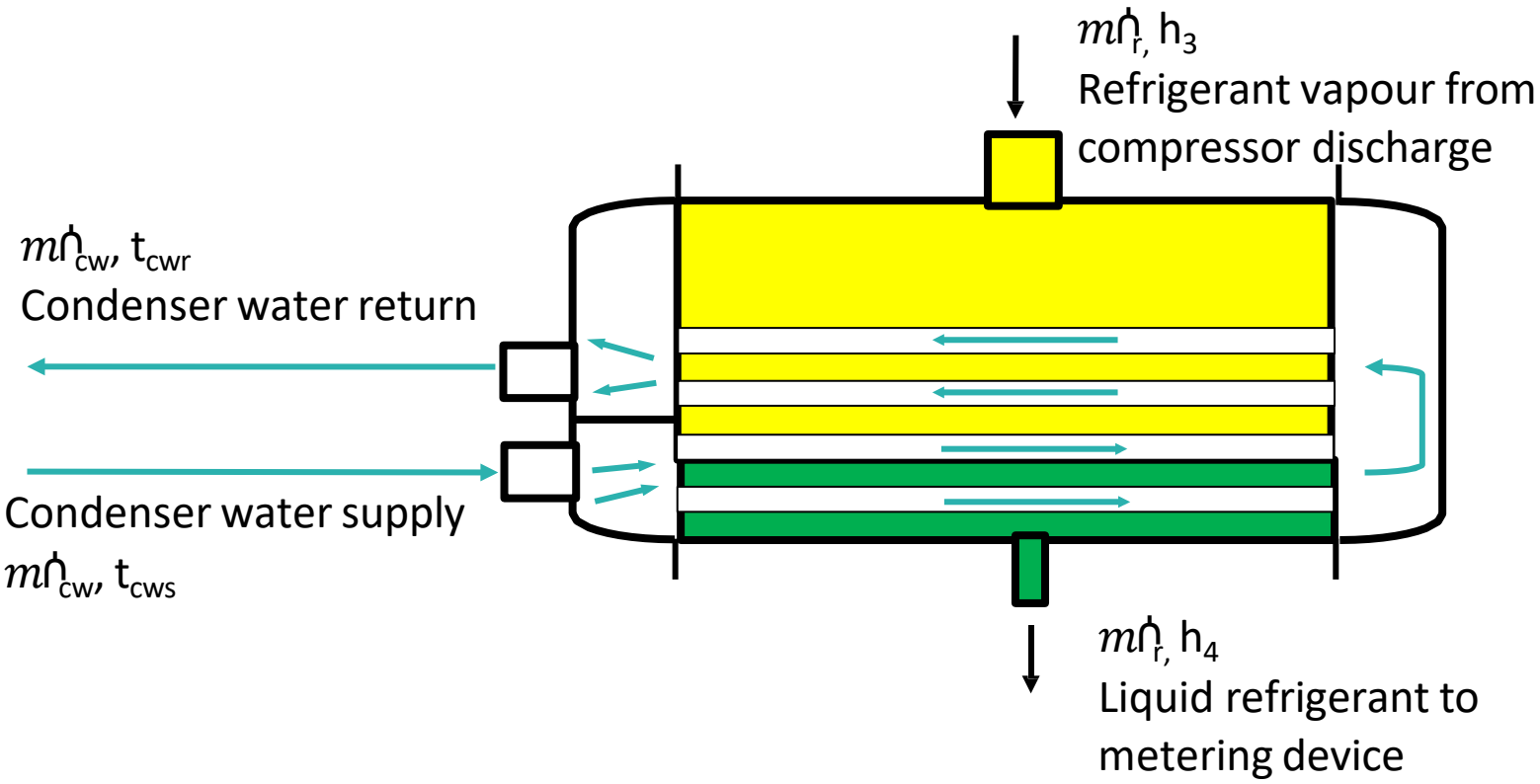
t_{chws} = chilled water supply temperature, °C

t_{chwr} = chilled water return temperature, °C

$\dot{V}_{chw}$ = volume flow rate of chilled water, m^3/s

ρ_w = density of water = 1000 kg/m^3

Chiller Heat Rejection Rate



Chiller Heat Rejection Rate

Based on the condenser water side,

Chiller heat rejection rate

$= m_{cw}\ c_w\ (t_{cwr} - t_{cws})$ or $\rho_w .\ V_{cw}\ c_w\ (t_{cwr} - t_{cws})$ kW

Where m_{cw} = mass flow rate of condenser water, kg/s

c_w = specific heat capacity of water = 4.2 kJ/kgK

t_{cws} = condenser water supply temperature, °C

t_{cwr} = condenser water return temperature, °C

V_{cw} = volume flow rate of chilled water, m^3/s

ρ_w = density of water = 1000 kg/m^3

Alternatively, Chiller heat rejection rate (kW)

= cooling capacity (kW) + power consumption of chiller (kW)

Chiller Efficiency

$$\text{Chiller efficiency} = \frac{\text{Chiller Electrical Input Power}}{\text{Refrigeration Capacity in RT}} \quad \text{kW/RT}$$

$$\text{COP} = \frac{\text{Refrigeration Capacity}}{\text{Compressor Power}}$$

For chillers with hermetic compressors, the relationship between chiller efficiency in kW/RT and COP is:

$$\text{kW/RT} = \frac{3.517}{\text{COP}}$$

Example

The following readings were taken at a water-cooled chiller:

Chilled water return temperature = 13°C
Chilled water supply temperature = 8°C
Volume flow rate of chilled water = 50 l/s
Condenser water return temperature = 34°C
Condenser water supply temperature = 29°C
Volume flow rate condenser water = 59 l/s
Chiller electrical power input = 189 kW

Determine:

(a) Chiller cooling capacity
(b) Chiller heat rejection rate
(c) Chiller efficiency in kW/RT
(d) Chiller efficiency in COP.

Example - Solution

(a) Chiller cooling capacity in kW = $\rho_w V_{chw} . c_w (t_{chwr} - t_{chws})$ kW

$$= 1000 \times \frac{50}{1000} \times 4.2 \times (13 - 8) = 1050 \text{ kW}$$

$$\text{Chiller cooling capacity in RT} = \frac{\text{Cooling Capacity in kW}}{3.517}$$

$$= \frac{1050}{3.517} = 298.5 \text{ RT}$$

Example - Solution

(b) Chiller heat rejection rate $= \rho_w . V_{cw} . c_w (t_{cwr} - t_{cws})$ kW

$= 1000 \times \frac{59}{1000} \times 4.2 \times (34 - 29)$

$= 1239$ kW

Alternatively,

Chiller heat rejection rate = cooling cap. + chiller elect. power input

= 1050 + 189 = 1239 kW

Example 8.1 - Solution

(c) Chiller efficiency in kW/RT = $\frac{\text{Chiller electrical power input}}{\text{Chiller cooling capacity in RT}}$

$= \frac{189}{298.5} = 0.63$ kW/RT

(d) Chiller efficiency in COP = $\frac{3.517}{\text{kW/RT}} = \frac{3.517}{0.63} = 5.58$

Chiller Efficiency

Efficiency Comparison for Different Chiller Types

	Reciprocating (kW/ton[2])	Centrifugal (kW/ton)	Rotary (kW/ton)	Absorption[3] (kW/ton)
Full load	0.84–1.00	0.48–0.65	0.70–0.80	3.2–5.6
Part load	0.84–1.00	0.55–1.00	0.75–0.90	Slightly higher than full load

[2] A lower number indicates a more efficient chiller.
[3] Because the input energy is not electricity, the units are converted to kW of heat energy demand.

Which type is more efficient?

Organisation	Recommended Minimum
ASHRAE	0.68 kW/ton
FEMP	0.56 kW/ton

Best Available 0.47 kW/ton

Air Handling Unit

Air Handling Unit

- Functions
  1. Mixes the return air with outdoor air
  2. Filters the mixed air
  3. Cools and dehumidifies the mixed air
  4. Delivers the cooled and dehumidified air to the rooms.

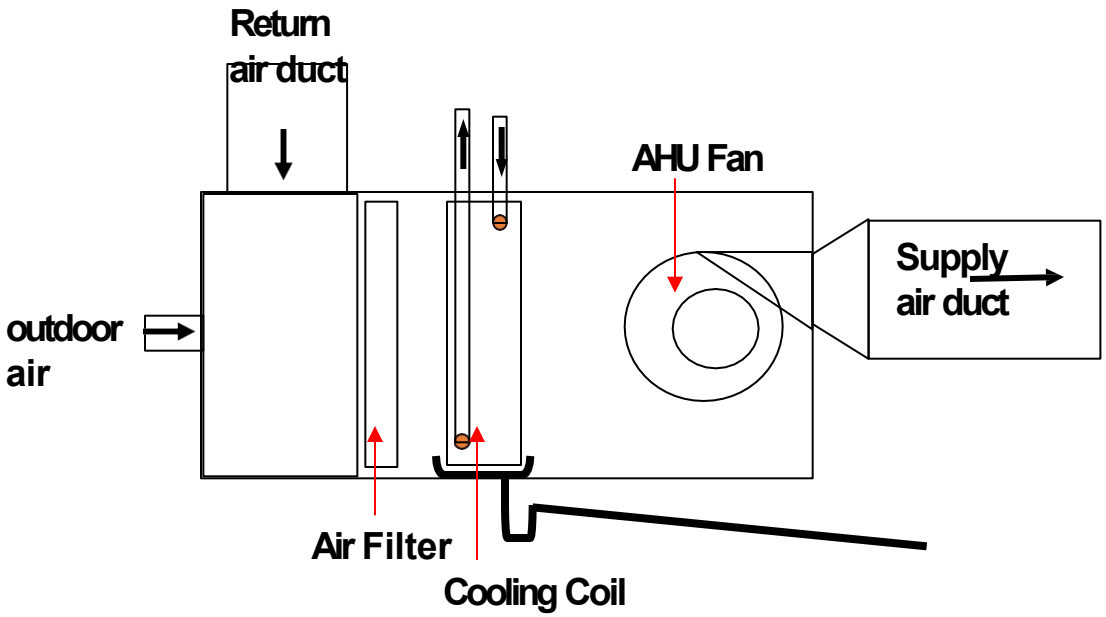
Air Handling Unit

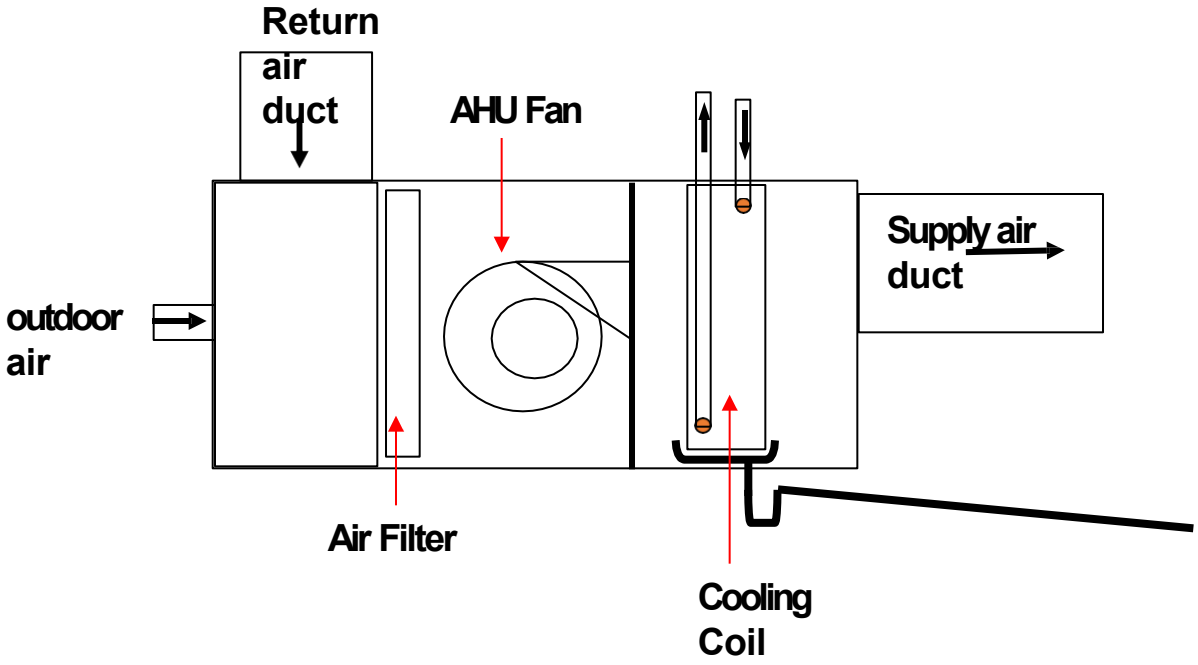
- Construction

  - Modular.

Typical unit will include:

- a mixing module
- a filter module
- a cooling coil module
- a supply fan module
  - blow thru' or draw thru',
  - backward curved centrifugal
  - belt and pulley transmission (CAV)
  - VAV-inlet guide vane/inverter control(VSD) (VAV)

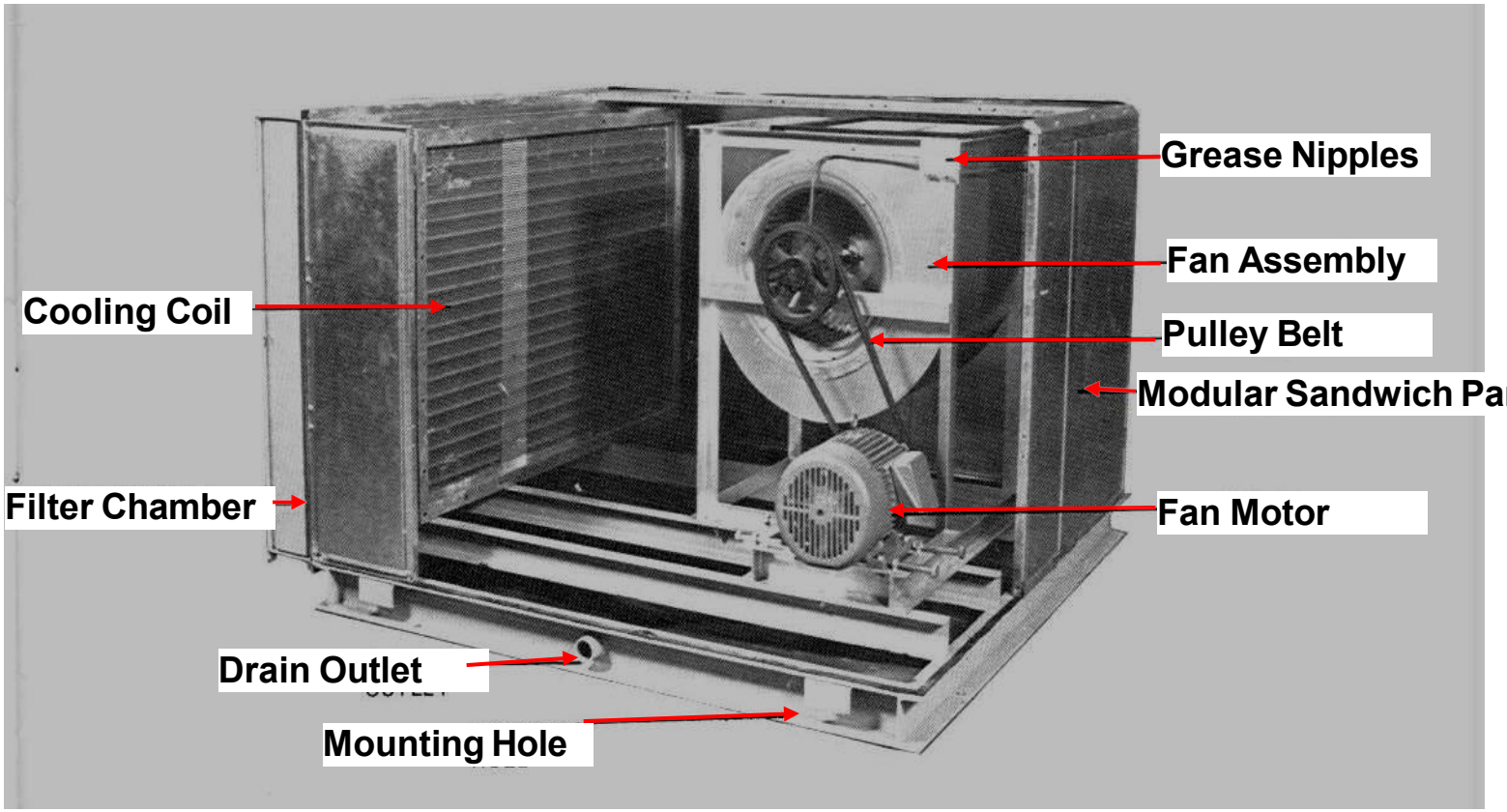

Return
air duct
AHU Fan
Supply
air duct
outdoor
air
Air Filter
Cooling Coil


Return
air
duct
AHU Fan
Supply air
duct
outdoor
air
Air Filter
Cooling
Coil

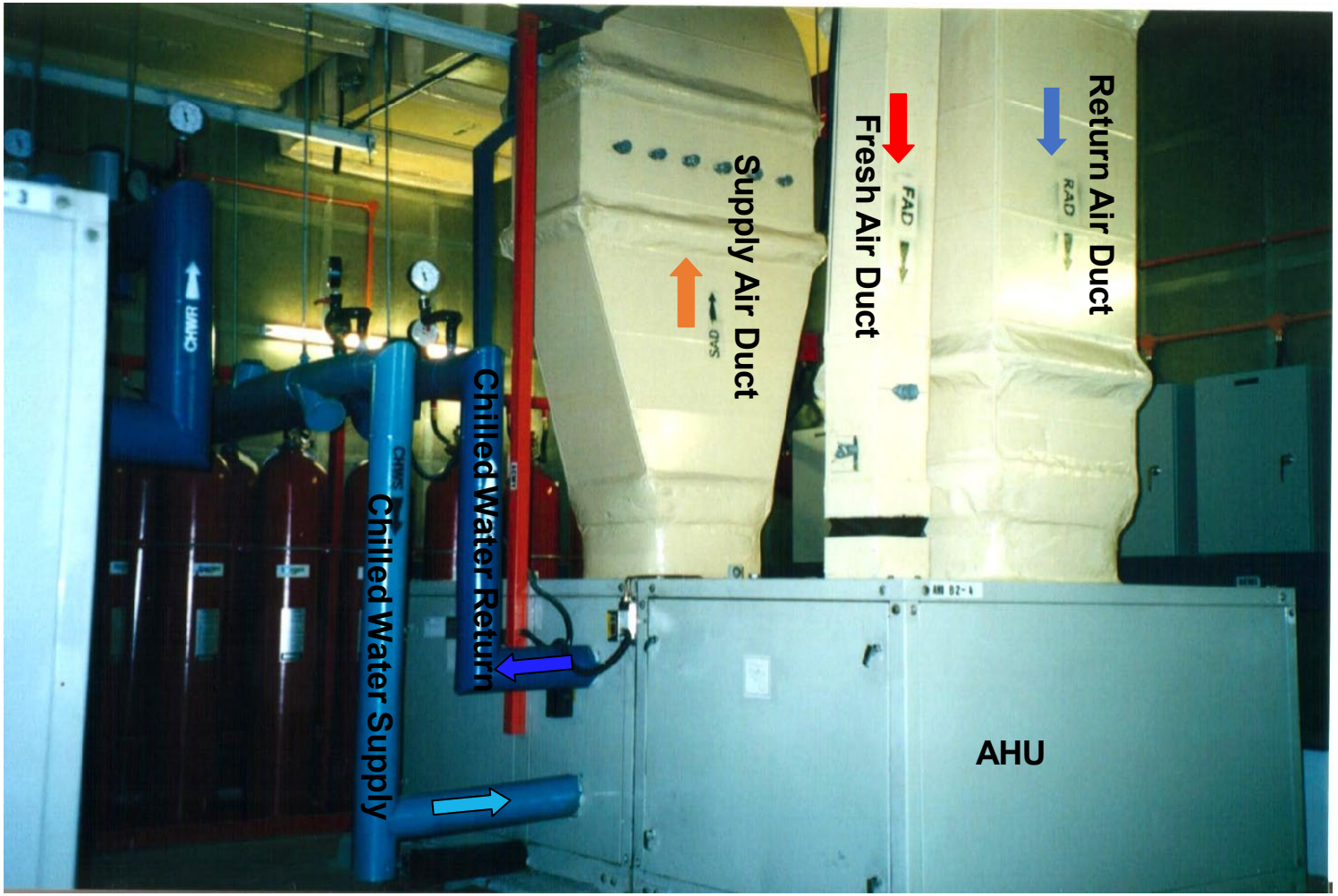
Air Handling Unit



Air Handling Unit



Air Handling Unit



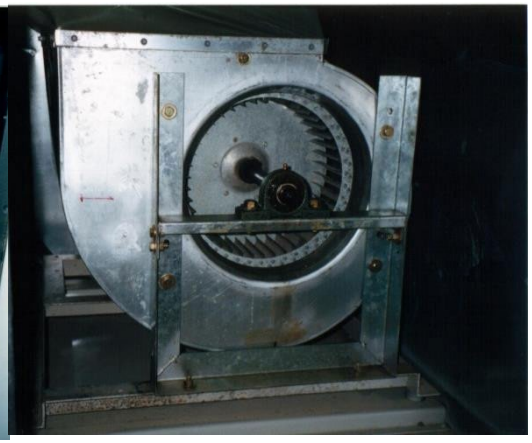
Air Handling Unit





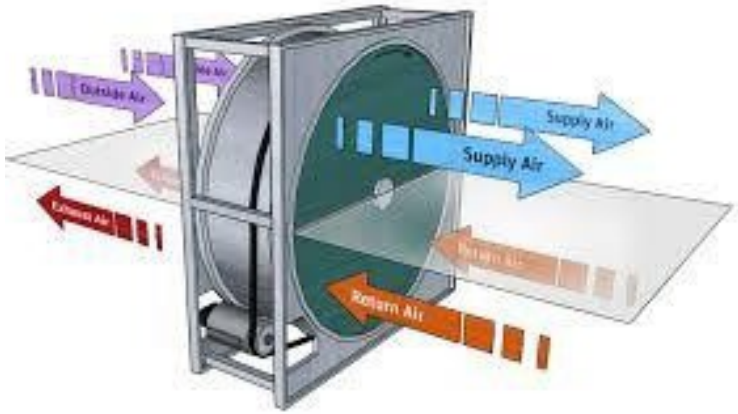






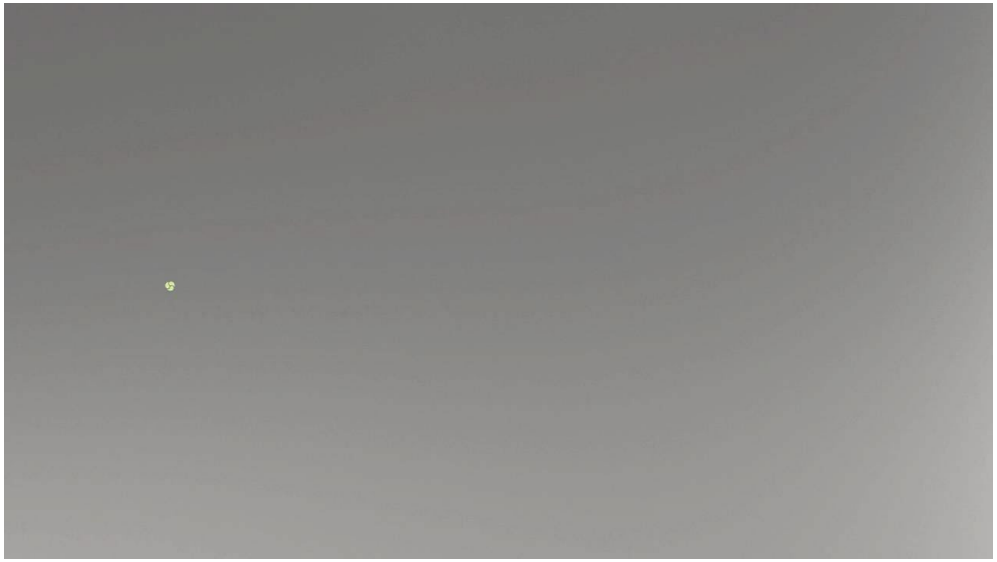
[Conversion to EC centrifugal]

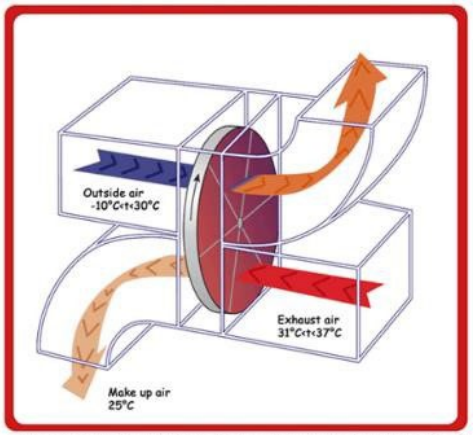
Heat Recovery



Heat Recovery Wheel







A diagram of a rotary heat exchanger, or "heat wheel" (From Uptime Technology BV)

Air Handling Unit

- Modules that can be added to AHU:
  - damper module
  - heating coil module
  - humidifier module
  - heat recovery module
  - bag or absolute filter module
  - sound attenuating module

Logging of Operating Parameters

- Parameters logged everyday:
  - Chilled water supply and return temps
  - Chilled water supply and return pressures
  - On-coil and off-coil dry-bulb and wet-bulb temperatures

Maintenance

- Daily Inspection
  - Check for any unusual noise and vibration.
  - Check for any water and air leaks.
  - Check pressure differential across air filters.

Maintenance

- Monthly Inspection
  - Check and adjust belt tension.
  - Lubricate fan and motor bearings.
  - Lubricate inlet guide-vane linkage.
  - Check and clean condensate pan.
  - Wash air filters if they are the washable type and replace if of disposable type.
  - Check operation of smoke detector
  - Check for obstruction to outdoor air intake.

Maintenance

- Every 6 months
  - Clean fan blades.
  - Clean cooling coil with chemicals or low pressure steam
  - Comb coil fins if necessary.
  - Check and adjust inlet guide-vane mechanism.
  - Test controllers and adjust set points if necessary.

Cooling Towers

Cooling Tower

- Performance of a cooling tower depends on wet-bulb temperature of air and is independent of its dry-bulb temperature.
- Efficiency of cooling tower is affected:
  - Velocity of air through tower
  - Area of wetted surface exposed to air
  - Enthalpy difference between air & water
  - Direction of air flow relative to water flow

Cooling Tower

Definitions

- Range

  Temperature difference between water entering and leaving cooling tower.

- Approach

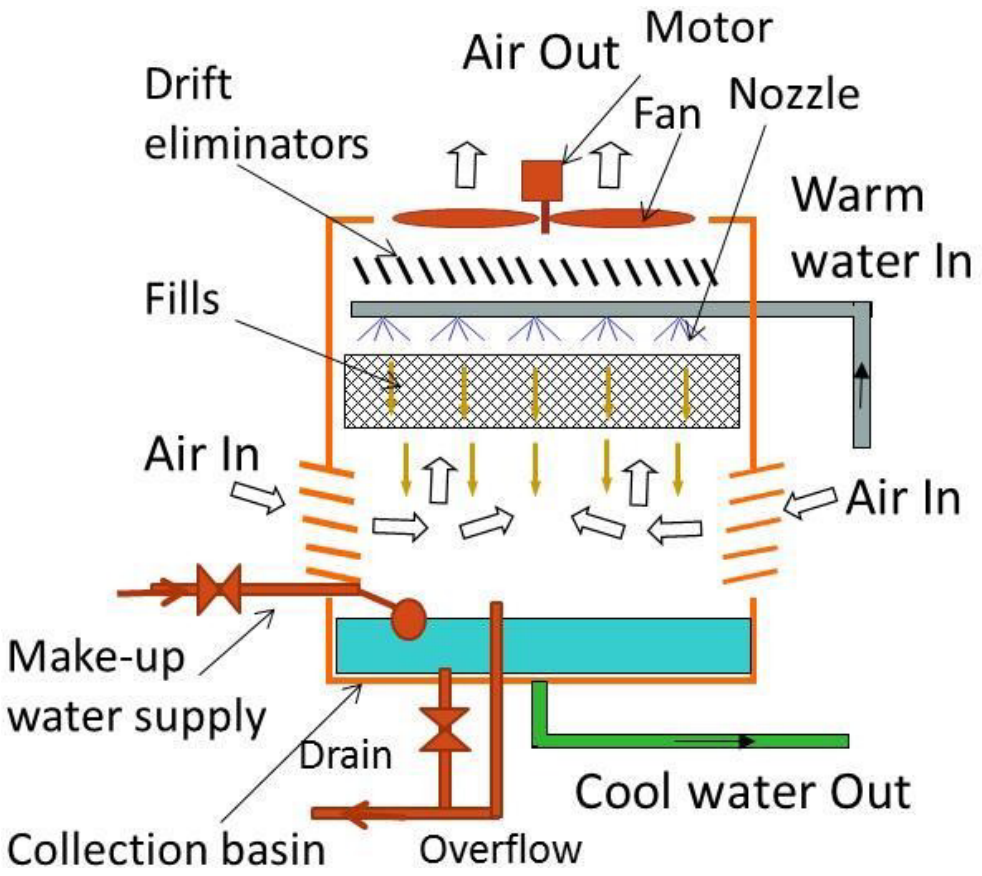
  Difference between leaving water temperature and entering air wet-bulb

Cooling Tower

- General Construction of Cooling Towers
  - Casing
  - Fill or packing
  - Nozzles or water troughs
  - Basin
  - One or more fans with pulleys/belts or gearboxes
  - Drain pipe
  - Bleed-off or blowdown pipe
  - Make-up water

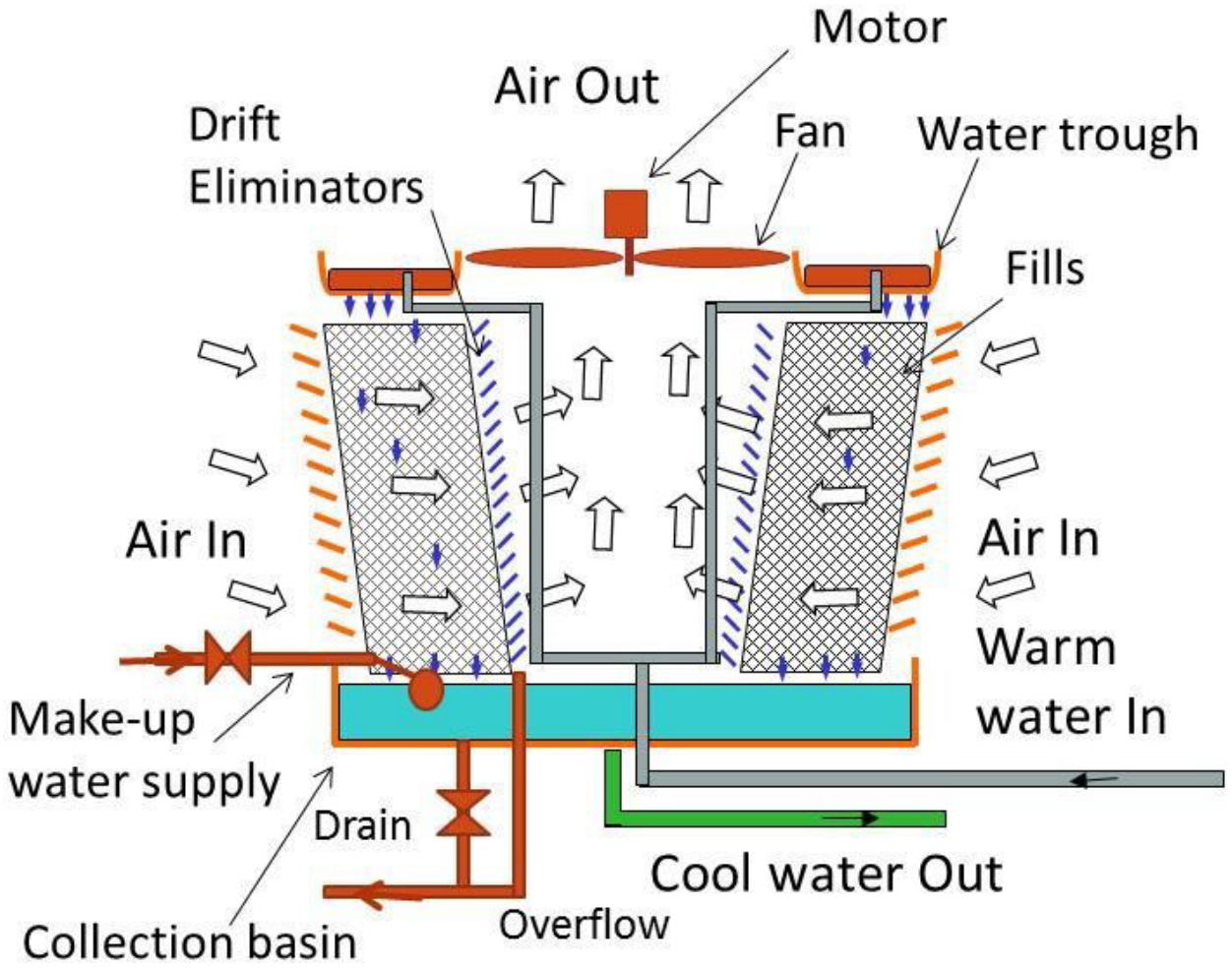
Types of Cooling Tower

Induced Draught (Draft)

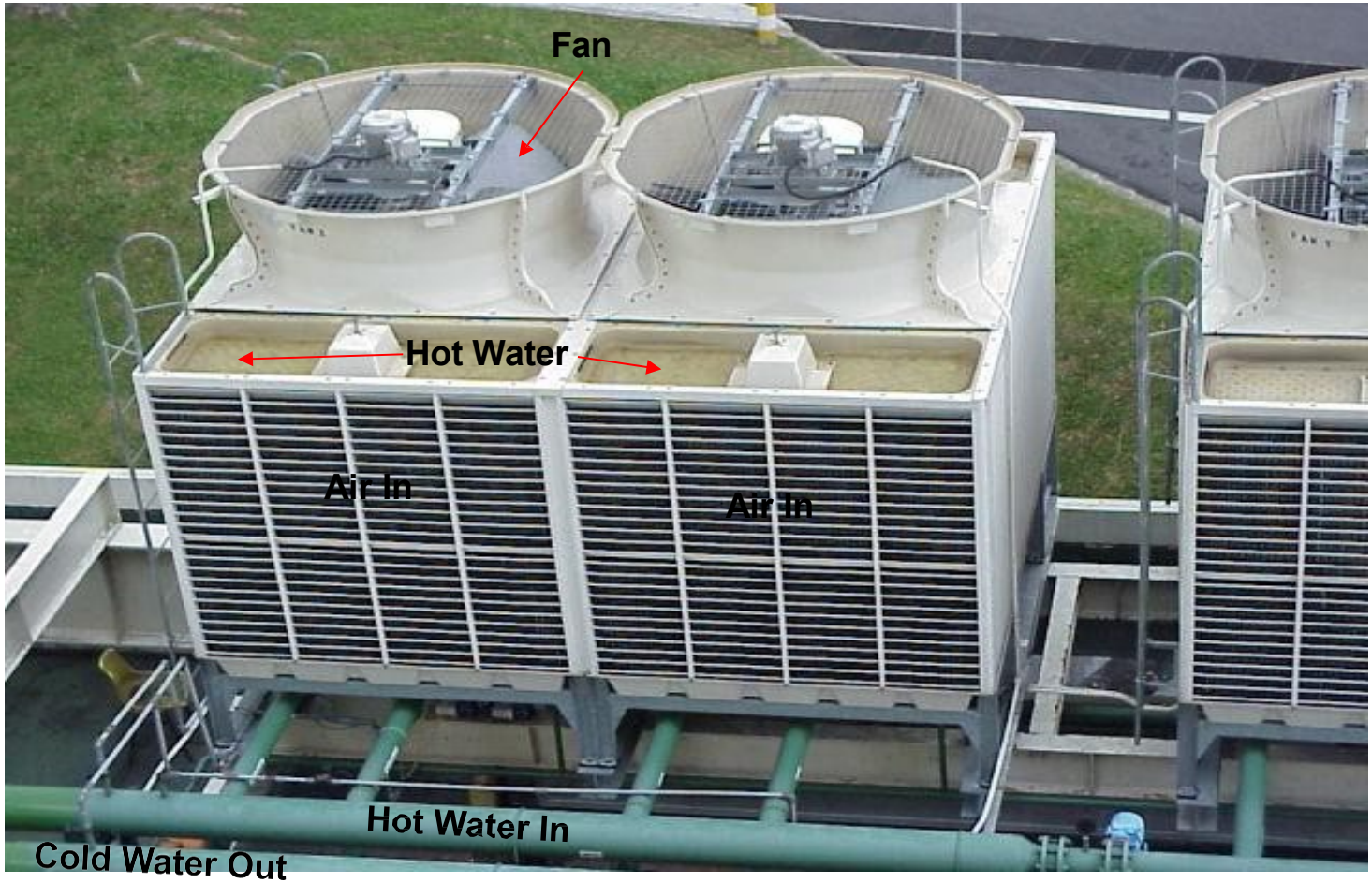


Types of Cooling Tower

Cross Draught (Draft)



Cooling Tower



Cooling Tower



Make-up Water

- Evaporation loss – ~ 1% for every 7°C range.
- Drift and windage loss ~ 0.2% .
- Bleed-off depends on concentration control ~0.3%.

Location of Cooling Tower

- Points to be taken into consideration:
  - Sufficient clearance for unrestricted airflow
  - Noise
  - Condensation on nearby walls & surfaces.
  - Air inlet located away from exhaust or contaminated air.
  - Air discharge located away from any outdoor air intake, doorways, windows

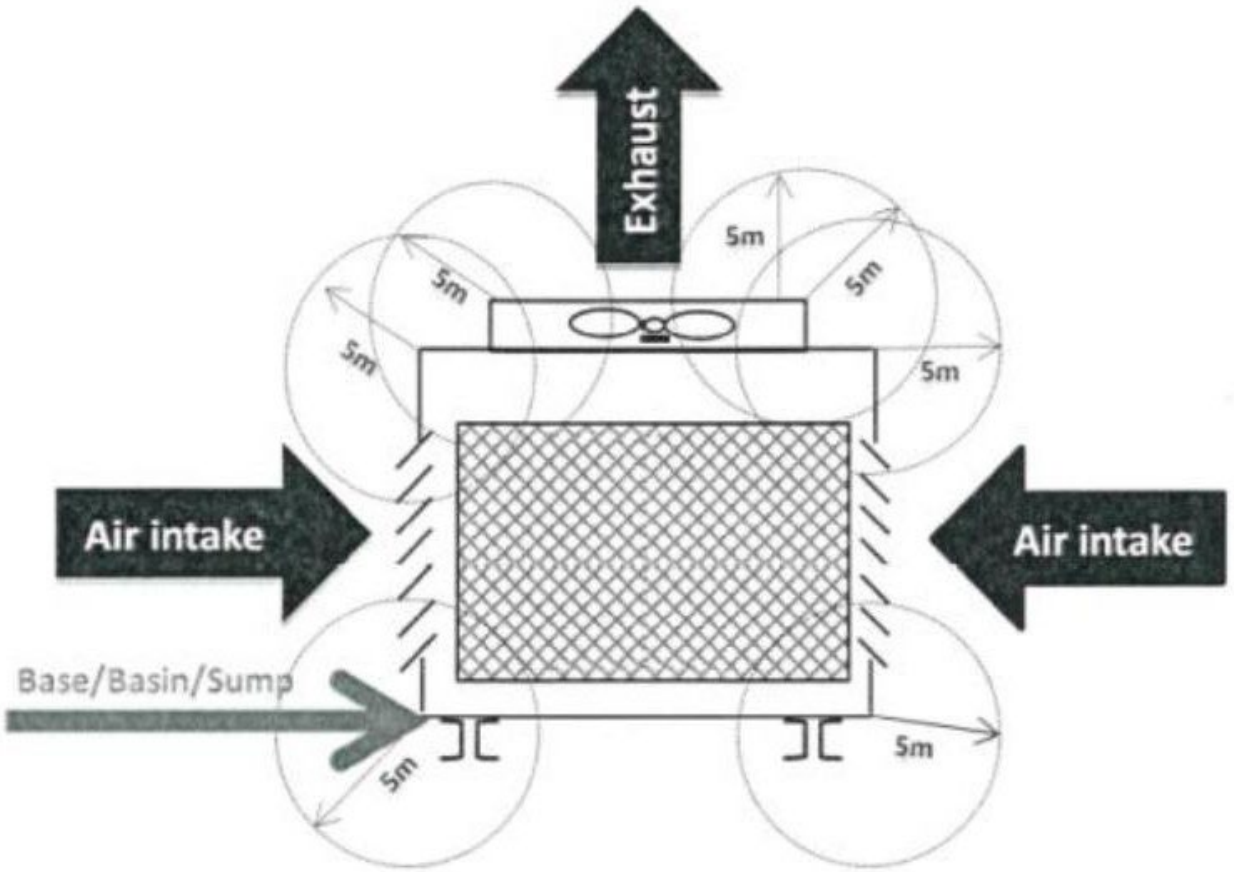
CP on Environmental Health (15th Edition)

- At least 5m from any air circulating and ventilating inlet, openable window, occupied area, pedestrian thoroughfare, trafficable area and any other areas of public access.

- At least 5 m from any kitchen exhaust discharge outlet, air handling system or other areas.

- 5m setback distance from nearest edge or structure of cooling towers, including base/basin/sump, packing exhaust and outlet point of exhaust hood, if any is installed.

CP on Environmental Health (15th Edition)



Source: CP on Environmental Health (15th Edition), NEA

CP on Environmental Health (15th Edition)

- Access routes such as doorways leading to areas within 5m setback distance - clear signage to inform and deter public from entry
- Accessible only by cooling tower maintenance staff who understand risk of Legionellosis and will work in accordance with their safe work procedures based on risk assessment done

Maintenance

Daily

- Inspect for obstruction to air flow.
- Check fan and motor for unusual noise and vibration.
- Check water-basin level and float.
- Check water treatment.

Maintenance

Monthly

- Inspect belts and pulleys, if applicable and adjust or replace
- Check oil in gear box if applicable.
- Check spray nozzles or water distribution holes for clogging.
- Check water-treatment effect.
- Clean intake strainer.
- Clean cooling tower of mud, silt, slimes, algae etc.

Maintenance

Six Monthly

- Lubricate fan and motor bearings as required.
- Replace oil in gear box, if applicable.

Yearly

- Check for corrosion or decay and execute repairs as necessary.

Water Treatment

- Scale Formation
- Corrosion
- Algae and Slime
- Dirt

Legionnaires' Disease

- Legionella pneumophila
- Incubation period is 2- 10 days
- Initial symptoms – malaise, headache followed by high fever and mental changes.
- Other symptoms - gastro-intestinal symptoms, dry cough.
- If untreated - extensive lung consolidation, respiratory failure and death.

Legionnaires' Disease

- Temperature of water - ideal for bacteria
- Water treatment is needed to inhibit multiplication
- Design and siting of towers to minimise likelihood of inhalation of fine water aerosols which are constantly ejected.

Legionnaires' Disease

- Assessment of risk of Legionnaires' disease, responsibility of building owner/Management Corporation
- Comprehensive maintenance programme including cleaning, disinfecting and water treatment is required.
- Formal record of maintenance
- Regular water sample tests
- Weekly visual inspections.
- Programme for routine disinfection, cleaning and manual desludging - twice yearly.

Procedure for Routine Cleaning and Disinfection

- Circulate water containing 5 mg/l of free chlorine with a biodispersant for 6 hours.
- Drain the system.
- Manually clean tower, sump, and distribution system.
- Refill with fresh water, and re-chlorinate to maintain a minimum level 5 mg/l for 6 hours, including a biodispersant;

Procedure for Routine Cleaning and Disinfection

- Recirculate to maintain a minimum level 5 mg/l for 6 hour;
- Drain and flush the system. Refill with fresh water and dose with the appropriate start-up level of treatment chemicals, including biocides;
- Recommission the system.

Formal Record of Maintenance

- Description of cooling tower - location, model and capacity.
- Name of person who ensures that proper records are kept.
- Person/company responsible for assessment of risk, and implementing and managing precautionary measures.
- Person/company carrying out water treatment

Formal Record of Maintenance

- Details of Maintenance:
  - Date and result of visual inspection
  - Date of cleaning and disinfection
  - Date of chemical treatment
  - Date of biocidal treatment
  - Monitoring for standard plate count and legionella bacteria count
  - Date and nature of any remedial works.

Water Sampling and Laboratory Test for Legionella Bacteria

- Standard Plate Count
  - At least once a month
- Legionella bacteria count
  - At least once every 3 months

SAC

Water Sampling and Laboratory Test for Legionella Bacteria

- Sample taken before dosing of biocide. If disinfection is carried out, sample taken no less than 3 days after disinfection.
- Sample taken from cooling tower pond. Systems working for at least one hour prior to collection of sample.

Water Sampling and Laboratory Test for Legionella Bacteria

- Not collected near a make-up water inlet or with sediments that have been deliberately stirred up.
- Transported at ambient temperature to approved laboratory within same working day.
- One water sample from each cooling tower.
- Individual report submitted for each cooling system to the Commissioner of Public Health, NEA, through e-submission.